Factors affecting inferences on natural mortality and associated environmental effects in state-space age-structured assessment models

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Abstract

Treatment of natural mortality is a major consideration in assessment models and state-space approaches allow estimation of temporal variation in this mortality rate as well as effects of specific covariates. However, there has been no investigation of the reliability of inferences made from state-space assessment models when there is process error or covariate effects on natural mortality. We conducted a large-scale simulation study with operating models simulating data assuming alternative levels of observation error, types and magnitude of process error, and magnitude of covariate effects on natural mortality. We fit estimating models to the simulated data with alternative assumptions regarding the inclusion of covariate effects, estimation of the median natural mortality rate, and type of process error. We found estimating covariate effects on natural mortality required ideal scenarios with lower observation uncertainty, contrast in fishing pressure, and higher covariate variability, but the source of process error could be mis-specified. Estimating annual natural mortality rate as random effects was also challenging, but spawning stock biomass and fishing mortality was reliable in a wide range of scenarios.

keywords: state-space assessment models, time-varying natural mortality, bias, AIC

Introduction

State-space population models are now used widely for fisheries stock assessment in Europe, the United States, and Canada (Nielsen and Berg, 2014; Cadigan, 2016; Pedersen and Berg, 2017; Stock and Miller, 2021). Because application of these methods are considered best practice and recommended for the next generation of stock assessment models (Hoyle et al., 2022; Punt, 2023), it is expected their use will only grow globally. An appeal of state-space models lies in their separation of sources of biological and measurement variability by treating temporal variation in population characteristics as latent stochastic processes with periodic observations measured with error. Through advances in computational capacity, we can use sophisticated numerical approaches to estimate parameters of these models as a combination of fixed and random effects (Thorson and Minto, 2015; Kristensen et al., 2016).

State-space stock assessment models with non-linear functions of latent processes and numerous observation types with different probability distributions, represent one of the most complex classes of state-space models. Our understanding of the effects of various factors on reliability of inferences from state-space assessment models is growing (Li et al., 2024; Miller et al., 2026). The importance of temporal contrast in population size and fishing mortality (F) and quality of data used to fit both traditional and state-space assessment models is known (Magnusson and Hilborn, 2007; Miller et al., 2026).

The effects of temporal variation in recruitment via undefined or explicit environmental factors have been extensively investigated in both traditional assessment models and state-space models (Myers, 1998; Haltuch and Punt, 2011; Johnson et al., 2016; Miller et al., 2016). Reliability of estimating environmental and spawning stock biomass (SSB) effects on recruitment in state-space assessment models requires a combination of strong effects, informative age-composition data, contrast in the environmental covariate, low variability of recruitment deviations, and contrast in SSB (Britten et al., 2026; Miller et al., 2026).

A critical aspect of fisheries assessment models and their use in management is short-term projections that are used to determine catch advice. While understanding drivers of recruitment is important, particularly for subsequent effects on reference points, recruitment in short-term projections typically has little impact on the exploitable biomass in the first few projection years. However, assumptions for M have immediate and larger effects on projected biomass because they affect the abundances at older age classes at the end of the data time series that constitute SSB and catch (Brodziak et al., 2008; Stock et al., 2021).

Estimation of the natural mortality rate (M) is possible in many scenarios (Lee et al., 2011; Miller and Hyun, 2018; Miller et al., 2026). Because of the effects of M on both biological reference points and short-term projections, better understanding sources of variation in M would provide more accurate estimation of abundance and productivity and, therefore, improved management. Temporal variation in M is less studied than recruitment, but its importance for explaining variability in observations has been demonstrated in state-space assessment models for stocks of Atlantic cod and yellowtail flounder (e.g., Cadigan, 2016; Stock et al., 2021). Furthermore, Deriso et al. (2008) demonstrated the importance of several factors affecting M annually for Pacific herring.

Assessment models could include temporal variation in many aspects of the population's dynamics or how observations are related to the population. For example, the Woods Hole Assessment Model (WHAM) can include process errors treated as random effects for annual changes in cohort abundance (hereafter referred to as apparent survival), catchability for indices of abundance, selectivity of fishing fleets or indices, movement between regions, or in M (Stock and Miller, 2021; Miller et al., 2025). However, mis-specifying the type of temporal variation could lead to biased estimation of population size and stock status and, therefore, poor fisheries management decisions (Legault and Palmer, 2016; Szuwalski et al., 2018). Studies of the reliability of identifying temporal variability in M are limited. Miller et al. (2026) found AIC could accurately distinguish process errors in apparent survival but not for those specifically due to M except when uncertainty in population observations (indices, catch, and age-composition data) was low and there was greater temporal variation in M. Li et al. (2024) found that including more sources of process error than existed in the operating model (OM) was a better model-building approach than excluding them *a priori*.

Here, we present results from a simulation study with OMs varying by degree of observation-error uncertainty, sources of process error, fishing history, temporal variation in environmental covariates, and magnitude of the covariate effect on M. We fit the simulated observations from these OMs with estimating models (EMs) that make alternative assumptions for sources of process error and whether median M and covariate effects were estimated. We evaluated the effects of these factors on 1) convergence of fitted models; 2) whether Akaike's information criterion (AIC) can determine the correct source of process error and correct assumption about covariate effects on M; 3) reliability of median M and covariate effect estimation; and 4) reliability of terminal year M and SSB estimation.

Methods

Our analyses used WHAM to construct both OMs and EMs (Miller and Stock, 2020; Stock and Miller, 2021; Miller et al., 2025). WHAM has been used extensively to configure OMs and EMs for several other simulation studies (Stock et al., 2021; Legault et al., 2023; Li et al., 2024; Britten et al., 2026; Li et al., 2026a) and is used to assess many commercially important stocks in the Northeast U.S. (e.g., NEFSC, 2022a,b, 2024). We used version 1.0.6.9000 (commit 77bbd94) to generate all results.

The factors defining the configuration of each of the 288 OMs, described in detail in subsequent sections, include source of process error (3 levels), index and catch observation uncertainty (2 levels), environmental covariate uncertainty (2 levels), temporal variation in the latent environmental covariate (4 levels), size of the covariate effect on M (3 levels), and fishing history (2 levels). We simulated 100 populations and covariates with process errors and data sets given the realized populations and covariates for each OM.

For each simulated data set we fit a set of 12 EMs. The factors that distinguish the EMs, also described in detail below, include source of population process error type (3 levels), whether median M was estimated or assumed known (2 levels), and whether the environmental covariate effect on M was estimated or not (2 levels).

The sources of population process error that were used in the OMs or assumed in the EMs were on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and M (R+M). We did not use the log-normal bias-correction feature for process errors or observations described by Stock and Miller (2021) for OMs and EMs. Using this correction with random effects and estimation of fixed effects via marginal likelihood can result in substantial bias in assessment output such as recruitment and SSB (Li et al., 2026b). Simulations were all carried out on the University of Massachusetts Green High-Performance Computing Cluster. Code for completing the simulations and summarizing results can be found at https://github.com/timjmiller/SSRTWG/tree/main/Ecov_study/mortality.

Operating models

Environmental covariate

In WHAM, environmental covariates are modeled as state-space processes with annual observations of the true latent covariate (Miller et al., 2016; Stock and Miller, 2021). In our simulations, we assumed the latent covariate to be a stationary first order autoregressive (AR1) process,

$$X_y|X_{y-1} \sim \mathrm{N}\left(\mu_E\left(1-\rho_E\right)+\rho_E X_{y-1}, \left(1-\rho_E^2\right)\sigma_E^2\right),$$

with marginal mean $\mu_E = 0$ and variance σ_E^2. The four configurations of the latent environmental covariate in the OMs assume one of two values for the marginal standard deviation $\sigma_E \in \{0.1, 0.5\}$ and for the autocorrelation parameter $\rho_E \in \{0, 0.5\}$.

WHAM treats the observations of the latent environmental covariate (x_y) as unbiased and normal conditional on the true covariate X_y,

$$x_y|X_y \sim \mathrm{N}\left(X_y, \sigma_e^2\right).$$

The standard deviation of the environmental observations in the OMs was one of two values $\sigma_e \in \{0.1, 0.5\}$. Figure S1 provides example simulations of the latent covariate and observations under the alternative configurations.

Population

Many of the characteristics of the population biology and structure including the age classes (10 age classes: 1 to 10+), time span (40 years), maturity (Figure S2, top left), growth (Figure S2, top right), time of spawning (1/4 of the year), and recruitment (Figure S2, bottom right) were identical to Miller et al. (2026). The maturity at age was a logistic function with age at 50% maturity (a_{50}) = 2.89 and slope = 0.88. Weight-at-age was derived from a von Beralanffy growth function where $t_0 = 0$, $L_\infty = 85$, and $k = 0.3$, and a length–weight relationship

$$W_a = \theta_1 L_a^{\theta_2},$$

where $\theta_1 = e^{-12.1}$ and $\theta_2 = 3.2$.

In WHAM, the general model for M in year y is a log-linear function of both covariate effects X_y and process errors $\varepsilon_{M,y}$ and a parameter β_M that defines median M,

$$\log M_y = \beta_M + \beta_E X_y + \varepsilon_{M,y},$$

where the process errors are modeled as random effects. The general model for the random effects is an AR1 Gaussian distribution,

$$\varepsilon_{M,y}|\varepsilon_{M,y-1} \sim \mathrm{N}\left(\varepsilon_{M,y-1}, (1-\rho_M^2)\sigma_M^2\right)$$

(Stock and Miller, 2021), but we assumed $\rho_M = 0$ in our R+M OMs such that there is no autocorrelation. We assumed the median M, $e^{\beta_M} = 0.2$, was constant across ages. For R and R+S OMs and EMs, $\varepsilon_{M,y} = 0$. For all R+M OMs, we assumed the standard deviation $\sigma_M = 0.3$, which was estimated in the R+M EMs. The covariate effect was one of 3 alternative values in the OMs, $\beta_{\mathrm{E}} \in \{0, 0.25, 0.5\}$. The parameters defining the simulated covariate time series, size of the covariate effect, and any M random effects resulted in a range of different levels of variation in annual M values (Figure S3).

We assumed expected recruitment each year was from a Beverton–Holt stock-recruit relationship (SRR)

$$R_y = \frac{a\mathrm{SSB}_{y-1}}{1 + b\mathrm{SSB}_{y-1}}.$$

All biological inputs to calculations of SSB per recruit (i.e., weight, maturity, and M at age) were constant in the R and R+S OMs without covariate effects on M. Therefore, steepness and equilibrium unfished recruitment were also constant over the time period for those OMs (Miller and Brooks, 2021). As in Miller et al. (2026), our assumed biological inputs and selectivity (defined below) with constant M resulted in the equilibrium F that reduces SSB per recruit to 40% of the unfished level being $F_{40\%} = 0.348$. With an assumed unfished recruitment of $R_0 = e^{10}$, setting $F_{\mathrm{MSY}} = F_{40\%}$ results in a steepness of 0.69 and $a = 0.60$ and $b = 2.4 \times 10^{-5}$. For R+M OMs and all OMs with covariate effects on M, steepness was not constant but we used the same a and b parameters as other OMs which equates to a steepness and R_0 at the median (or mean on log-scale) of the time series models for M and the covariate.

We also used the same two fishing scenarios as Miller et al. (2026) for OMs. In the first scenario, the stock experienced overfishing at $2.5F_{\mathrm{MSY}}$ for the first 20 years followed by fishing at F_{MSY} for the last 20 years (denoted $2.5F_{\mathrm{MSY}} \rightarrow F_{\mathrm{MSY}}$). In the second scenario, the stock was fished at F_{MSY} for the entire time period (40 years). The magnitude of the overfishing assumptions was intended to reflect estimates of overfishing for Northeast US groundfish stocks from Wiedenmann et al. (2019).

We configured all R, R+S, and R+M OMs with uncorrelated random effects on recruitment with standard deviation on log(recruitment) $\sigma_R = 0.5$. Miller et al. (2026) used this same assumption for R+M OMs and other OMs with fishery selectivity and index catchability process errors. For R+S OMs, apparent survival process errors were uncorrelated with $\sigma_{2+} = 0.3$. These standard deviations were on the lower end of the range of values estimated for groundfish stocks in the NE US (see Table S1 in Britten et al., 2026).

Catch and index observations

We defined the generation of observations of aggregate (total combined across ages) catch and indices and corresponding age-composition observations identical to Miller et al. (2026). There was a single fleet operating year round for catch observations with logistic selectivity defined with $a_{50} = 5$ and slope $= 1$ (Figure S2, bottom left). Observations were generated for all 40 years of the model. There were two index time series intended to represent fishery-independent surveys occurring in the spring (0.25 way through the year) and the fall (0.75 way through the year). We assumed catchability of both surveys to be 0.3.

WHAM treats the aggregate catch and index observations as log-normal and we assumed the standard deviation parameter was 0.1 for all annual catch observations. We assumed catch and index age-composition observations were generated from a logistic-normal distribution where errors on the multivariate normal scale were independent (Aitchison and Shen, 1980; Stock and Miller, 2021; Trijoulet et al., 2023). The standard deviation parameter was also constant across ages. There were two levels of observation uncertainty for aggregate indices and age-composition data for both indices and catch. A low-uncertainty specification assumed the standard deviation parameter was 0.1 for all annual aggregate index observations and the standard deviation of the logistic-normal for all age-composition observations was 0.3. In the high-uncertainty specification the standard deviation parameter for the annual aggregate index observations was 0.4 and that for the age-composition observations was 1.5.

Estimating models

There were three factors defining the configuration of each EM: 1) whether β_M was estimated or assumed known; 2) whether an environmental effect β_E was estimated or not; and 3) whether the process errors were assumed on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and M (R+M). We fit each EM to each of 100 simulated data sets from each OM. The configuration of the process errors in the EMs generally matched the corresponding options in the OMs. For example, random effects were assumed uncorrelated for R EMs and R OMs and for R+S EMs and R+S OMs. Although unintended, R+M EMs were configured to estimate the correlation of M random effects although they were simulated without correlation in the R+M OMs. The environmental covariate observations were included in all EMs, whether effects on M were estimated or not, to ensure comparability of AIC. All fixed-effects parameters for selectivity, catchability, fully-selected F, mean recruitment, initial abundance at age, and standard deviations for logistic-normal age-composition distributions were estimated. Any process-error variance parameters for recruitment, apparent survival, and M were also estimated. For all EMs, the standard deviation parameters for environmental covariates and aggregate catch and indices were assumed known at the true value.

Measures of reliability

EM convergence

We measured the frequency of convergence when fitting each EM to the simulated data sets for each OM. There are various ways to assess convergence of the fit (e.g., Carvalho et al., 2021) but we defined successful convergence as the Hessian of the marginal log-likelihood being invertible and providing variance estimates for the fixed effects parameters as recommended by Miller et al. (2026).

AIC for model selection

We measured the frequency of correct model selection using marginal AIC, known to be appropriate for mixed-effects models specifically when the analyst is interested in fixed effects rather than random effects (Vaida and Blanchard, 2005). Other studies have shown the utility of marginal AIC in distinguishing process error and other model structures for state-space assessment models applied to a particular stock (Miller and Hyun, 2018; Stock and Miller, 2021).

For a given OM, the set of EMs that were compared all made the same assumption about β_M (estimated or treated as known at the true value). For a given EM, the marginal AIC is a function of the marginal log-likelihood maximized with respect to the estimated fixed effects ($\boldsymbol{\theta}$) in the EM and the number of estimated fixed effects $n(\boldsymbol{\theta})$,

$$\text{AIC} = -2\left[\text{argmax}_{\boldsymbol{\theta}} \log L(\boldsymbol{\theta}) - n(\boldsymbol{\theta})\right].$$

All EM fits that successfully completed the optimization were used for this set of analyses. We did not condition on convergence as defined above because some lack of convergence would be expected for the correct behavior of more complicated models that include process errors that did not exist in the OM. For example, R+M EMs fit to R OMs would be expected to estimate no variance in the M random effects and the estimated variance parameter going to zero would cause poor convergence but have the same marginal log-likelihood (with more fixed effects) and therefore higher AIC as expected. The correct EM made the correct assumption for the source of process errors (R, R+S, R+M) and either included a covariate effect on M when an effect was simulated ($\beta_E \in \{0.25, 0.5\}$) or did not include an effect when one was not simulated ($\beta_E = 0$).

Parameter estimation bias and accuracy

All results here were based on OM simulations with fits that satisfied the Hessian-based convergence criterion described above. We used this conditioning to reflect how practitioners would proceed in analyses of model fits with real assessment data.

We focused on statistical behavior of estimators of the covariate effect on M ($\widehat{\beta}_E$); the median M parameter ($\widehat{\beta}_M$); and terminal year M, SSB, and fully-selected fishing mortality (F). In preliminary analyses we examined results for the estimators of all the annual values for M, SSB and F over the whole time series, but we found no appreciable differences in patterns across the various factors defining the OMs and EMs. Furthermore, because results for terminal year F were generally inversely related to those for SSB, they are not discussed further but provided in the Supplemental Materials.

For each EM fit to a given simulated data set, We calculated errors of $\widehat{\beta}_E$ and $\widehat{\beta}_M$, and the Hessian-based standard error estimators of these parameters ($\widehat{SE}\left(\widehat{\beta}_E\right)$ and $\widehat{SE}\left(\widehat{\beta}_M\right)$) as the difference between the estimate and the true value. We defined the true standard errors as the standard deviation of estimates across converged fits of a given EM and OM combination. We calculated relative errors for terminal year M, SSB, and F. For a given attribute θ_i, the relative error for an estimate from an EM fit to a given simulated data set i,

$$\mathrm{RE}_i\left(\widehat{\theta}_i\right) = \frac{\widehat{\theta}_i - \theta_i}{\theta_i}. \tag{1}$$

To measure accuracy of terminal year M, SSB, and F, we calculated the root mean square error (RMSE). Again defining θ as one of these attributes,

$$\mathrm{RMSE}\left(\widehat{\theta}\right) = \sqrt{\frac{1}{n_c}\sum_{i=1}^{n_c}\left(\widehat{\theta}_i - \theta_i\right)^2},$$

where $n_c \leq 100$ is the number of simulations with converged fits of a given EM and OM combination. The true values for terminal year SSB and M (in R+M OMs and any OM with covariate effects on M) varied among simulations. Finally, we calculated 95% confidence intervals (CIs) for $\widehat{\beta}_E$ and $\widehat{\beta}_M$ using Hessian-based standard error estimates for converged EMs that estimated these parameters in each OM simulation and recorded whether they included the true value.

Estimation of terminal year M could appear to be unbiased even for EMs that assumed no temporal variation in M because the assumed or estimated β_M parameter can be equal or near the median of the true terminal year values. This centering of the true values at the constant value of M in the EM happens because the mean of the M process error and covariate distributions were assumed equal to 0. Therefore, we also calculated the correlation of estimates and simulated values for terminal year M for a given EM and OM combination,

$$\text{Cor}\left(\widehat{M}, M\right) = \frac{\text{Cov}\left(\widehat{M}, M\right)}{\text{SD}\left(\widehat{M}\right)\text{SD}\left(M\right)}.$$

For R+M OMs and any other OMs where there were covariate effects on M, high correlation (approaching 1) along with median relative errors close to 0 indicates ability of the EM to estimate annual M. If correlation and median relative errors are both near 0, this indicates no ability to estimate annual M. Note also that the correlation is undefined when either the estimated or true value are constant across simulations because the corresponding standard deviations are 0. This applies to R and R+S EMs that assumed β_M to be known and $\beta_E = 0$, and R and to R+S OMs where $\beta_E = 0$. We also did not use calculated correlations for EMs when there were only two converged fits because they are uninformative (can only take on values of -1 or 1). We made the same correlation calculations for SSB because these estimates and true values also varied across simulations. There may be correspondence of RMSE and the correlation results for M and SSB, but RMSE is a function of both bias and variance whereas higher (or better) correlation can occur regardless of bias and has a unitless scale (-1 to 1) that facilitates comparison across parameters and EM-OM combinations.

Summarizing results across OM and EM attributes

There were numerous OM and EM attributes across all simulation scenarios and, like Miller et al. (2026), we used two methods to summarize the major factors explaining differences in results for each measure of reliability. First, we fit regression models with the responses as the measures of reliability described above and the predictor variables were defined based on OM and EM attributes (e.g., MacKinnon et al., 1995; Wang et al., 2017; Harwell et al., 2018). We used logistic regression for binary indicators of convergence, AIC-based selection of the appropriate assumption of the covariate effect on M, and indicators of CI inclusion. For indicators of AIC-based selection of EM process error source (multiple categories), we performed multinomial regressions. For other measures of reliability we fit ordinary linear regression models. Relative errors of terminal year M, SSB, and F (Eq. 1) are bounded below at -1, so we used a transformation of these values

$$f\left(\widehat{\theta}_i\right) = \log\left[\mathrm{RE}\left(\widehat{\theta}_i\right) + 1\right]. \tag{2}$$

We omitted fits where estimated terminal year SSB, M, or F was equal to zero (RE $= -1$) because of the undefined transformation. RMSE is bounded below at 0 so we used a log transformation for responses in corresponding regressions. We did not use any transformation for the correlation of estimated and true values for terminal year M and SSB. Note there was only one RMSE and correlation calculated for an EM from all simulation fits that converged from a given OM scenario. For all regressions we fit separate models with just individual OM and EM factors included, all factors combined, with all second order interactions, and with all third order interactions. For the multinomial regression, we used the `vglm` function from the VGAM package (Yee, 2008, 2015). We calculated percent reduction of residual deviance (from the null model with no factors) for each of the regression fits. As Miller et al. (2026) noted, there is a lack of independence of the results for each EM fitted to the same simulated data set of a given OM and therefore we did not perform formal statistical analyses of effects of OM and EM attributes.

For the second summarization method, we used classification and regression trees (Breiman et al., 1984) to illustrate the primary OM and EM attributes and interactions that partition the values for each measure of reliability (e.g., Gonzalez et al., 2018; Collier et al., 2022). We used classification trees for categorical measures (convergence, AIC model selection, and indicators of CI inclusion) and regression trees for the other measures with continuous scales (errors, relative error, RMSE, and correlation). The response variables were the same as the regressions for the deviance reduction analyses. We used the `rpart` function in the rpart package (Therneau and Atkinson, 2025) to fit trees. Full trees were determined using default settings except that we increased the number of cross-validations to 100. For one classification tree fitted to AIC selection of process error configuration results for R+S OMs, we had to make a small change to the default prior probabilities (the proportions of each class in the data) used in the criterion on whether to allow branches from a given node because the default priors would not produce any branches. For clarity, we manually pruned the full trees to show just the primary branches.

All OM and EM factors we used to examine reductions in deviance and regression trees are given in Table S1. We provide more detailed results for each EM and OM configuration in the Supplemental Materials using similar methods to Miller et al. (2026). The detailed results present probability of convergence, probability of each EM having lowest AIC, median error of β_M and β_E and corresponding standard error estimates, and median relative error and RMSE for terminal year SSB, F, and M. There are also detailed results for correlation of estimates and true values for terminal year M and SSB.

Results

For many of the measures of reliability we found that the factors that provided the greatest reduction in deviance for fitted regression models were the same as those that defined the primary branches of the classification and regression trees. The trees also explain which levels of a given factor provide better or worse measures of reliability. Therefore, we provide all the results for deviance reduction of regression models in the Supplemental Materials and do not discuss them further.

EM Convergence

EM process error assumption determined the primary branches of trees for each OM process error configuration (Figure 1). Overall convergence was best for R+S OMs, but better convergence occurred when the process error was correct for R and R+S OMs . For R+M OMs, R EMs converged well (88%), but good convergence of R+M EMs occurred for any true process error source if the OMs also had constant fishing pressure and EMs assumed median M parameter (β_M) to be known. All branches based on observation error of covariates or other assessment data (indices and age composition) showed better convergence with more precise observations. Better convergence also occurred when EMs assumed no covariate effects (regardless of whether the OM simulated them), when EMs assumed median M was known, and when OMs assumed greater temporal variability in the true environmental covariate.

AIC performance

Process error source selection

AIC performed poorly in selecting the correct process error for R+M OMs where lowest AIC occurred for the simpler R EMs (Figure 2). However, accuracy of AIC-based selection was high for R and R+S OMs. For R+S OMs, mis-classification occurred to R EMs with the lowest AIC when OMs had higher error in population observations. For R OMs, there was a small decrease in accuracy when the OMs included the largest covariate effect on M, but in those scenarios, AIC was more accurate for the process error assumption when the EMs also incorrectly assumed no covariate effect.

Covariate effect selection

The factors defining the primary branches of classification trees for accuracy of AIC-based model selection depended on the magnitude of the true effect assumed in the OM (Figure 3). When OMs had no covariate effect ($\beta_E = 0$) the OM process error type determined the primary branches whereas level of covariate observation error determined the primary branches when $\beta_E > 0$. Accuracy was high overall for OMs with no covariate effect (88%) (low Type I error) and low when there was a covariate effect (23% and 35% for OMs with $\beta_E = 0.25$ and 0.5, respectively) (high Type II error), but higher accuracy occurred for certain combinations of OM attributes. When the true covariate effect was greatest ($\beta_E = 0.5$), AIC accuracy was 91% for EMs fit to OMs with high temporal variability in the covariate and low error in population and covariate observations regardless of whether the EM process error was correct. However, when the covariate effect was weaker ($\beta_E = 0.25$), high accuracy only occurred for a subset of those OMs with R or R+M process errors (76%), and the further conditioning on OMs with high autocorrelation of the covariate provided higher accuracy (86%). When there was no covariate effect, AIC accuracy was best for R OMs (95%), and for R+S and R+M OMs, accuracy was improved if the OMs also had lower error in population observations and the EM process error assumption was correct (85%).

Median natural mortality rate inferences

Estimation bias

Median errors for $\widehat{\beta}_M$ were close to 0 (-0.06 to -0.03) across all simulations for R and R+M OMs, but not for R+S OMs (-0.45; Figure 4). Median errors were closer to zero when there was a change in fishing pressure for R+S and R+M OMs. Median errors closer to zero occurred with more precise population observations for all OM process error sources. For R OMs with constant fishing pressure, EMs that assumed no covariate effect and the correct process error type indicated worse bias when the OM simulated the strongest covariate effect ($\beta_E = 0.5$). For R+S and R+M OMs, there was indication of less bias when EMs estimated covariate effects (rather than assume none) and when the correct process error type was assumed.

Detailed results revealed that bias of $\widehat{\beta}_M$ was negligible for all EM process error assumptions fitted to R and R+M OMs when there was contrast in fishing pressure and low error in population observations, but the matching EM process error assumption was also important for EMs fit to R+S OMs (Figures S4 to S6). For those OMs, any attributes related to covariate process error and effects on M had little effect on bias.

Standard error estimation bias

Median errors indicated strong negative bias across all OM process error types (Figure S7). Lower error in population observations provided better median errors (closer to 0). For R+S OMs, better median errors occurred when the EM process error assumption was correct.

Detailed results for each OM and EM configuration indicated little bias in certain scenarios where OMs had temporal contrast in fishing pressure and low error in population observations (Figures S8 to S10). For R and R+M OMs with those conditions, all EM process error assumptions provided median errors close to zero except for OMs with the strongest true covariate effect ($\beta_E = 0.5$) and autocorrelation and high variability of the true covariate time series. For R+S OMs with those conditions, only EMs with the correct process error assumption exhibited low (absolute) median error.

Confidence interval bias

Across all simulations, rates of 95% CI coverage were generally poor and indicated a negative bias with a range of 83-87% for the different OM process error types (Figure 5). Contrast in OM fishing history and lower error in population observations generally indicated improved CI coverage. However, better coverage occurred for R EMs fit to R+S OMs with higher error in population observations.

Detailed results indicated reliable coverage for some scenarios (Figures S11 to S13). For R OMs with contrast in fishing history and lower error in the covariate observations, EMs with any process error assumptions and that estimated the covariate effect showed reliable coverage. The same result occurred for R+S OMs as long as the EMs assumed the correct process error type. However, coverage was generally unreliable for any of the EMs fit to R+M OMs.

Covariate effect inferences

Estimation bias

Like $\widehat{\beta}_M$, median errors of $\widehat{\beta}_E$ across all simulations for each OM process error configuration were negative but close to 0 (-0.1 to -0.07; Figure 6). Better estimation (median error closer to 0) occurred with lower uncertainty in population or covariate observations and higher temporal variation in the true covariate. For R OMs with the largest true covariate effect, estimation error was poor unless there was also high temporal variation and low observation error for the covariate. For R+S OMs, we also found lower estimation error when the EM process error assumption matched. We observed low median error across all simulations for R+M OMs with low error in covariate observations, but there were some combinations of factors that provided median errors near zero when the R+M OMs had greater error in the covariate observations.

The detailed results for all OM and EM configurations demonstrated that the median errors get worse with increased values for the true β_E for many scenarios without low error in population and covariate observations and contrast in fishing pressure, indicating that the estimates remain close to zero when the true $\beta_E > 0$ (Figures S14 to S16). The detailed results also indicated little bias in covariate effect estimation regardless of the OM and EM process error configuration when EMs assumed median M was known and OMs had lower error in covariate observations and greater temporal variability of the true covariate.

Standard error estimation bias

Median errors across all simulations for a given OM process error configuration were strongly negative (-0.8 to -0.6), but combinations of OM and EM attributes resulted in improved values closer to zero (Figure S17). For all OM process error types, lower error in covariate observations provided better estimation (median error closer to 0).

Examining the detailed results for each OM and EM configuration showed little indication of bias in the same OM scenarios where there was little indication of bias for $\widehat{\beta}_E$ (Figures S18 to S20). However, in the R+S OMs with lower error in covariate observations and greater temporal variability of the true covariate, the EMs that had the correct process error as well as known median M performed best.

Confidence interval bias

Over all simulations, rates of 95% CI coverage indicated negative bias with rates ranging 85-88% for the different OM process error configurations (Figure 7). For R OMs, higher rates of CI coverage occurred with lower error in covariate observations, but when that error was higher, higher rates were associated with OMs where no covariate affect was assumed ($\beta_E = 0$). For R OMs with some covariate effect and higher error in covariate observations, higher rates of coverage occurred with higher error in the population observations. For R+S

OMs, higher rates of CI coverage were observed for EMs that assumed R+S or R+M process errors.

The more detailed results for each OM and EM configuration showed that good CI coverage can be obtained for all of the investigated covariate effect sizes when there was low error in covariate and population observations, high covariate temporal variability and contrast in fishing pressure for R and R+S OMs. The EM process error configuration was not a factor for the R OMs, but R+S or R+M EM configurations were necessary for R+S OMs (Figures S21 to S22). However, CI coverage was negatively biased for all EM process error configurations fit to R+M OMs with these conditions even though little bias was observed in estimation of β_E or its standard error (Figure S23). Examining the estimates for each simulated data set when R+M EMs were fit to R+M OMs where we observed little estimation bias, there appears to be a positive correlation of the covariate and standard error estimates such that estimates less than the true value had negatively biased standard error estimates which would make CIs too small (Figure S24).

Terminal year natural morality rate

Estimation bias

For R and R+S OMs, the median estimation error is zero for terminal year M when EMs assumed β_M was known because of a large number of fits for OMs with no temporal variability in M and the EMs assumed no covarate effects were estimated and median M was known (Figure 8). However, median error for the corresponding subset of fits in R+M OMs where there was always temporal variability in true terminal M was also close to zero. When β_M was estimated, median errors were near zero for many EMs in R OMs. Median error was problematic for this subset of EMs in R+S OMs, but there was indication of lower bias when R+S or R+M EMs were fit. The detailed results for each OM and EM attribute (Figures S25 to S27) further show that there was little evidence of bias for R+S OMs when error in population observations was low and there was contrast in fishing pressure and the EM process error configuration was correct. In R+M OMs, median errors indicated little bias overall with temporal contrast in fishing pressure, although detailed results demonstrated the best reliability when there was also lower error in population observations.

Estimation accuracy

Like bias, regression trees for RMSE showed better accuracy when EMs assumed β_M was known rather than estimated (Figure S28). Better accuracy was also generally associated with lower error in population observations and temporal contrast in fishing pressure. For R+S OMs, R+S EMs provided better accuracy than incorrect process error assumptions.

Using correlation to measure accuracy of terminal year M estimation provides a different perspective with true covariate variability (R and R+S OMs) and population observation uncertainty (R+M OMs) defining the primary branches (Figure S29). For R and R+S OMs, better correlation occurred with higher covariate variability, lower error in covariate observations, and when the EM estimated covariate effects for those OMs. Like RMSE, median correlation improved when median M was known.

Terminal year spawning stock biomass

Estimation bias

Although primary branches for all OM process error types were defined by fishing history type, differences between them in median errors was small (Figure 9). For R+S and R+M OMs, there was indication of less bias when EMs assumed β_M was known rather than estimated like terminal year M. For R+S OMs with constant fishing pressure and β_M estimated, less bias was indicated when the EM process error assumption was correct.

Estimation accuracy

Whether accuracy was measured by RMSE or correlation of estimated and true values, regression trees demonstrated better accuracy with contrast in fishing pressure, lower error in population observations, and when EMs assumed β_M was known (Figures S30 and S31). Unlike terminal year M, SSB estimates were generally highly correlated with the true SSB values when there was contrast in fishing pressure and low error in population observations regardless of the OM and EM process error and covariate effect assumptions a(Figures S32 to S37).

Discussion

Our simulation study demonstrated that reliable inferences about covariate effects on natural mortality require low observation error, contrast in fishing pressure, and large temporal variability in the environmental covariate. Under these conditions, reliable AIC-based selection and estimation of environmental effects on M can be possible even when the process error was mis-specified (e.g., R and R+S EMs fit to R+M OM). Under those same conditions, reliable estimation of standard errors and CI coverage was also possible for R and R+S OMs especially when the EM process error type was correct. However, there was poor CI coverage for R+M OMs even though there was little bias in estimation of the effect or standard errors, possibly related to correlation of the estimated effects and standard errors. Cadigan et al. (2024) found CI coverage to be biased for SSB and F estimation in a state-space model in some scenarios but they attributed the poor coverage to bias in Hessian-based standard error estimation and their simulations held any process error random effects constant. Consideration of other methods of calculating CIs may be warranted (e.g., Boyko and O'Meara, 2024). Previous research has shown that estimation of a constant M parameter requires contrast in time series and informative data (Lee et al., 2011), so it is unsurprising that estimation of covariate effects on M also requires relatively good information via more precise observations and higher contrast in the covariate time series.

We found AIC unreliable for determining the source of process error for EMs fit to R+M OMs where marginal temporal variability was specified as $\sigma_M = 0.3$. Miller et al. (2026) investigated R+M OMs with two levels of M process-error variability ($\sigma_M \in \{0.1, 0.5\}$) and only found AIC able to accurately distinguish R+M process errors with the higher level of process-error variability. If AIC accuracy increases with σ_M, our results together would suggest the minimum level of variability for reliable detection may be with σ_M somewhere between 0.3 and 0.5. In our results and those by Miller et al. (2026), inaccurate selection of AIC typically chose R EMs which would indicate the fitted R+M EMs would estimate no variability in the M process errors. Future studies like ours where R+M OMs are simulated with greater variability in M process errors would better inform reliability of covariate effect inferences under such scenarios.

Deriso et al. (2008) attempted to estimate process errors as well as covariate effects with M for Pacific herring but similarly found no variability in M, suggesting there was too much uncertainty in the available observations relative to the true temporal variability in M. Our results showed AIC-based selection of covariate effects could be accurate even when the EM process error assumption was incorrect and that R EMs were often selected for fits to R+M OMs probably because of low variability simulated in M process errors. This suggests that the findings of Deriso et al. (2008) on covariate effects for Pacific herring may also be robust to true low variability in M. However, they did not account for uncertainty in covariate observations, some of which would probably have substantial uncertainty (e.g., competition and predation covariates). We found higher covariate observation uncertainty resulted in true covariate effects to be less detectable using AIC but we did not investigate the implications for incorrectly assuming no covariate uncertainty for covariate inferences.

It was important to consider RMSE and correlation of estimates with the true values for reliable estimation of terminal year M because summaries of relative errors could suggest unbiasedness that is not relevant due to the distribution of true values being centered at an assumed constant value or near an estimated constant value in the EM (e.g., R and R+S EMs that have a temporally constant M fit to OMs with temporally varying M). Despite the inability of AIC to distinguish M process errors, reliable estimation of terminal M was possible in some scenarios regardless of the EM process error assumption. To achieve little bias and high correlation of terminal M estimates required larger variability in the covariate (when OMs and EMs included the effect on M) along with lower population and covariate observation error. It was also more difficult when EMs estimated β_M rather than treating it as known which suggests relative changes in M may be estimated more reliably than the absolute scale of M (c.f., Zhang et al., 2020). However, there was little correlation of terminal year M estimates and true values for R+M OMs without covariate effects for any combination of OM and EM attributes (Figure S34). Therefore, our results suggest estimating terminal M is easier when the temporal variation is due to effects of highly variable covariates than unexplained variation in M. Like the results for AIC-based selection, this may be a function of the level of variability in M we assumed for R+M OMs.

Accurate estimation of terminal year SSB was more robust than terminal year M. When OMs had contrast in fishing pressure and low error in population observations, there was little bias and high correlation of terminal year SSB estimates regardless of whether OMs and EMs included covariate effects. However, for R+S OMs, it was important for the EM process error configuration to be correct. Fortunately, our results and those by Miller et al. (2026), suggest AIC-based model selection is accurate for this type of process error. However, the reliability of the SSB estimation does break down in the less ideal scenarios when there is higher error in population observations, and lack of contrast in fishing pressure (e.g., Figures S38 to S40).

The R+S and R+M EMs both include process error for the apparent survival of cohorts and would be expected to perform similarly, and they did in our simulations when the OM and EM matched. However, we found that the accuracy for model output like SSB and F that are important for management was generally worse using R+M EMs for R+S OMs than using R+S EMs for R+M OMs in the high information OMs with contrast in fishing pressure and low error in population observations. Additionally, questions of the appropriate M for biological reference points raised when using R+M EMs because M explicitly is varying over time (e.g., Legault and Palmer, 2016). We therefore recommend following the suggestion from Li et al. (2024) to prefer R+S EMs over R+M EMs unless strong biological evidence is present to support a particular R+M EM. Such support could be found through both biological understanding (e.g., Trijoulet et al., 2020) as well as statistical properties such as a large delta AIC for R+M compared to R+S associated with greater temporal variability in M (Miller et al., 2026). We note that our results are based on OMs simulated without autocorrelation in the process errors, although it has often been found in applications to specific stocks (e.g., Cadigan, 2016; Stock et al., 2021). Therefore, studies examining selection of models with autocorrelation and their evaluation via management performance in closed-loop simulations are important next steps.

The pattern of improved inferences using state-space models in the presence of higher contrast in fishing pressure, lower observation error and greater process error variability that emerges from our work here and by Miller et al. (2026) is consistent with expectations from statistical theory. State-space models are more reliable when observation error is lower than process error (Auger-Méthé et al., 2016), and, more generally, maximum likelihood estimation is less biased and more accurate as the information in the data increases (Neyman and Scott, 1948; Kiefer and Wolfowitz, 1956). However, the greater complexity of state-space assessment models over the more well-studied linear and Guassian state-space models makes determining sufficient levels of precision in observations difficult. Statistical information available to the state-space assessment model could be increased with longer time series of observations, more time series (e.g, multiple indices of abundance and corresponding age-composition data sets), or more precise observations of a given time series of observations (e.g., greater index observation precision or more sampling for age composition). Indeed, there are many commercially important stocks where there may be just a single survey time series or temporal gaps in surveys or age-composition observations. We expect less reliability of state-space assessment models in these scenarios with lower statistical information similar to our simulation scenarios with higher observation error. Our simulations also assumed that the variance of aggregate population observations and covariate observations was known in the EMs whereas estimates of these parameters are often used in practice. Further research of these "data-poorer" scenarios is required and it will be important to conduct self-test simulations (where the OM and EM are the same) on a case-by-case basis to ensure the reliability of models used to manage commercially important fish stocks.

The ability to accurately infer covariate effects on M in some situations indicates that such investigations may be fruitful. The ability to make inferences could improve further when WHAM is extended to incorporate tagging data (Miller et al., 2025). Tagging data can greatly inform M estimation (Pollock et al., 1991; Hampton, 2000) and it should also have a similar effect on estimation of covariate effects or unexplained temporal variation in M. Given our findings and planned future WHAM development, we expect investigations of, and accounting for, temporal variation in M to become more common within the fisheries stock assessment process. At the same time, it will be equally important to conduct research that will improve our understanding of how best to measure the depletion of stocks and determine catch advice for these stocks where we can no longer assume M to be constant.

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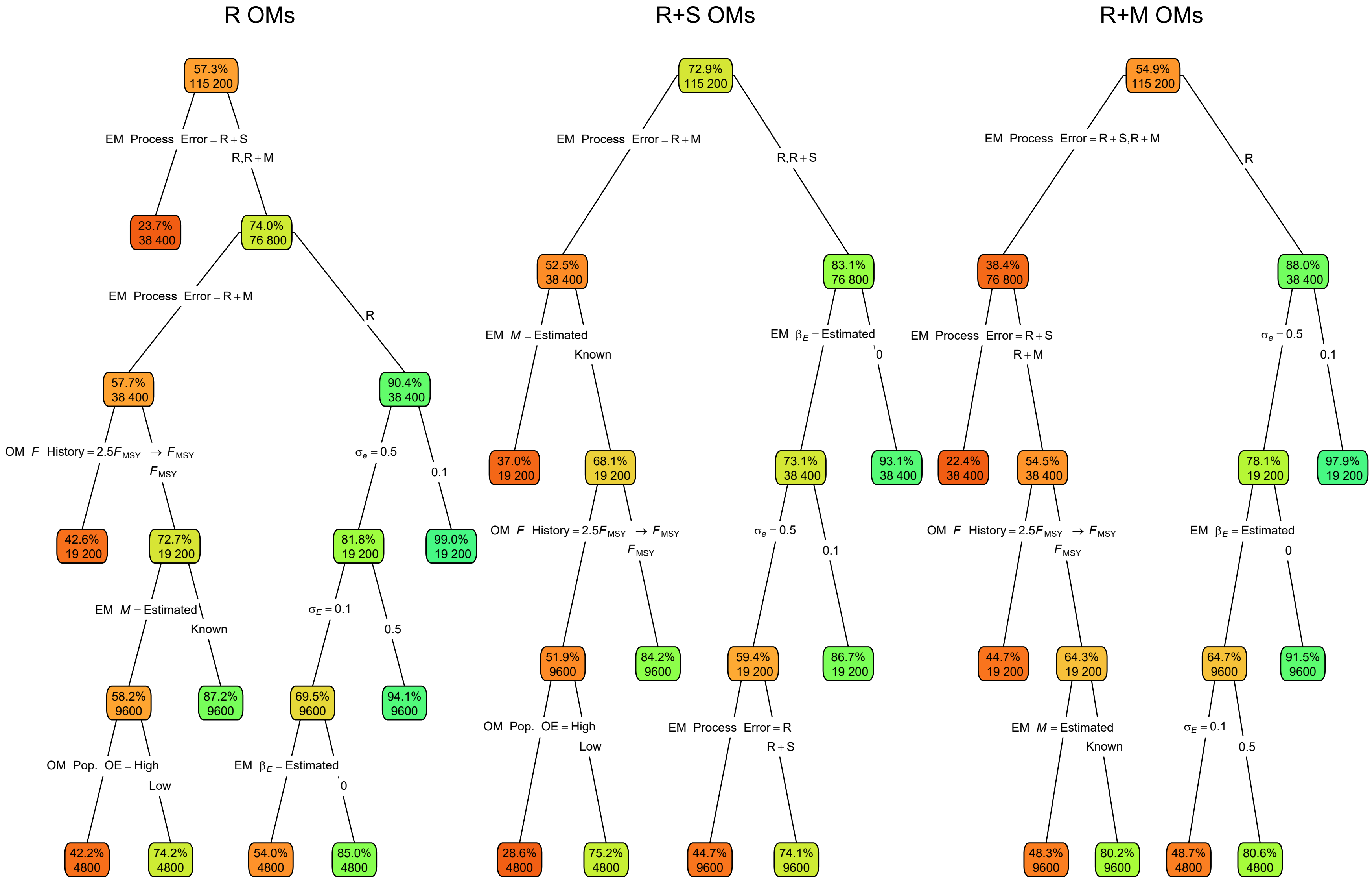

Fig. 1. Classification trees indicating primary factors determining convergence as defined by providing Hessian-based standard errors for operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). Nodes denote percent convergence (top) and number of fits (bottom) for the corresponding subset. Lower or higher convergence rates are indicated by more red or green polygons, respectively. See Table S1 for description of OM and estimating model (EM) factors defining nodes and branches.

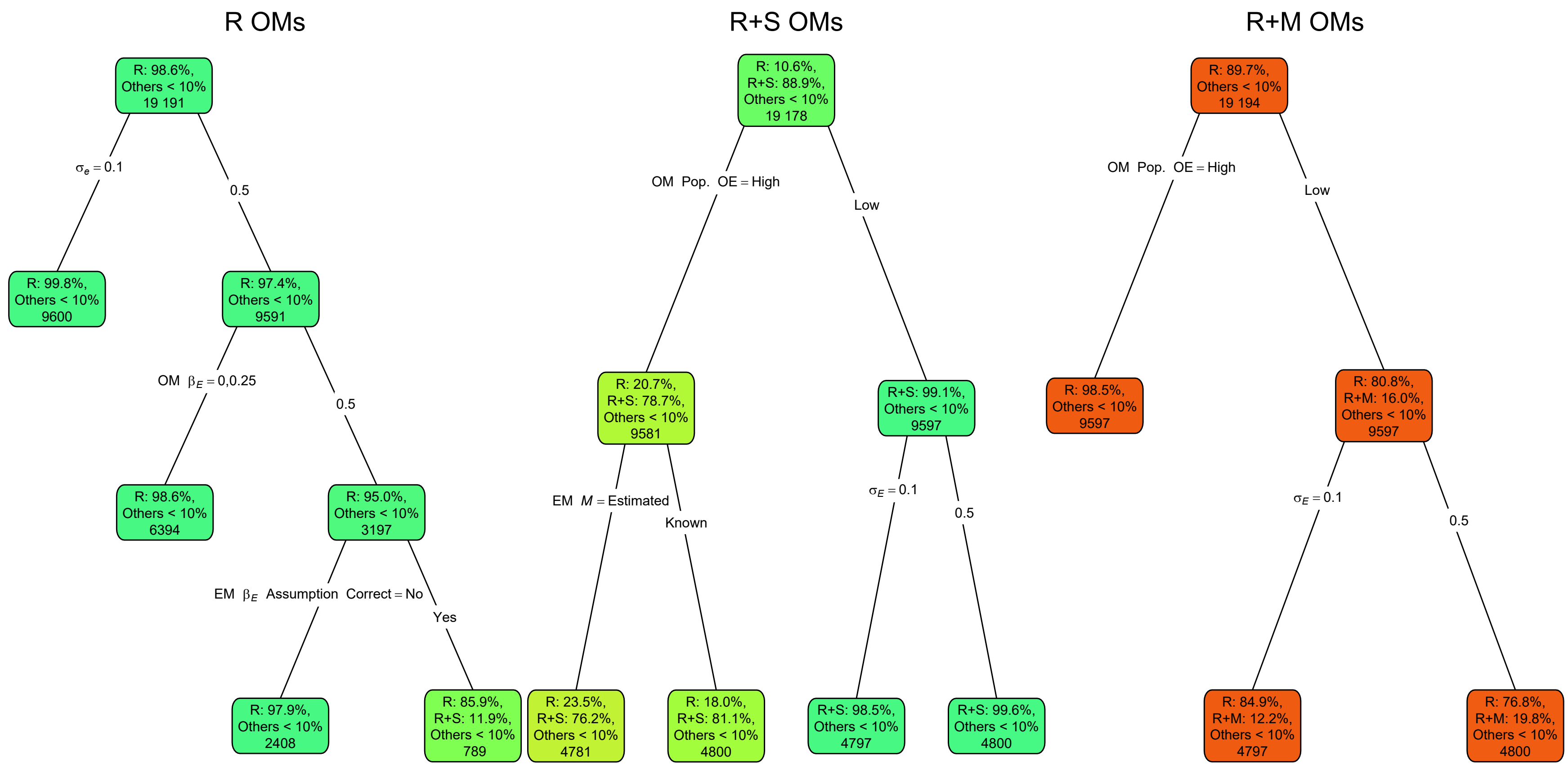

Fig. 2. Classification trees for which estimating model process error assumption provides the lowest AIC. Each node shows the percentages (if greater than 10%) of estimating model fits with a given process error assumption having the lowest AIC for the corresponding subset. Process error assumptions with percentages less than 10% are grouped as “other.” Lower or higher accuracy are indicated by more red or green polygons, respectively. See Figure 1 caption and Table S1 for further details.

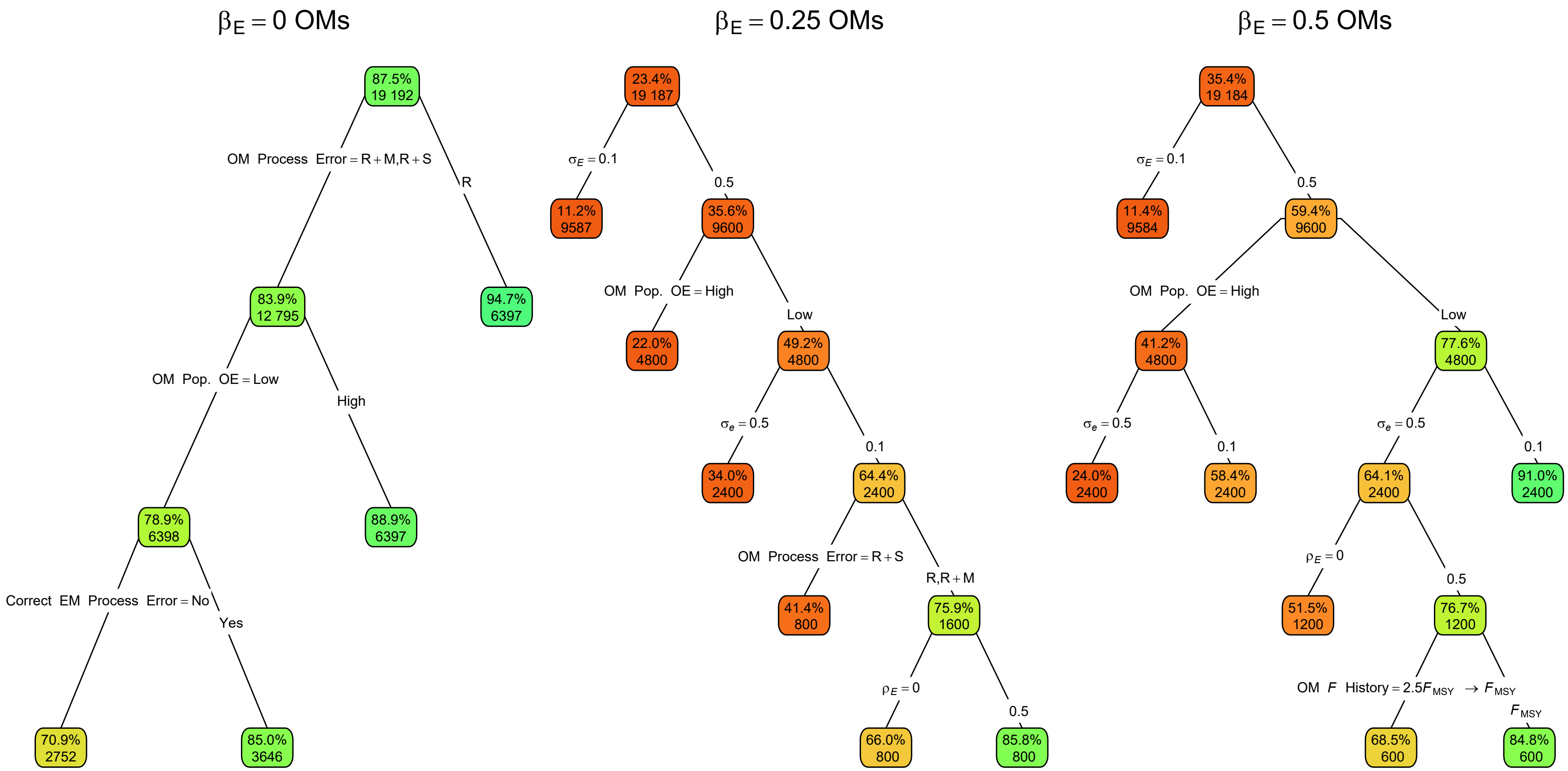

Fig. 3. Classification trees for whether the correct estimating model assumption for covariate effect on M (no effect with OM $\beta_E = 0$ or estimated effect when OM $\beta_E > 0$) provides the lowest AIC for OMs assuming values for β_E of 0, 0.25, and 0.5. Nodes denote the percentage of estimating models with the correct assumption and lowest AIC for the corresponding subset. Lower or higher accuracy are indicated by more red or green polygons, respectively. See Figure 1 caption and Table S1 for further details.

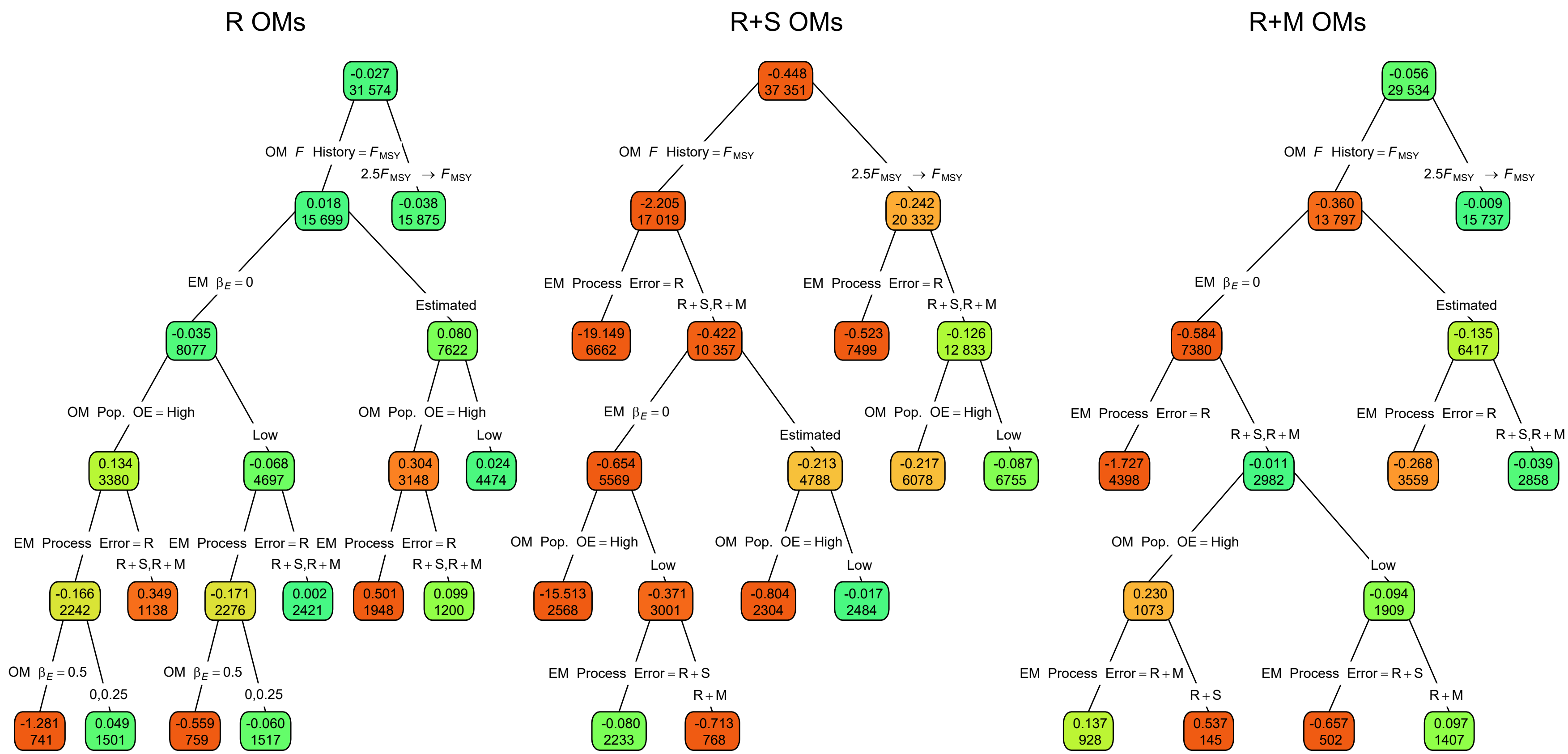

Fig. 4. Regression trees for estimation error of the median M parameter (β_M). Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

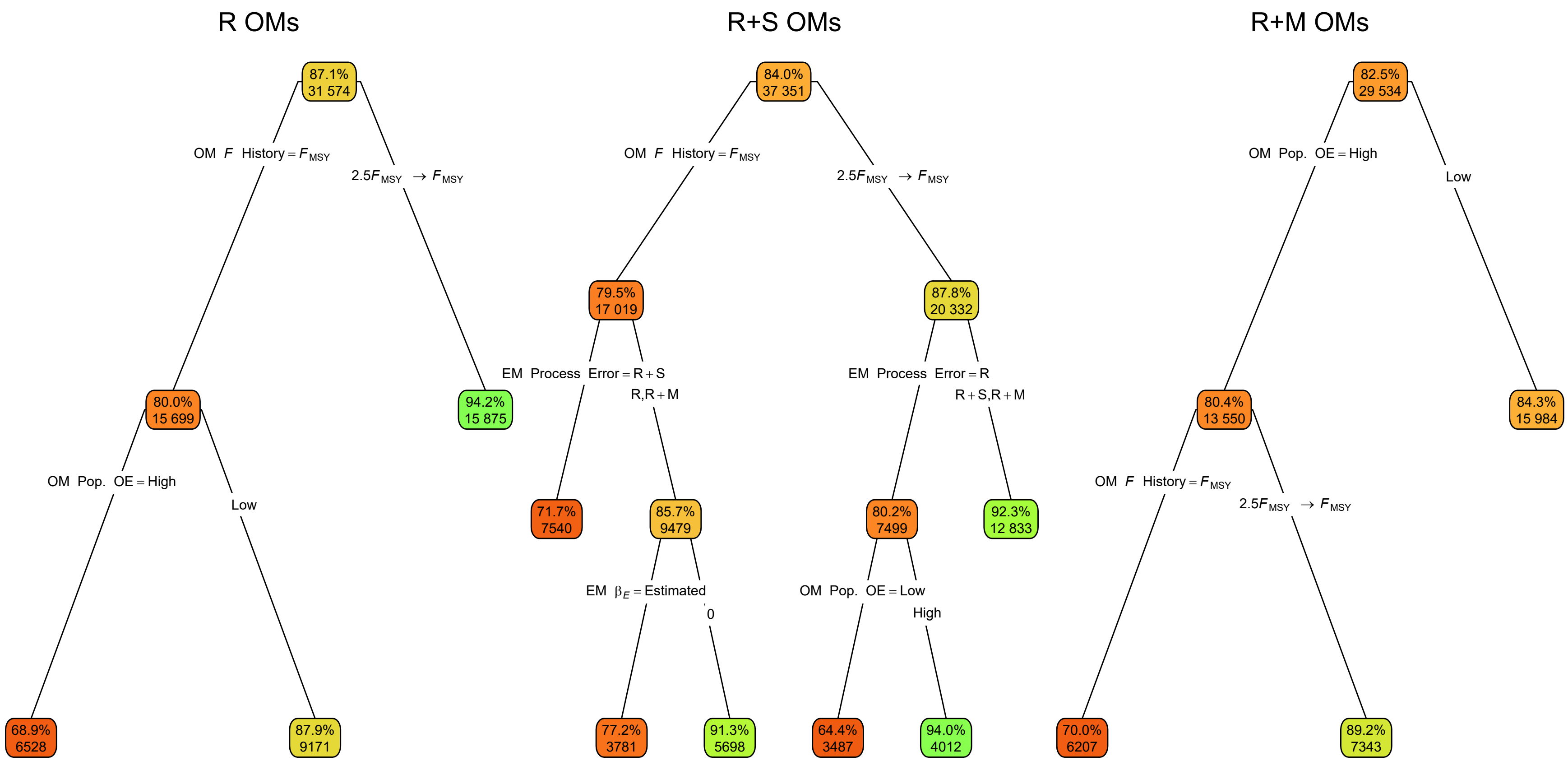

Fig. 5. Classification trees for whether confidence intervals (CIs) for the estimated median M parameter (β_M) included the true value. Each node shows the percentage of CIs that include the true value for the corresponding subset. Higher or lower percentages are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

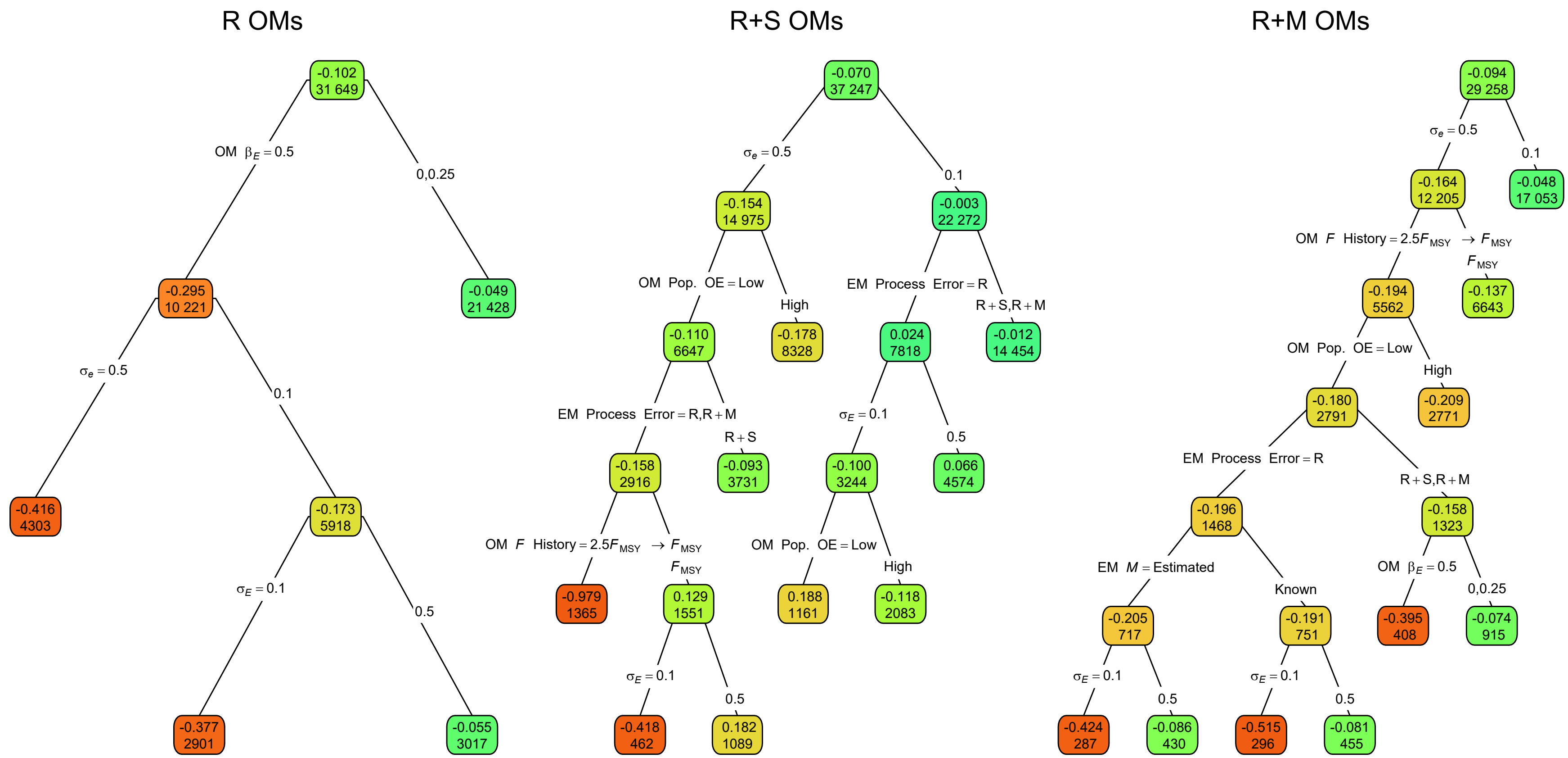

Fig. 6. Regression trees for estimation error of the covariate effect on M ($\widehat{\beta}_E$). Each node shows the median error (top) for the corresponding subset. Lower or higher absolute median error are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

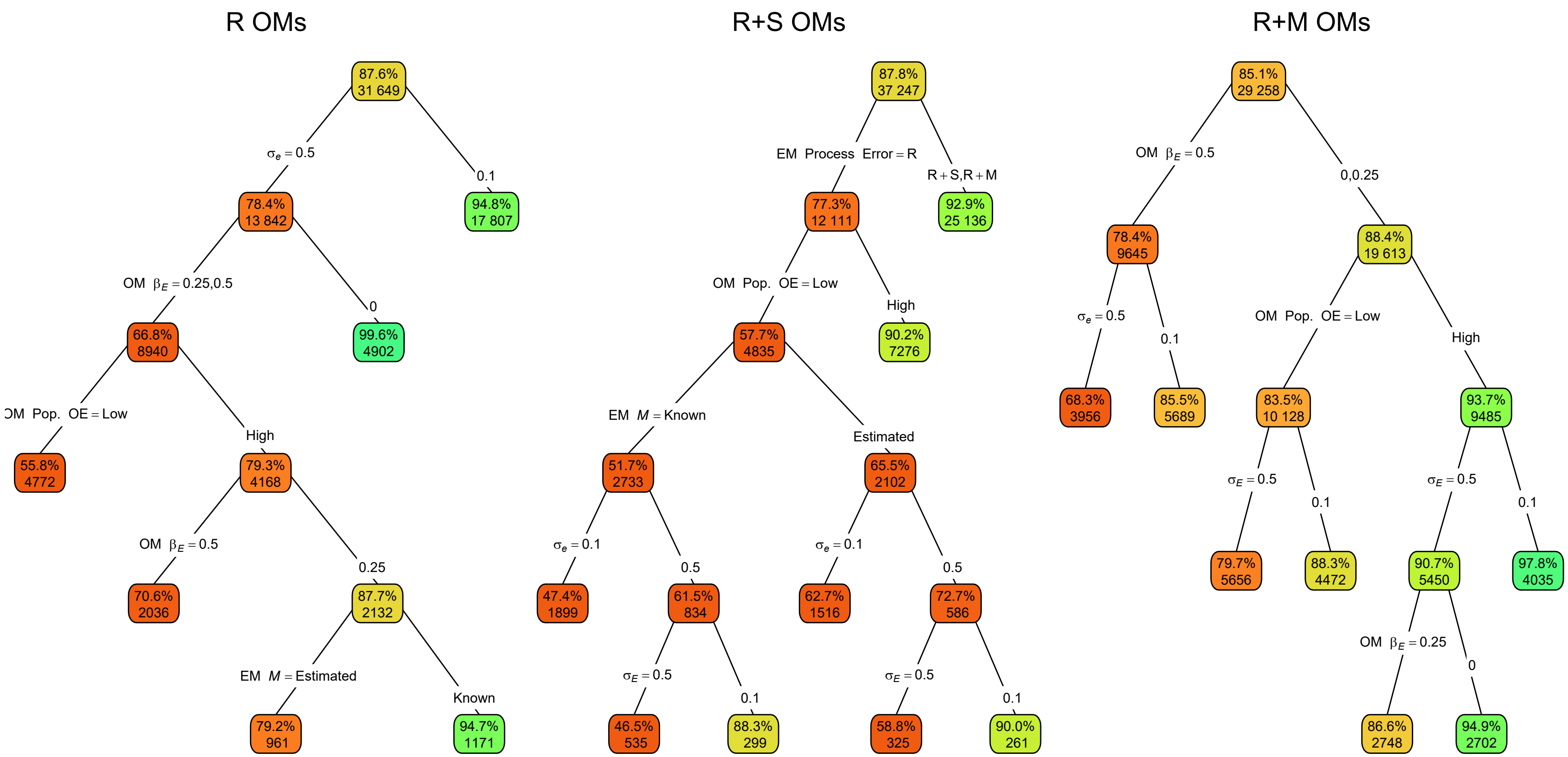

Fig. 7. Classification trees for whether confidence intervals (CIs) for the estimated covariate effects on M (β_E) included the true value. Each node shows the percentage of CIs that include the true value for the corresponding subset. Higher or lower percentages are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

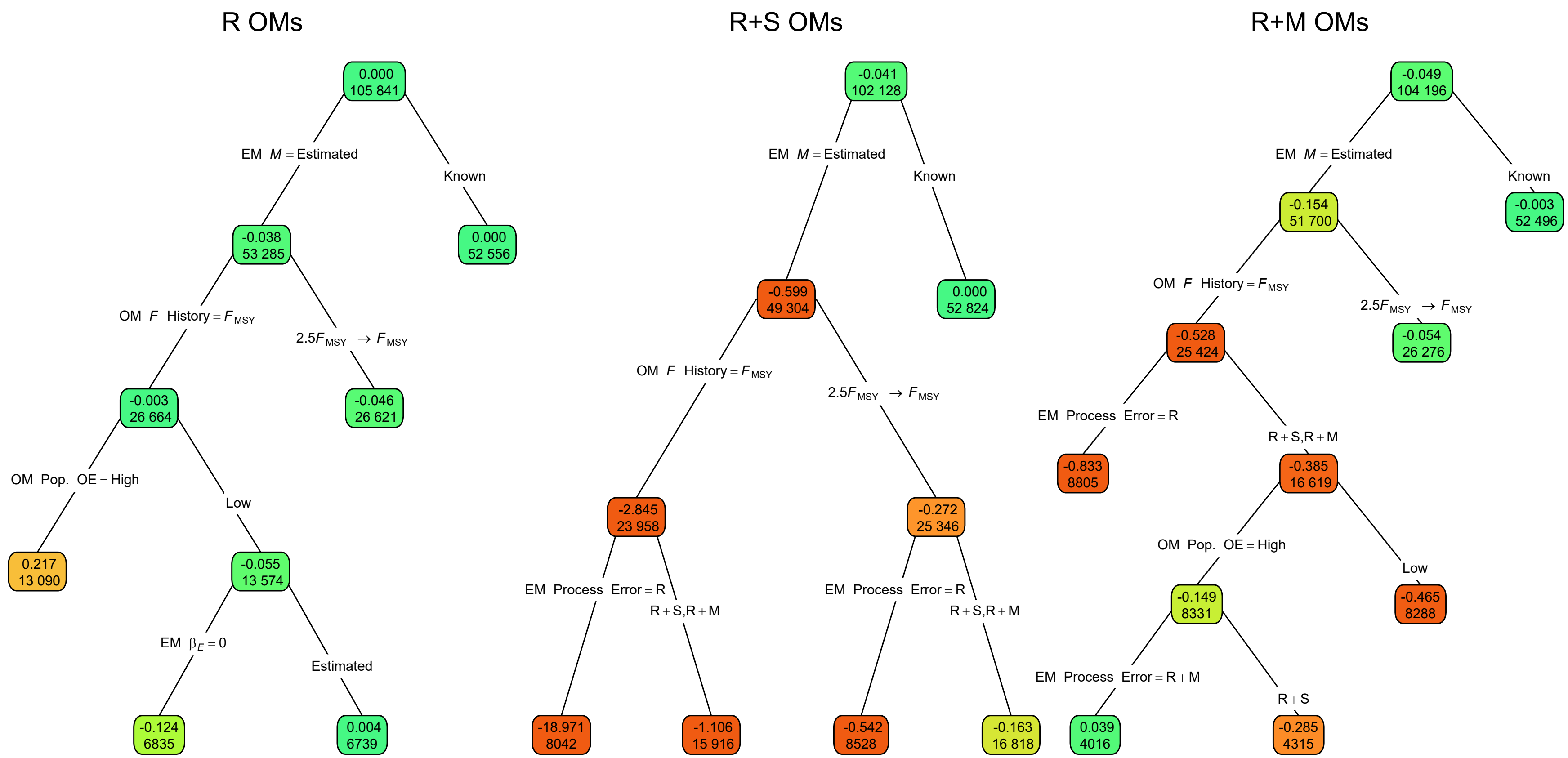

Fig. 8. Regression trees for estimation error of terminal year M as measured by Eq. 2. Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

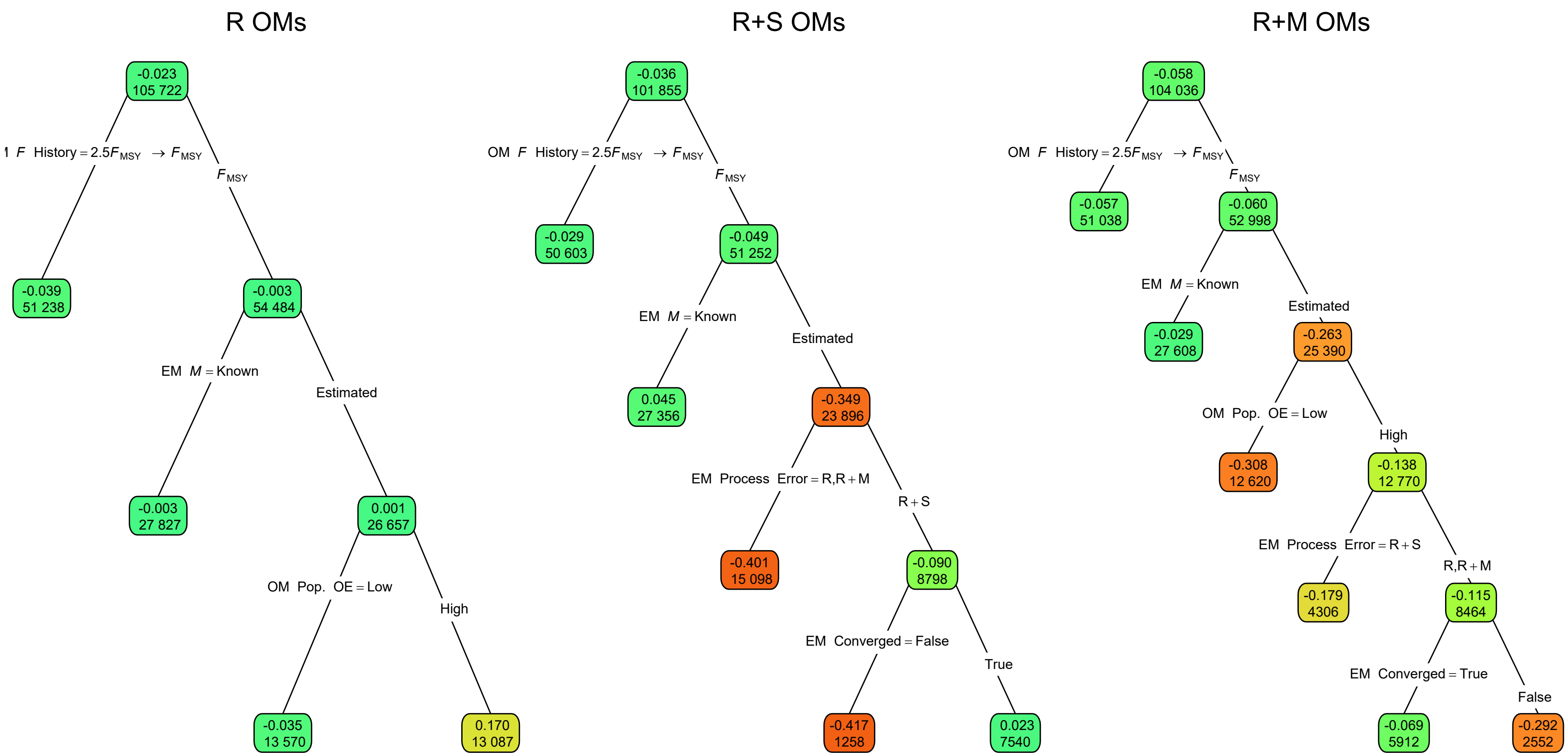

Fig. 9. Regression trees for estimation error of terminal year SSB as measured by Eq. 2. Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

Supplemental Materials

Referenced Figures and Tables

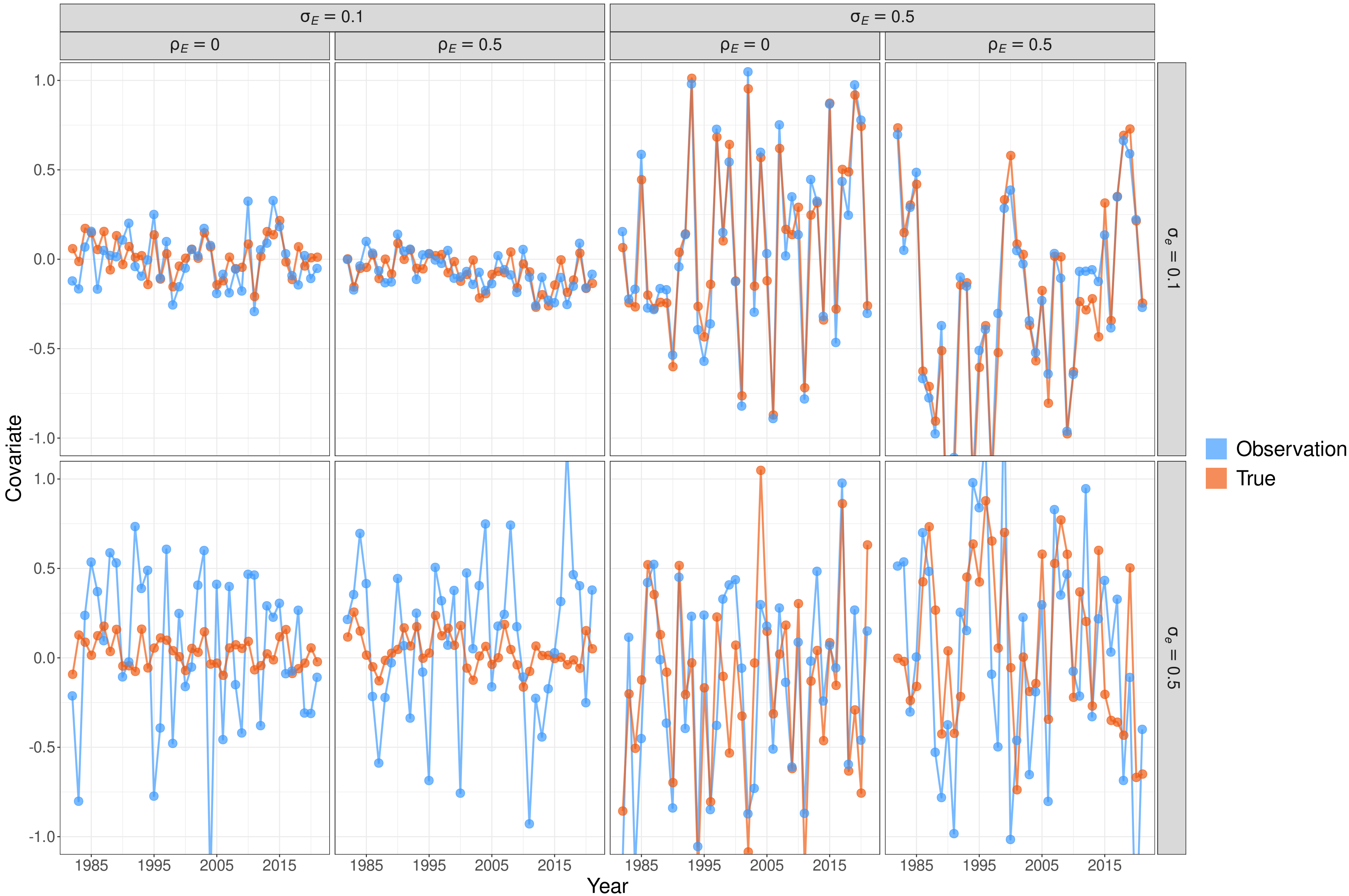

Fig. S1. Example simulations of environmental covariate latent processes and observations with different levels of observation error, and different assumptions about variability of the latent process.

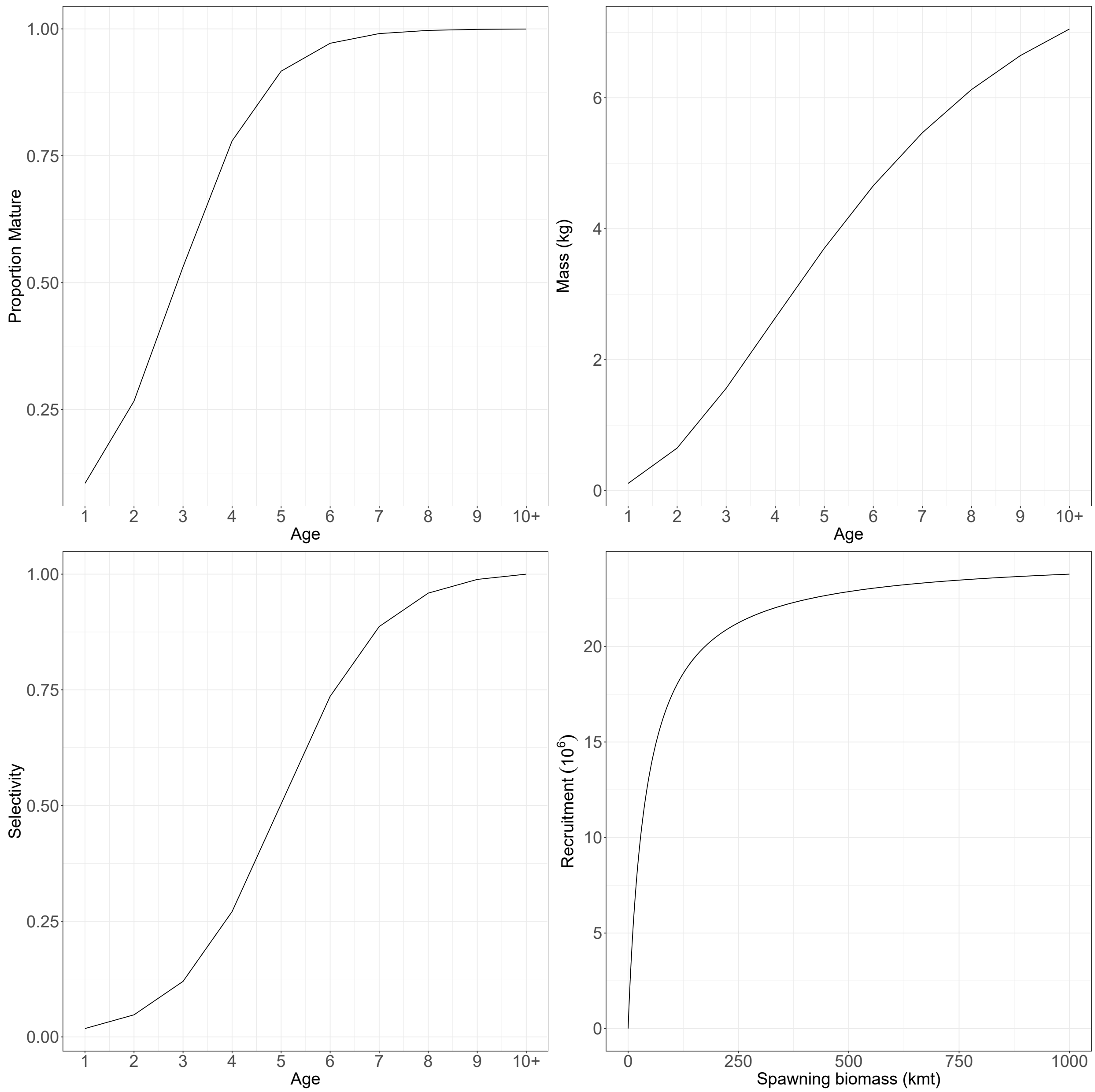

Fig. S2. The proportion mature at age, weight at age, fleet and index selectivity at age, and Beverton-Holt stock-recruit relationship assumed for the population in all operating models. For operating models with random effects on fleet selectivity, this represents the selectivity at the mean of the random effects.

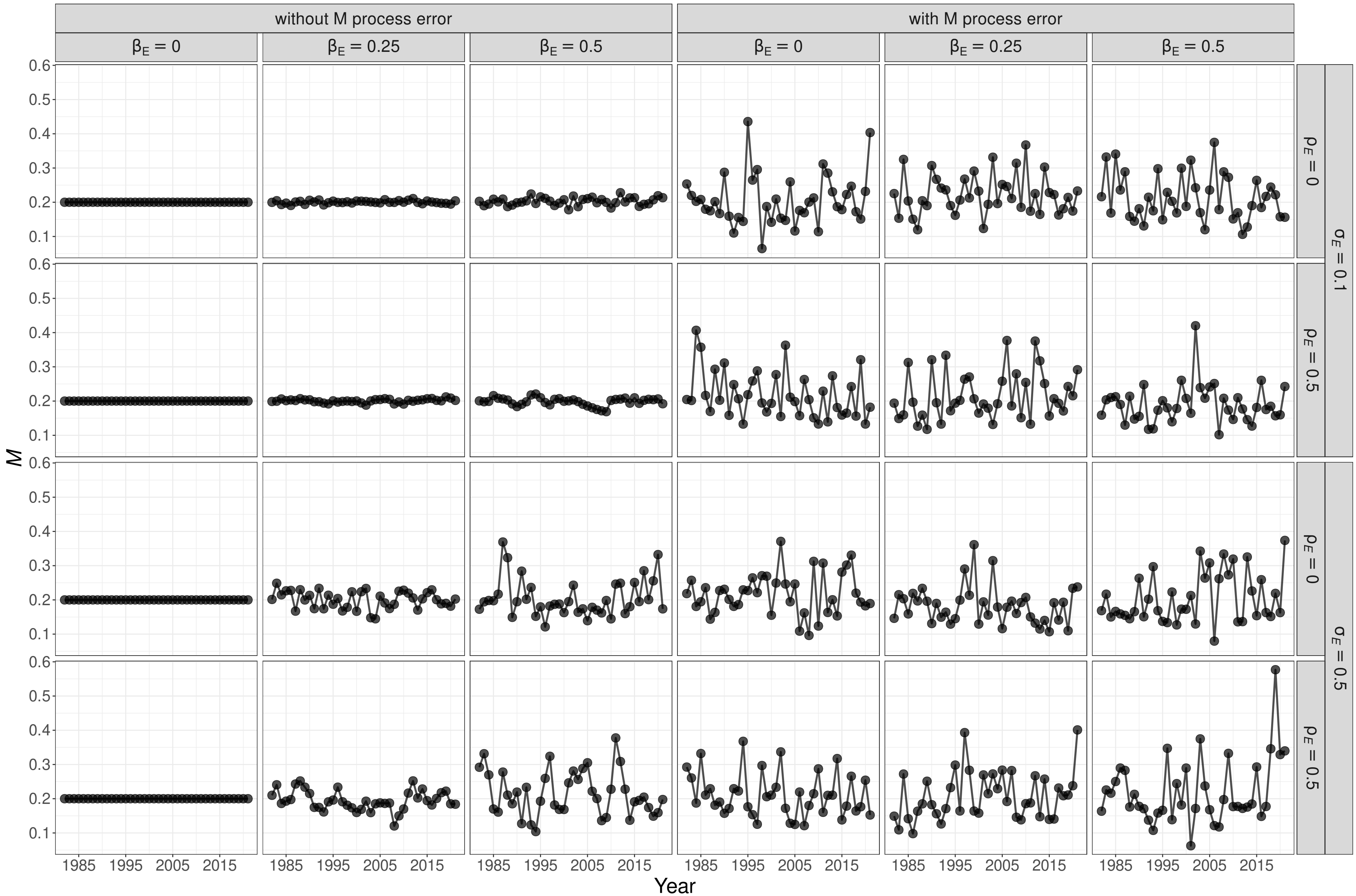

Fig. S3. Example simulations of annual M that may be a function of a temporally varying environmental covariate and autoregressive random effects. The marginal standard devation of random effects is $\sigma_M = 0.3$.

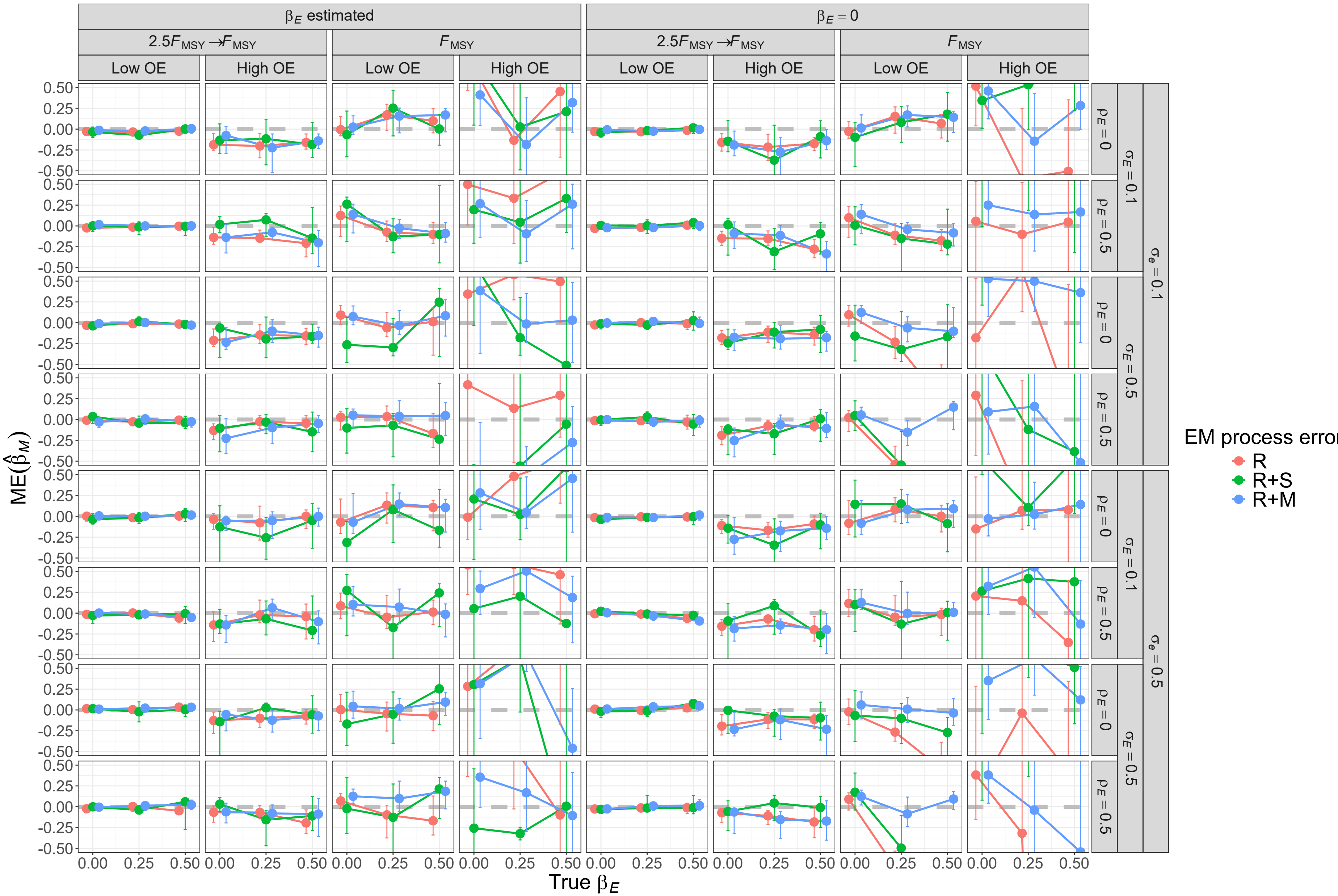

Fig. S4. For R operating models, median error (ME) of estimates of β_M from fitting estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

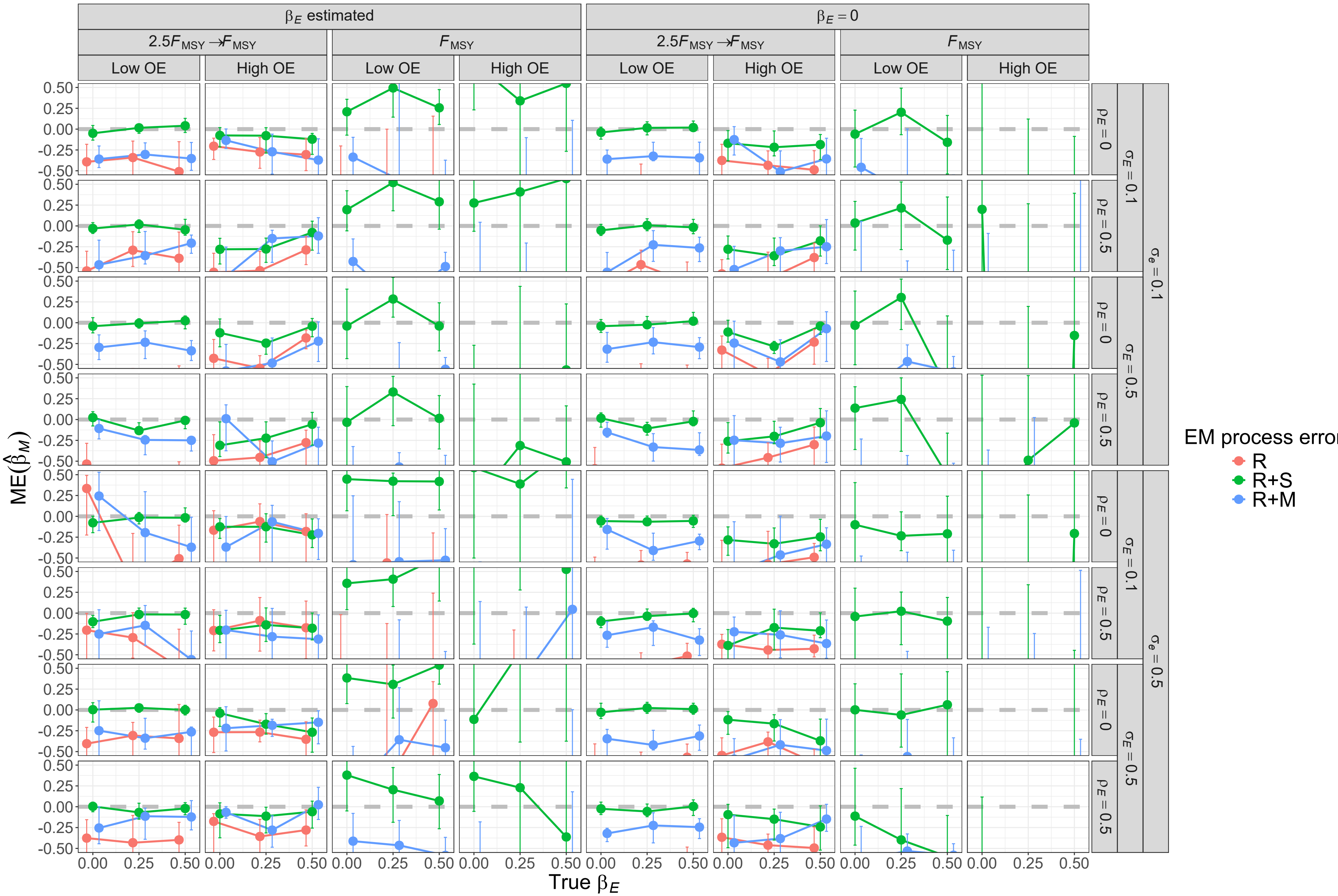

Fig. S5. For R+S operating models, median error (ME) of estimates of β_M from fitting estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

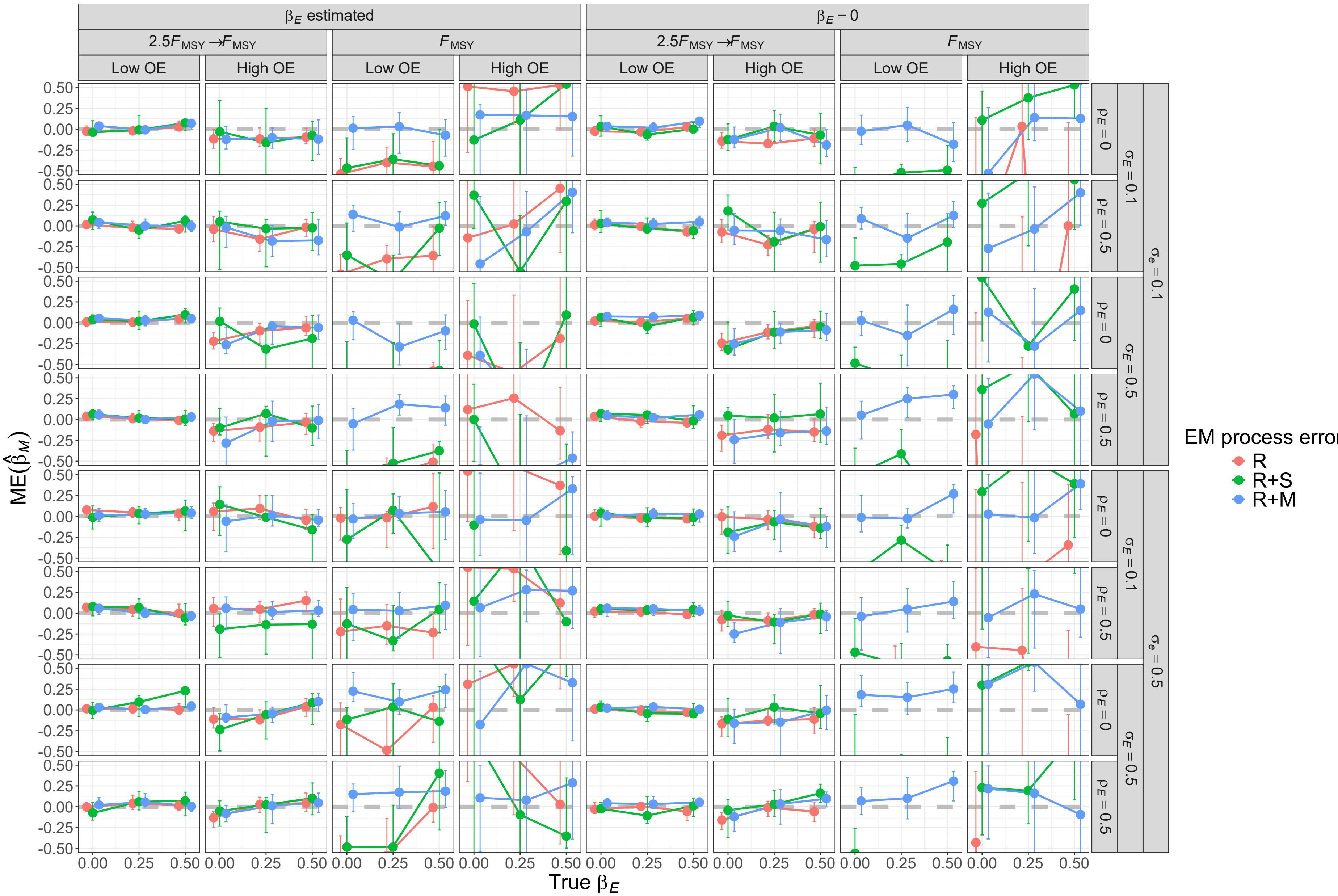

Fig. S6. For R+M operating models, median error (ME) of estimates of β_M from fitting estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

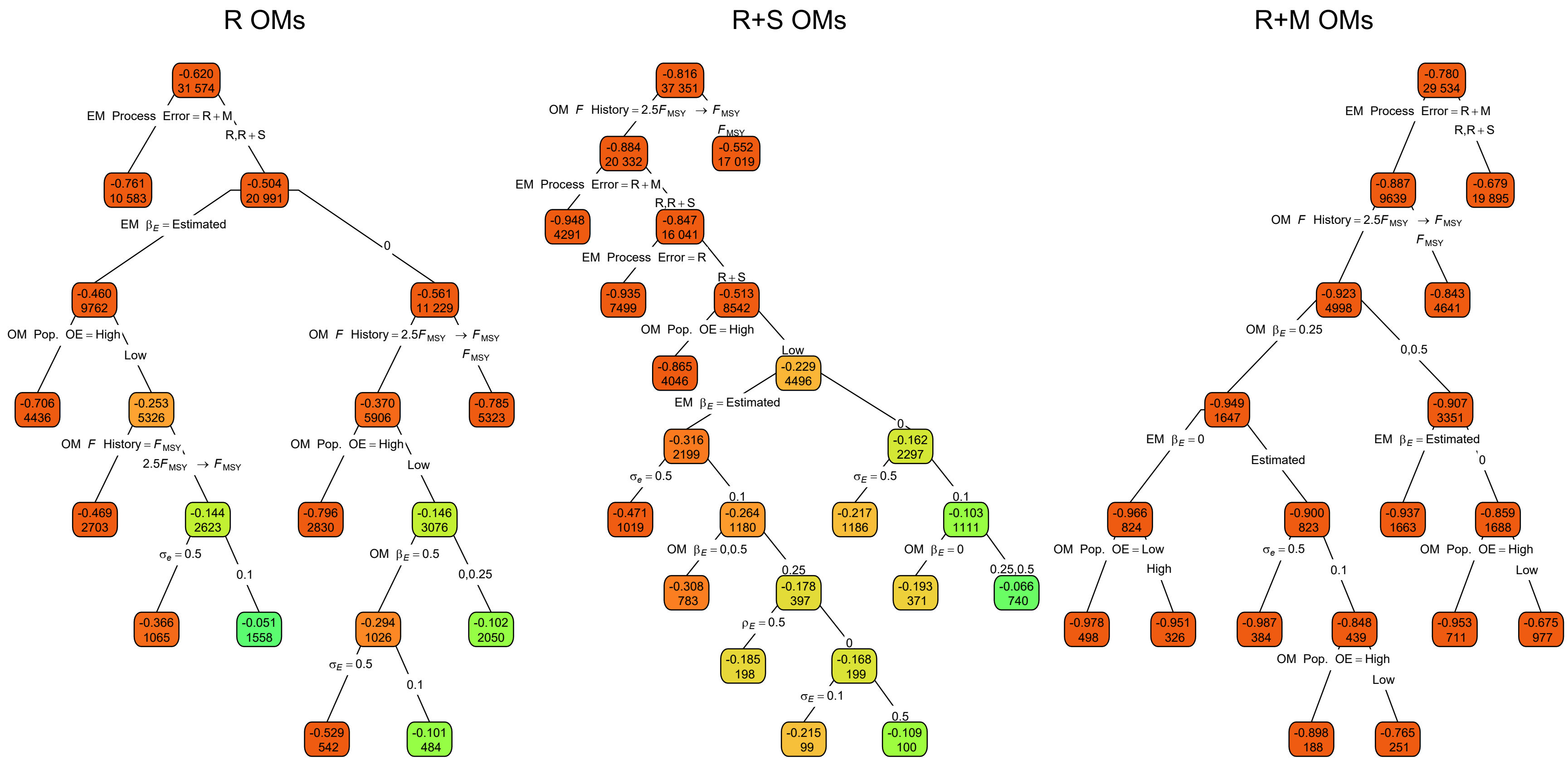

Fig. S7. Regression trees for errors of the Hessian-based standard error estimates for median M parameter $(\widehat{\mathrm{SE}}\left(\widehat{\beta}_M\right))$. Each node shows the median error for the corresponding subset. Lower or higher absolute median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

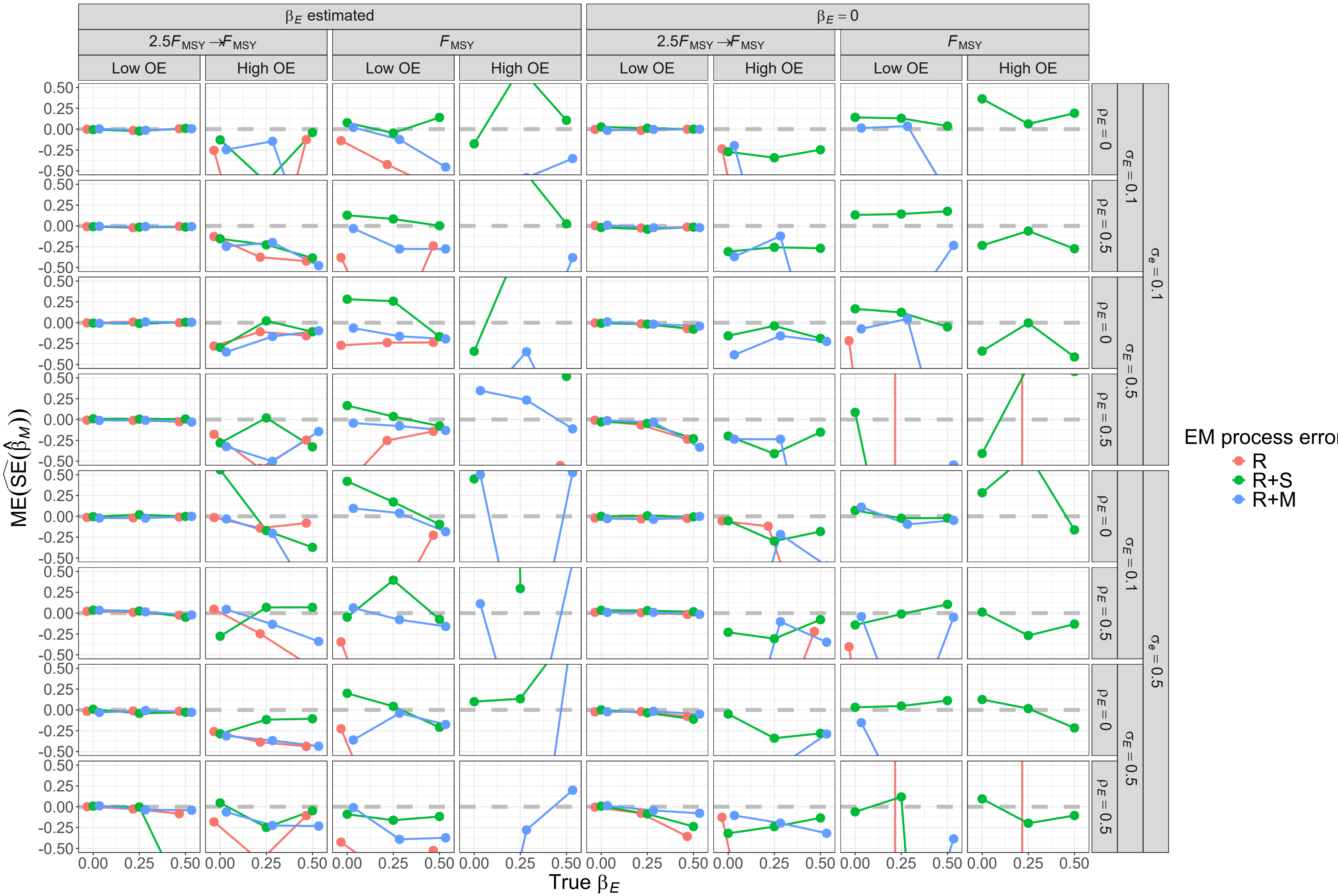

Fig. S8. For R operating models, median error (ME) of Hessian-based estimates of standard error for median M parameter (β_M) from fitting estimating models with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates accross converged fits to simulated data sets for a given operating model scenario.

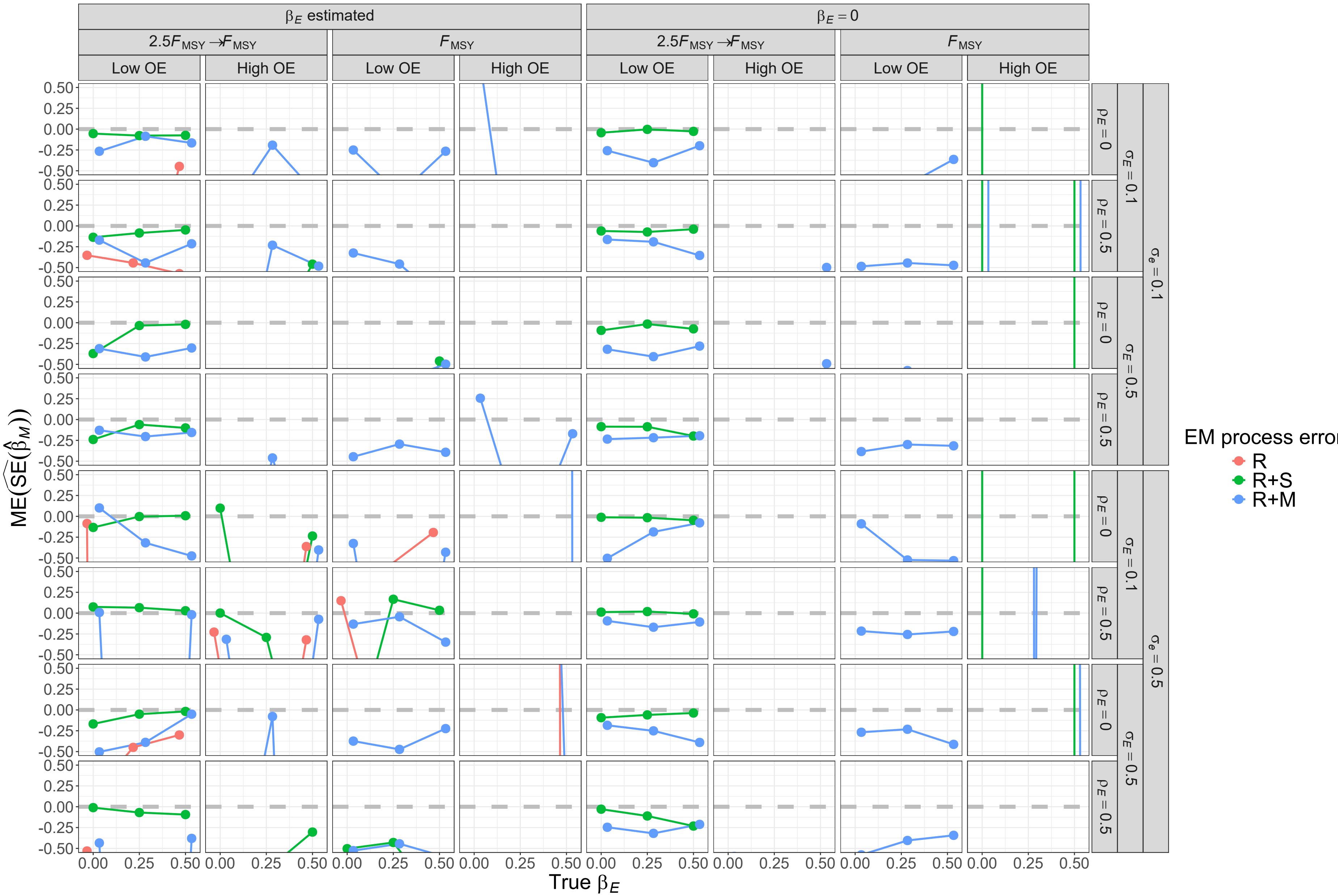

Fig. S9. For R+S operating models, median error (ME) of Hessian-based estimates of standard error for median M parameter (β_M) from fitting estimating models with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates accross converged fits to simulated data sets for a given operating model scenario.

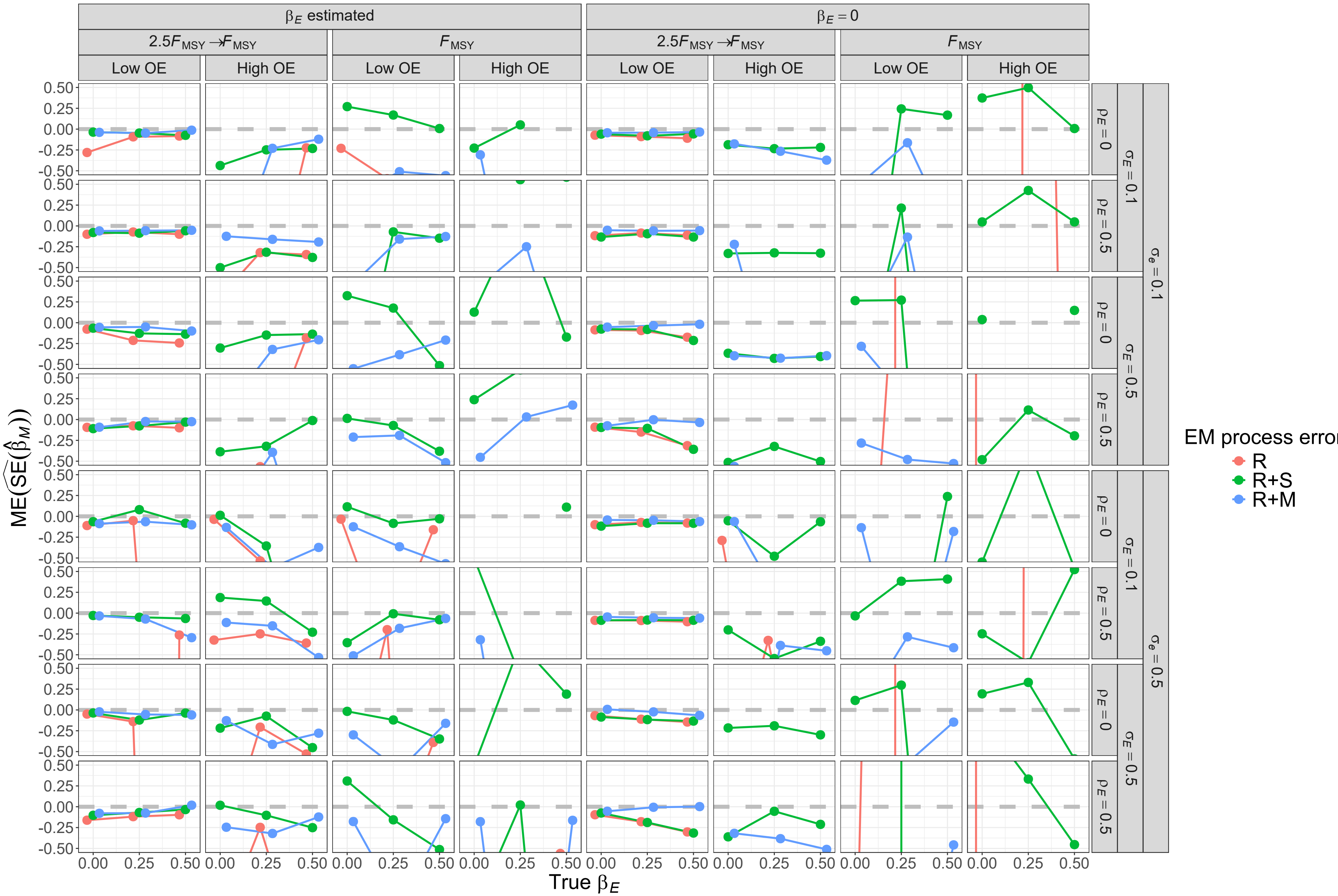

Fig. S10. For R+M operating models, median error (ME) of Hessian-based estimates of standard error for median M parameter (β_M) from fitting estimating models with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates accross converged fits to simulated data sets for a given operating model scenario.

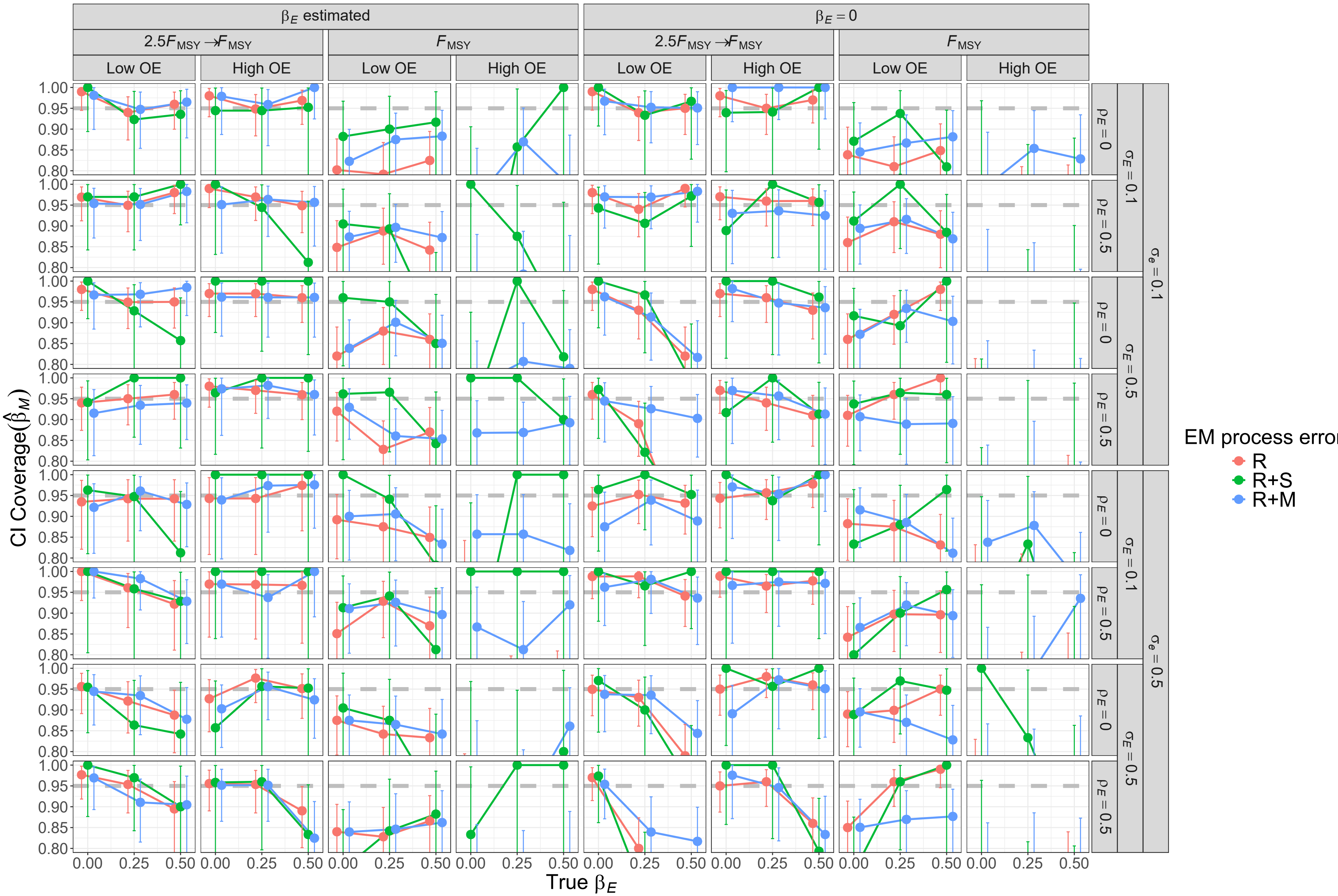

Fig. S11. For R operating models, probability of 95% confidence interval for β_M containing the true value for estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

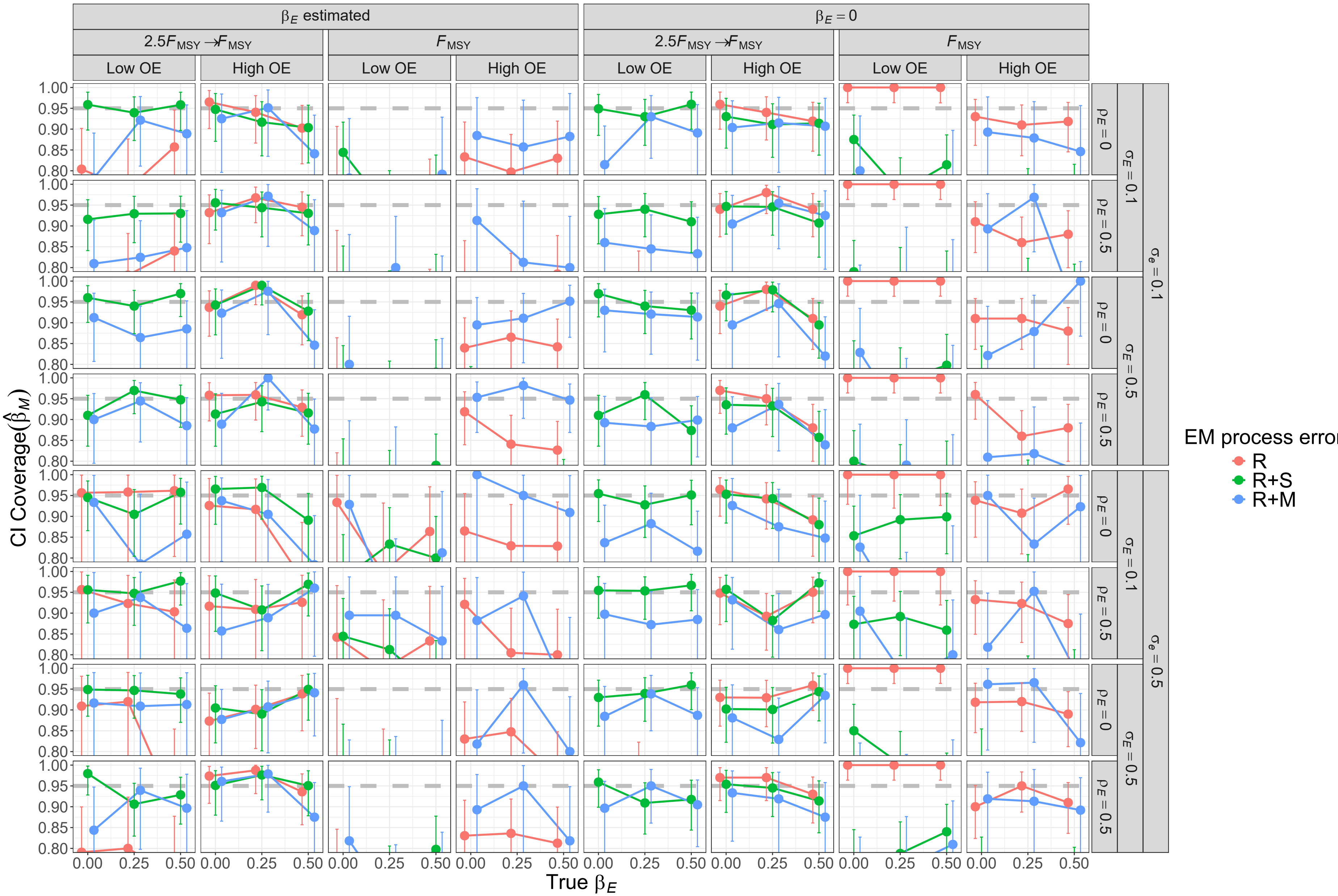

Fig. S12. For R+S operating models, probability of 95% confidence interval for β_M containing the true value for estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

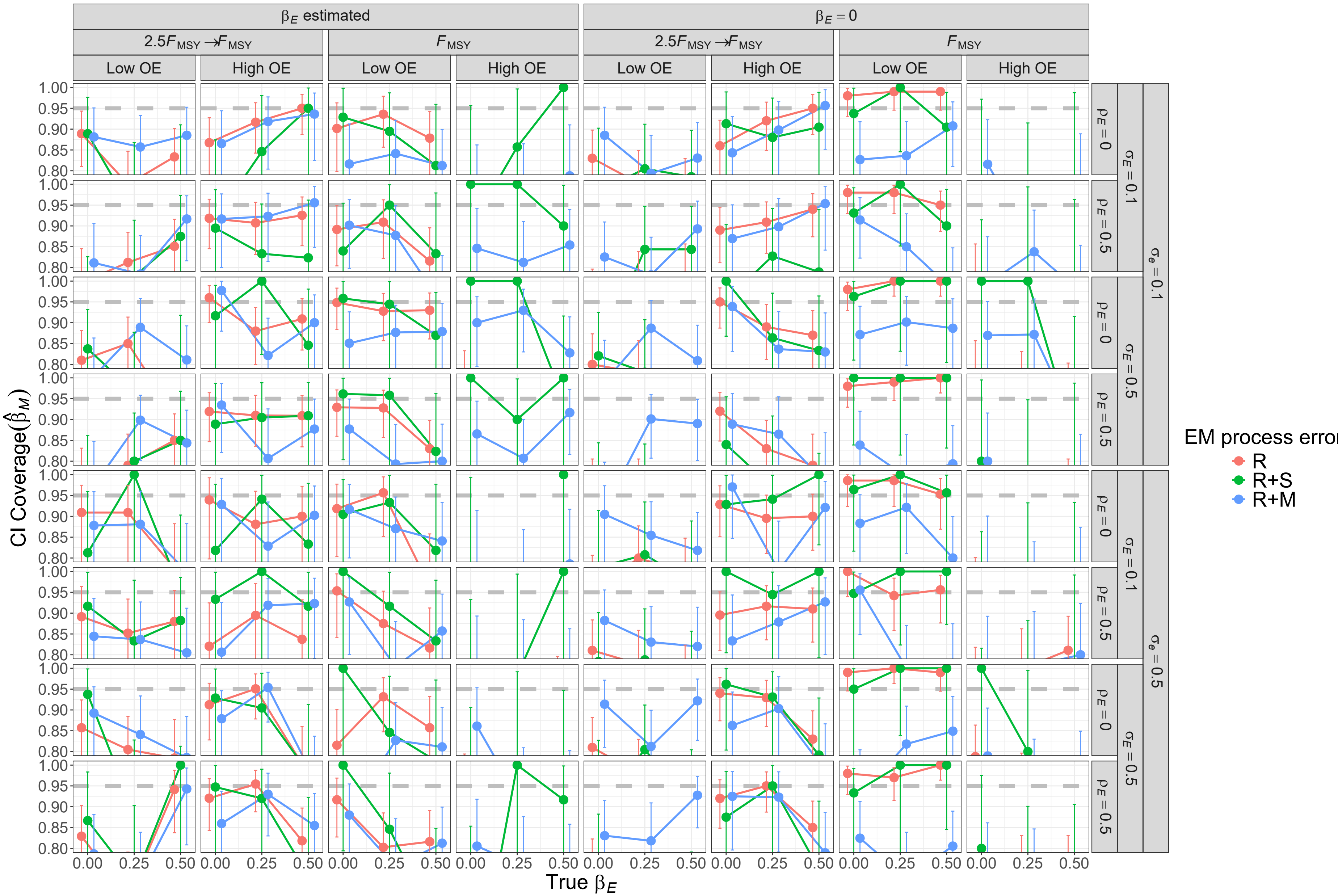

Fig. S13. For R+M operating models, probability of 95% confidence interval for β_M containing the true value for estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

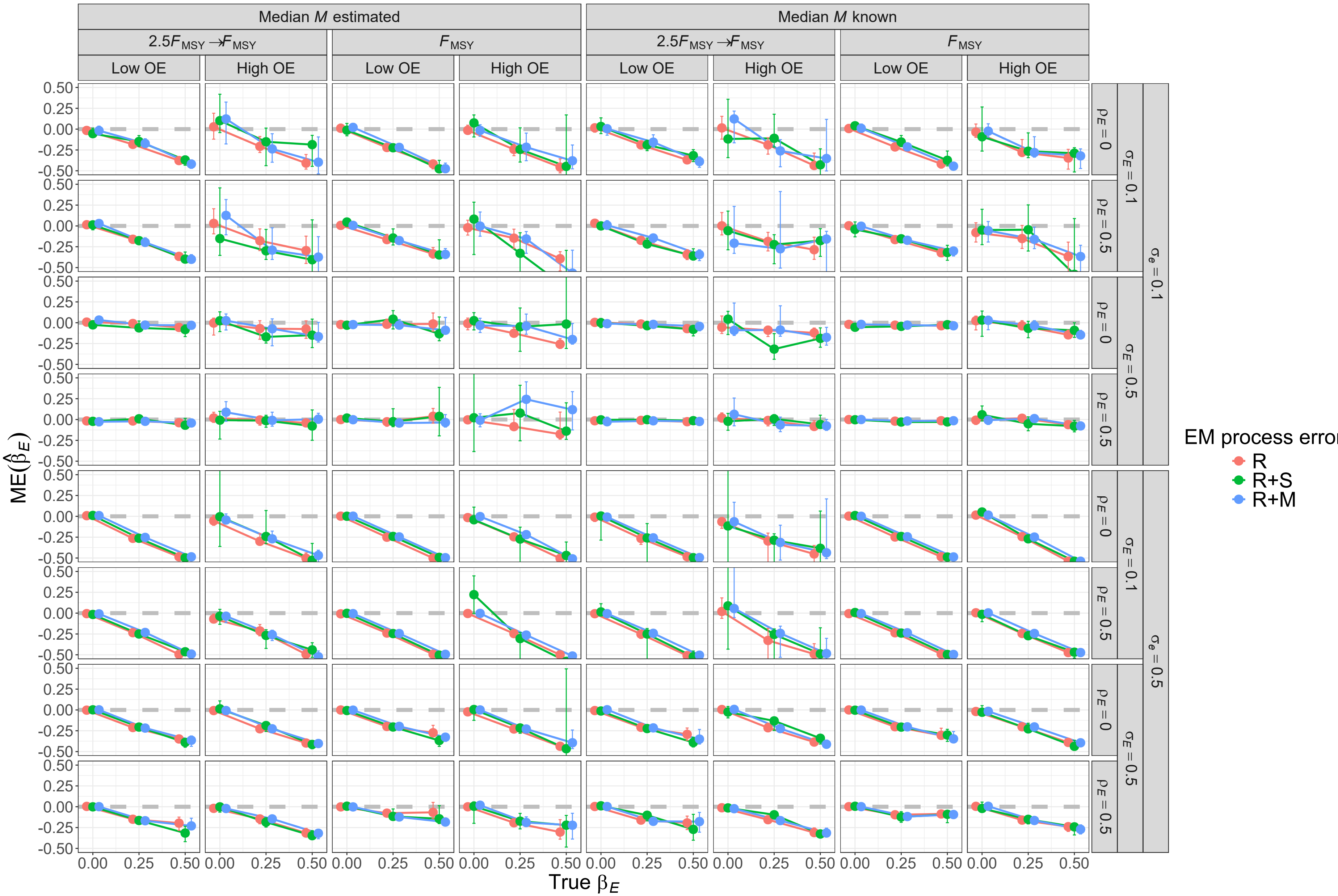

Fig. S14. For R operating models, median error (ME) of estimates of environmental effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

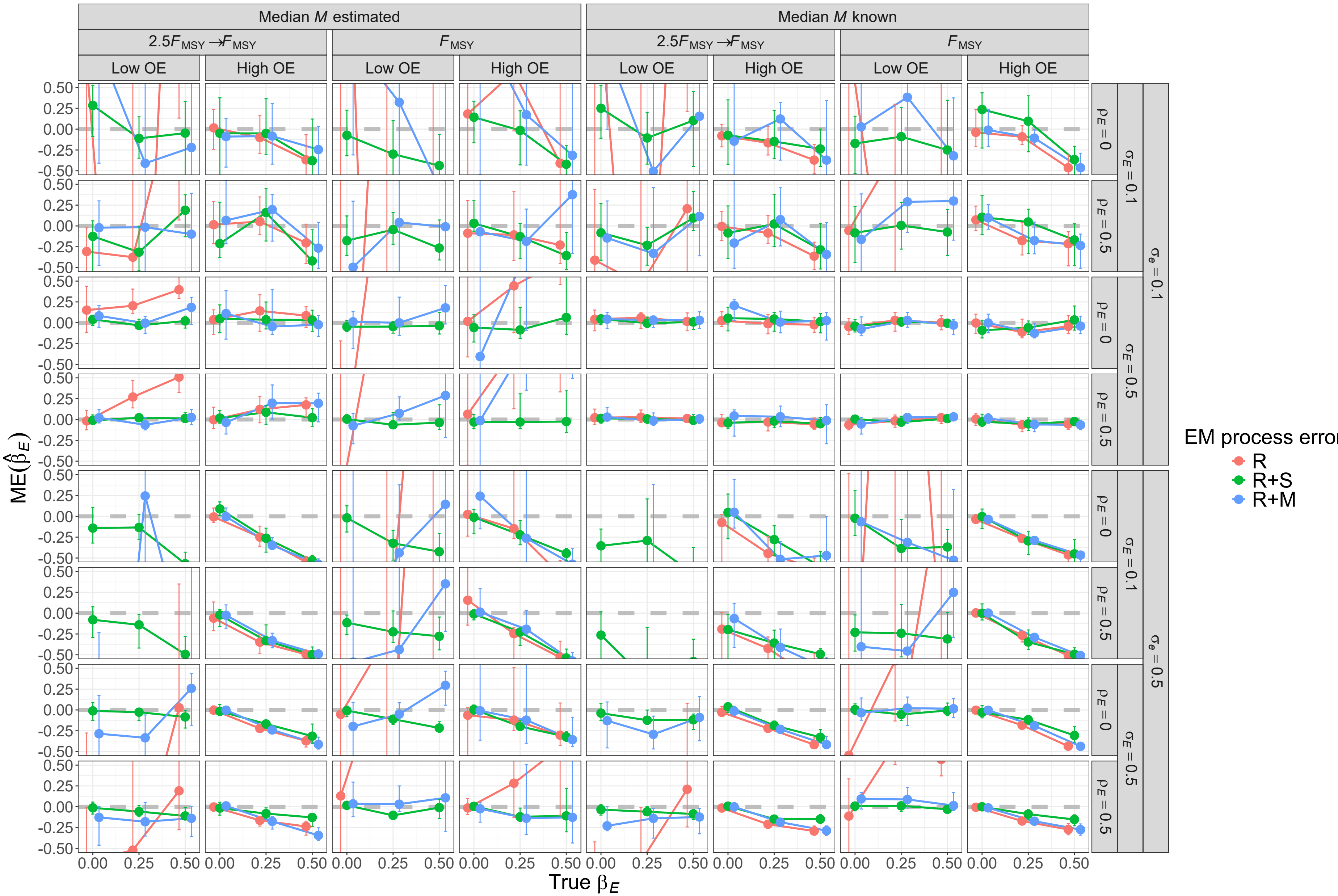

Fig. S15. For R+S operating models, median error (ME) of estimates of environmental effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

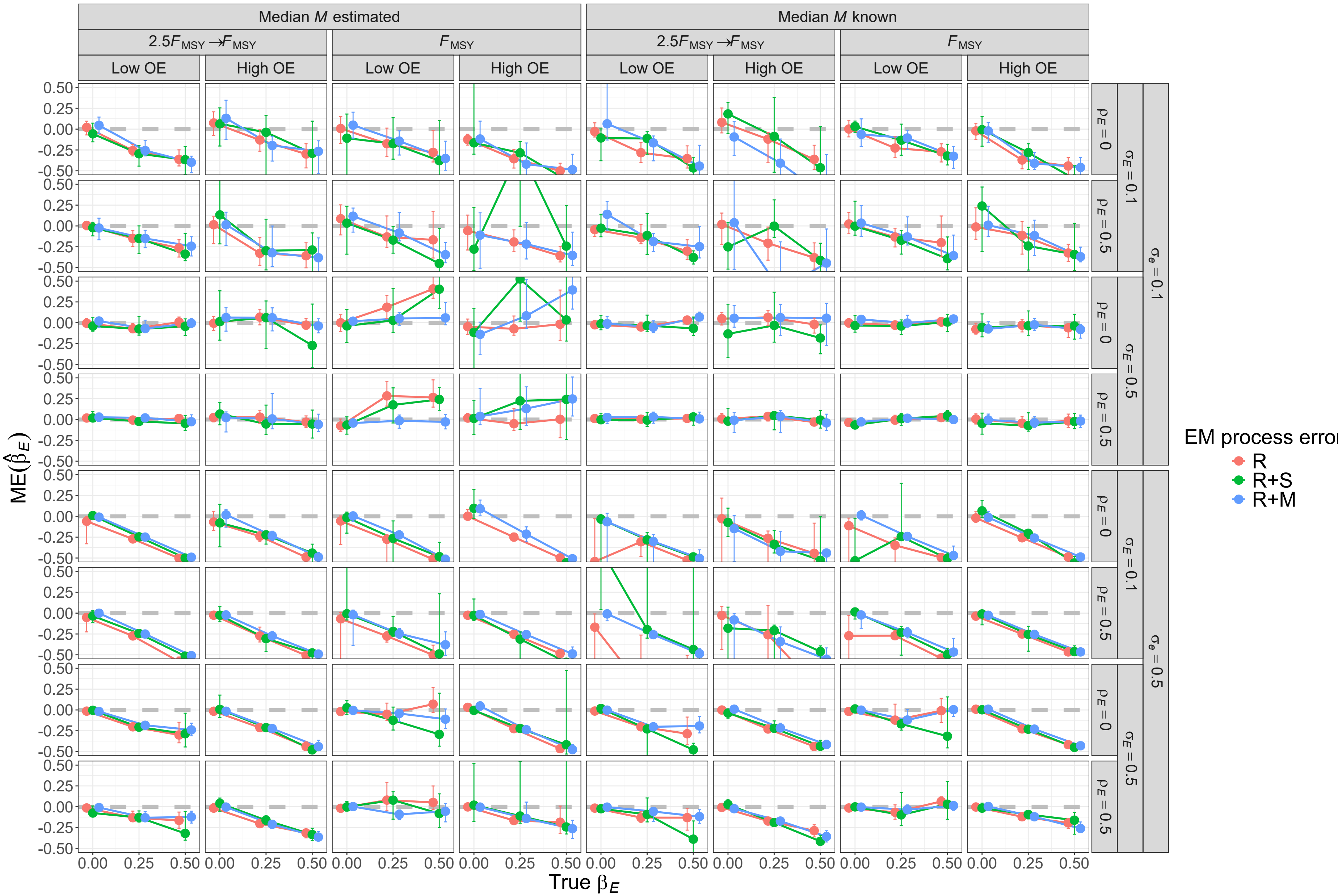

Fig. S16. For R+M operating models, median error (ME) of estimates of environmental effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

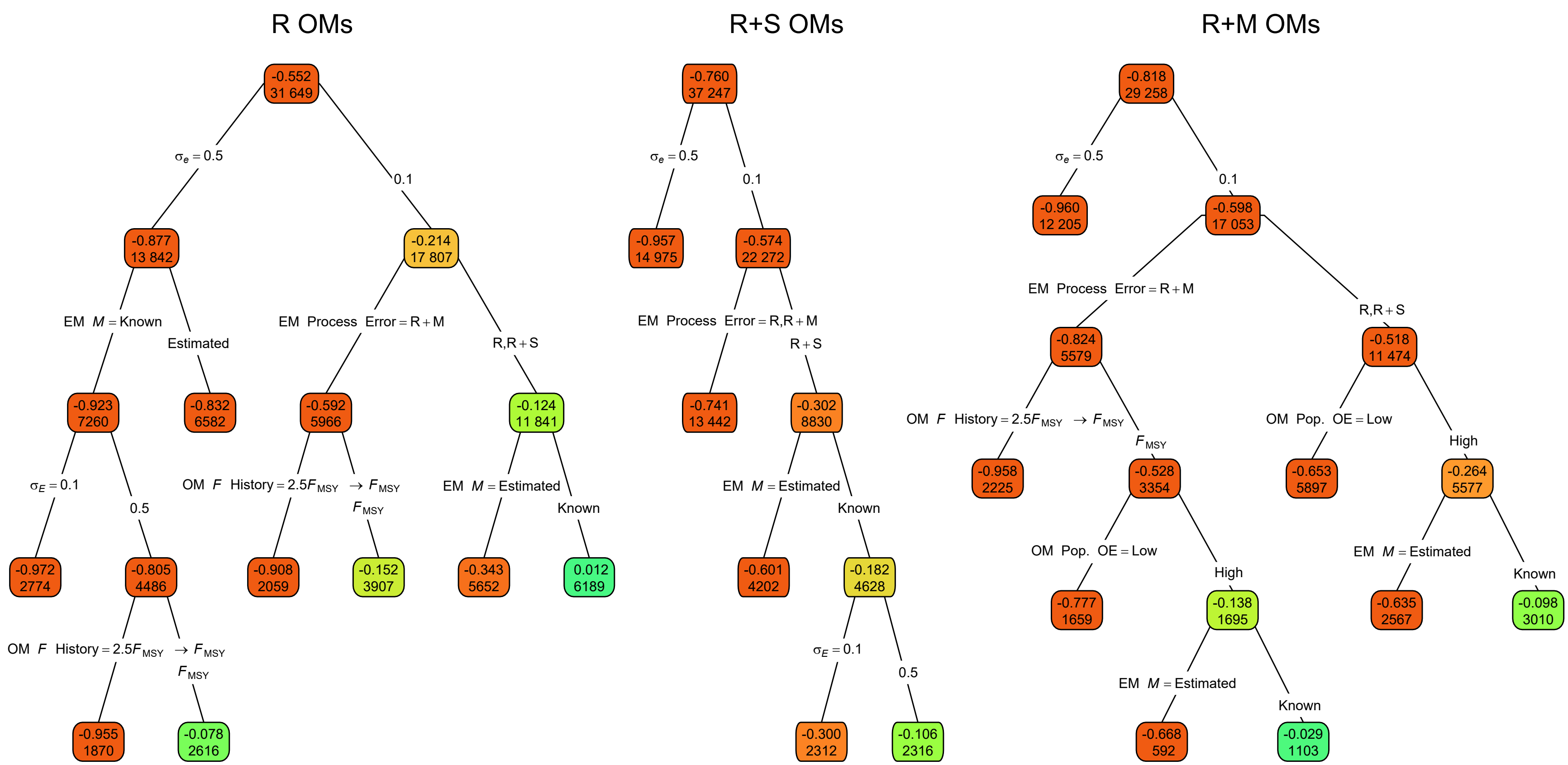

Fig. S17. Regression trees for error of the Hessian-based standard error estimates for covariate effect on M $\left(\widehat{\mathrm{SE}}\left(\widehat{\beta}_E\right)\right)$. Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

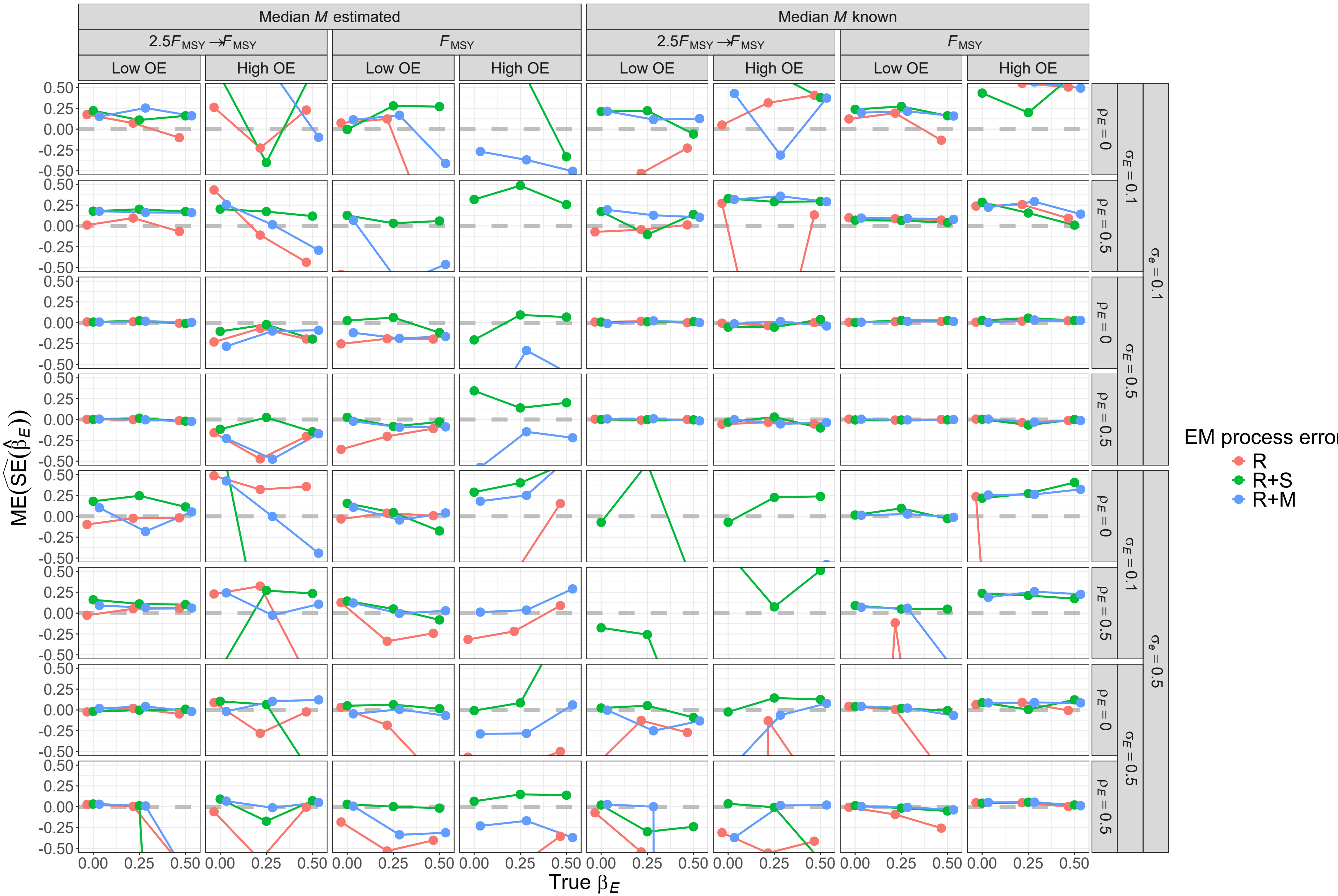

Fig. S18. For R operating models, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). True standard error is defined as the mean of the standard error estimates accross converged fits to simulated data sets for a given operating model scenario.

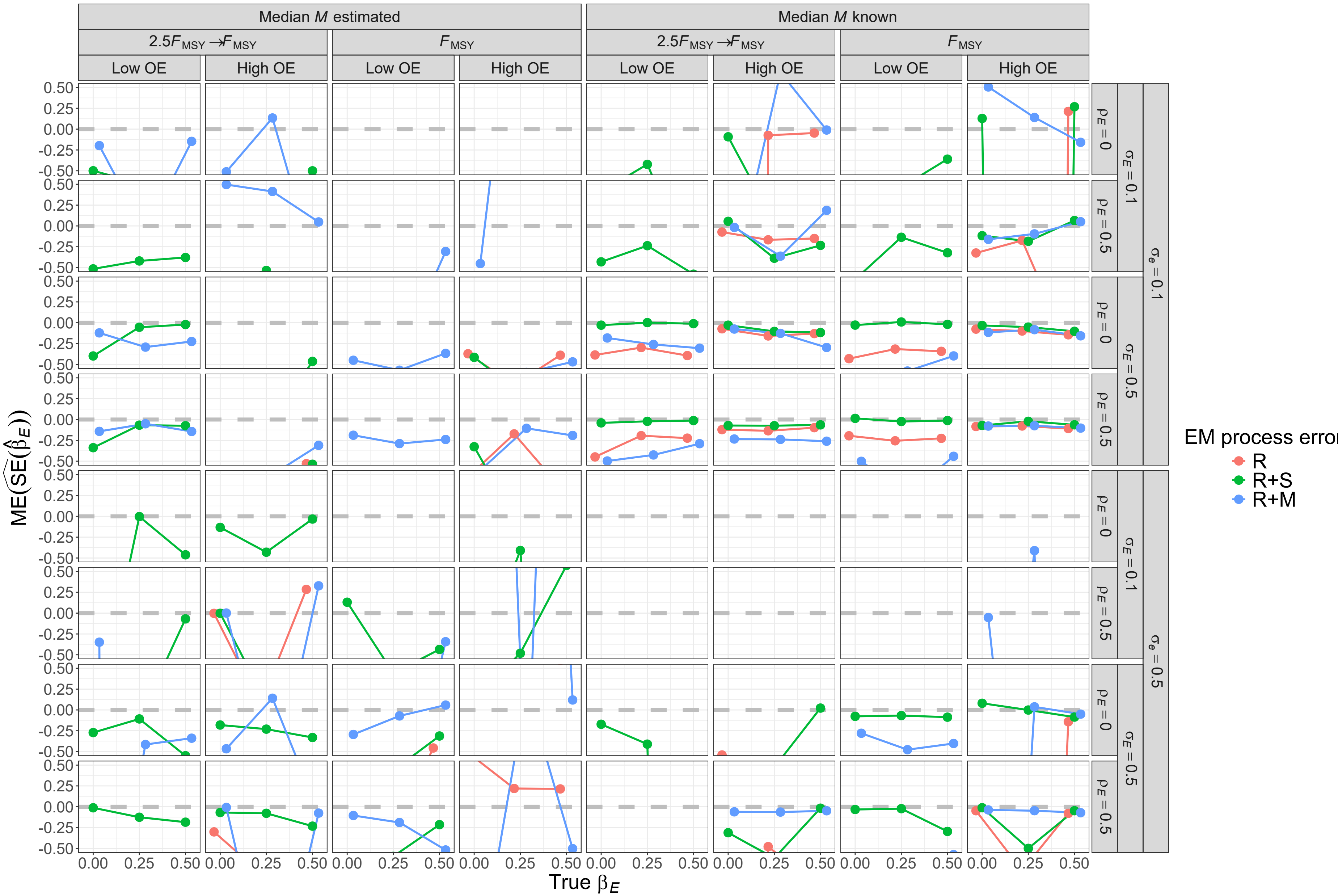

Fig. S19. For R+S operating models, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). True standard error is defined as the mean of the standard error estimates accross converged fits to simulated data sets for a given operating model scenario.

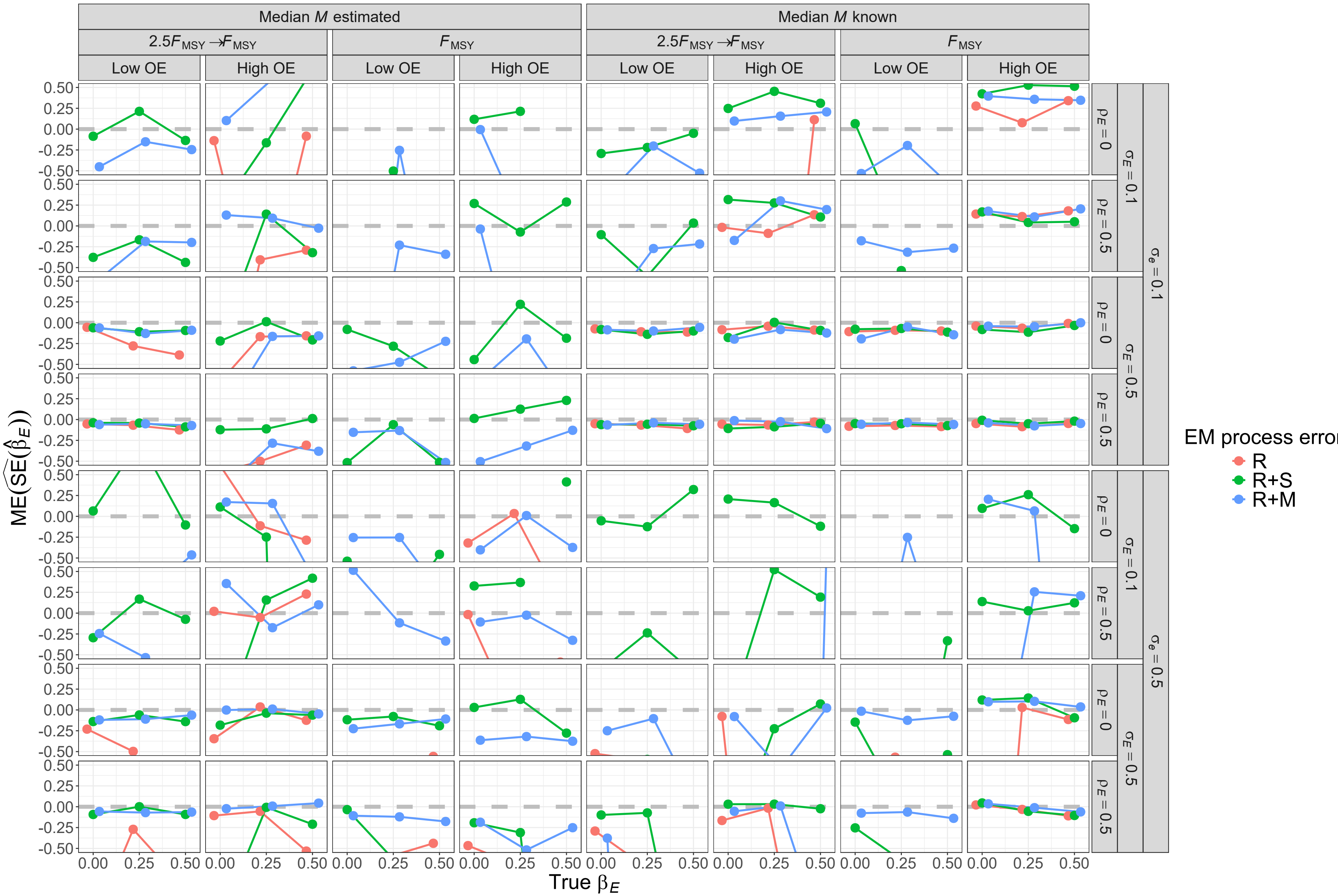

Fig. S20. For R+M operating models, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). True standard error is defined as the mean of the standard error estimates accross converged fits to simulated data sets for a given operating model scenario.

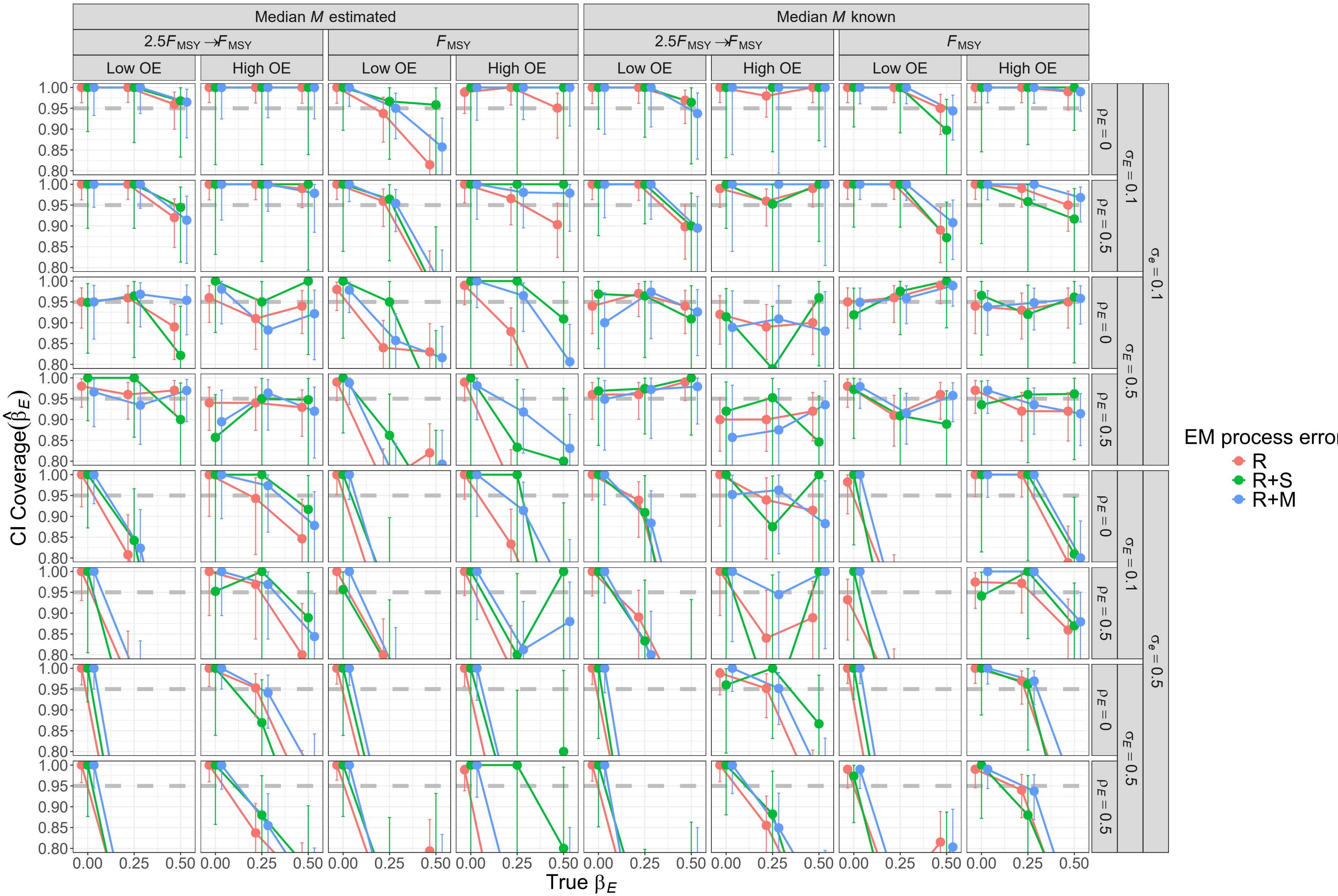

Fig. S21. For R operating models, probability of 95% confidence interval for β_E containing the true value for estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

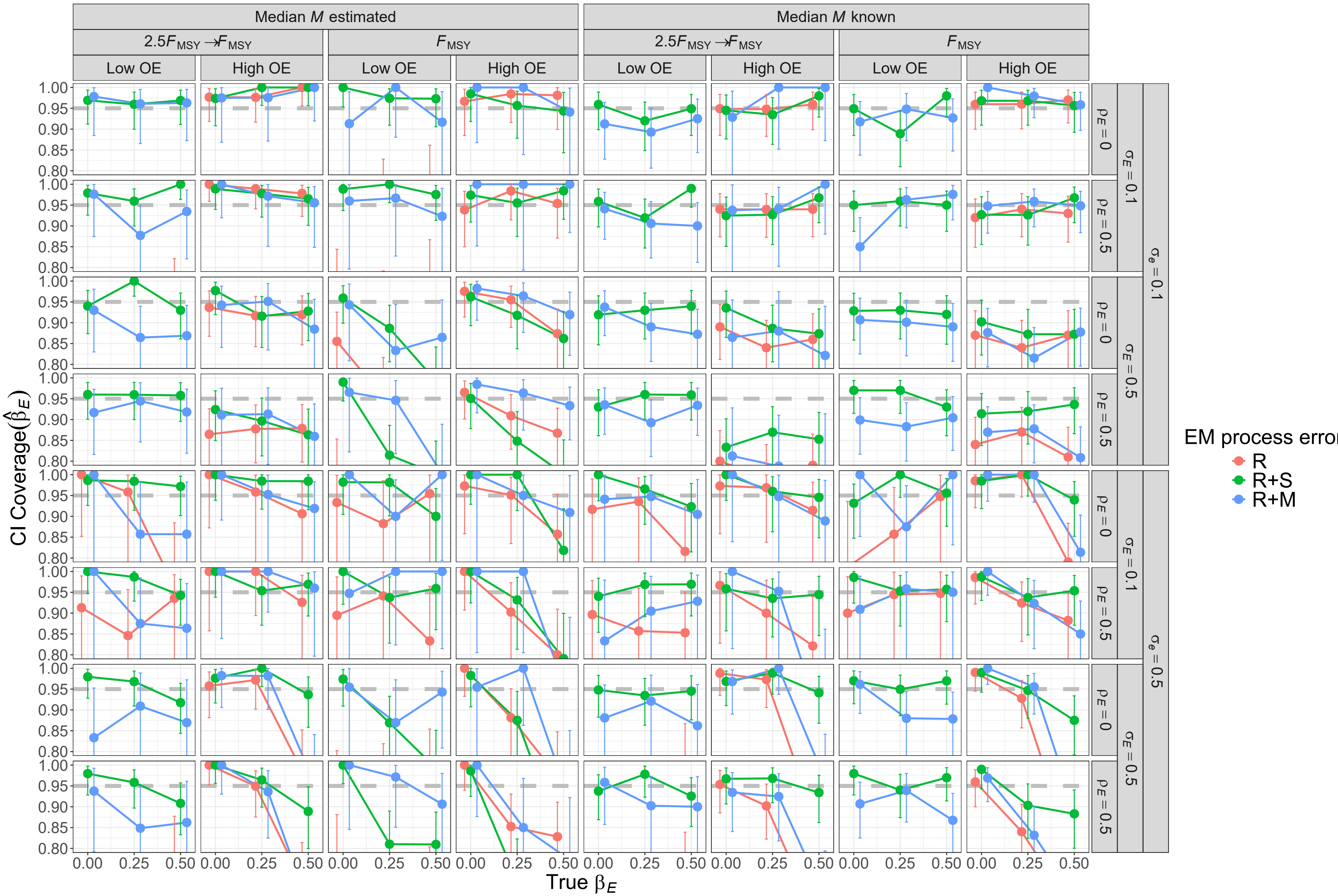

Fig. S22. For R+S operating models, probability of 95% confidence interval for β_E containing the true value for estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

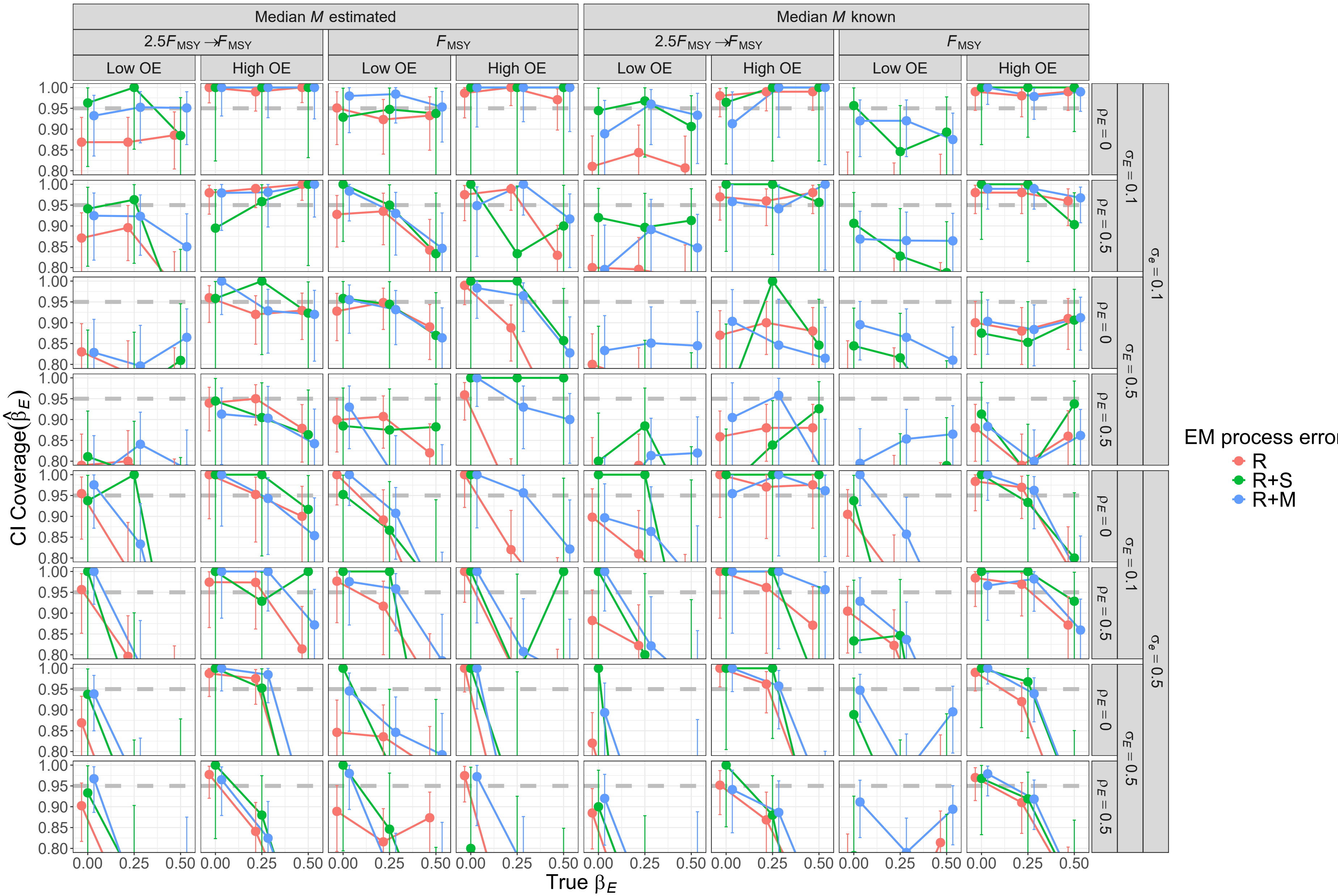

Fig. S23. For R+M operating models, probability of 95% confidence interval for β_E containing the true value for estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

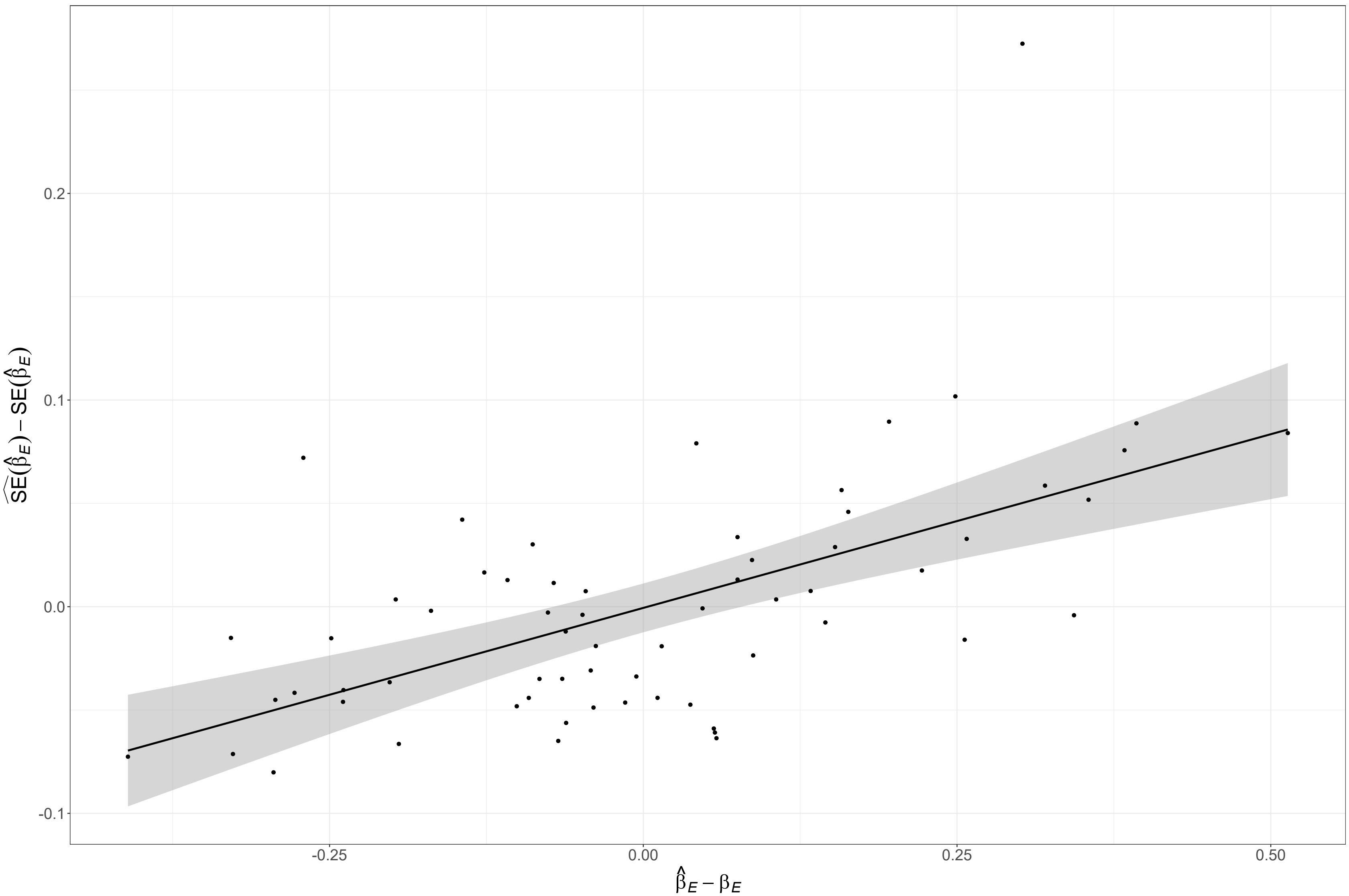

Fig. S24. Positive correlation of covariate effect estimates and Hessian-based standard error estimates for estimating model that also estimates the median M parameter (β_M) and has correct R+M process error assumption fitted to simulated data from the operating model with R+M process errors, temporal contrast in fishing pressure, low observation uncertainty for both population ($LowOE$) and covariate observations ($\sigma_e = 0.1$), high and uncorrelated temporal variability in the true covariate ($\sigma_E = 0.5$ and $\rho_E = 0$), and the strongest covariate effect on M ($\beta_E = 0.5$).

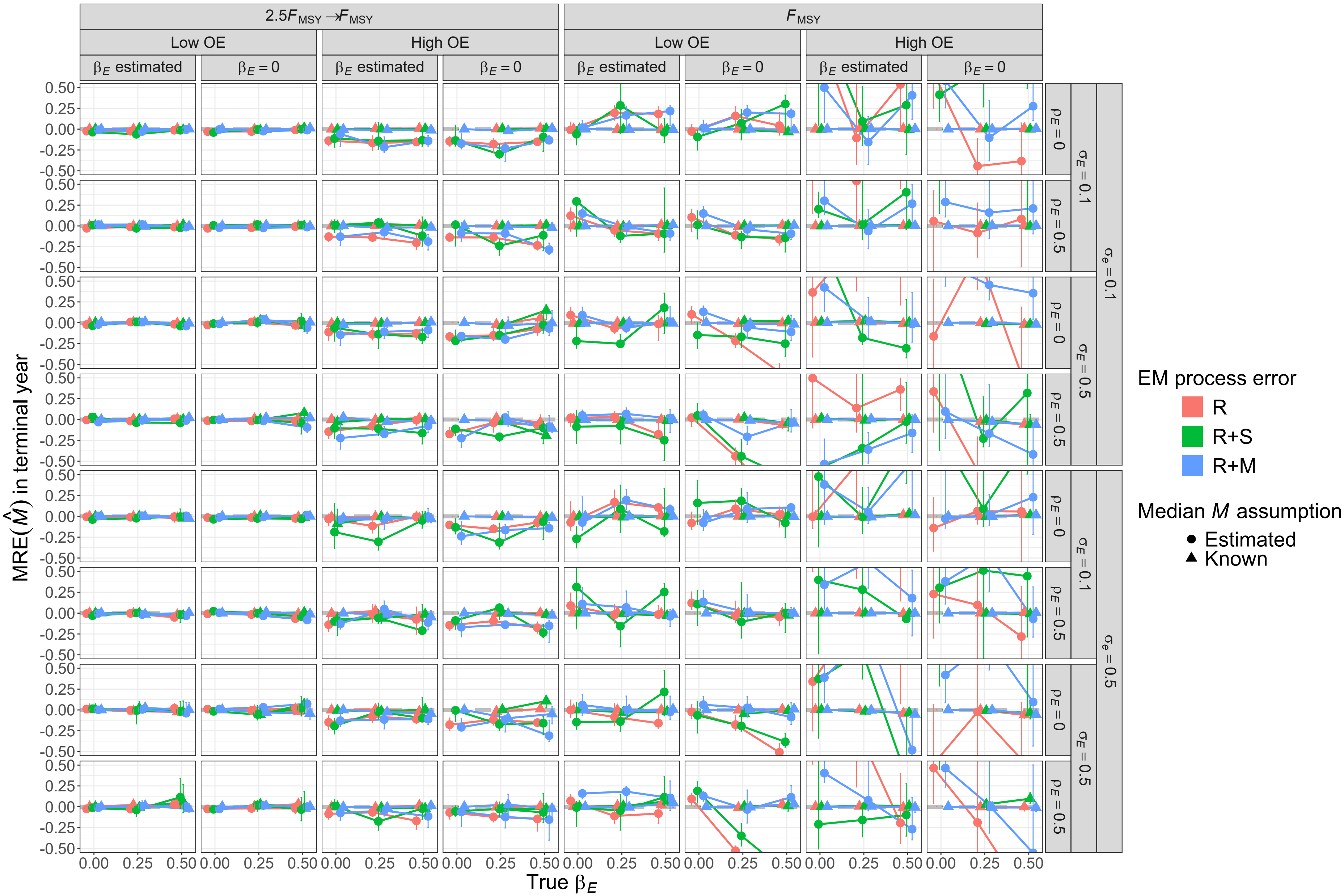

Fig. S25. For R operating models, median relative error (MRE) of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

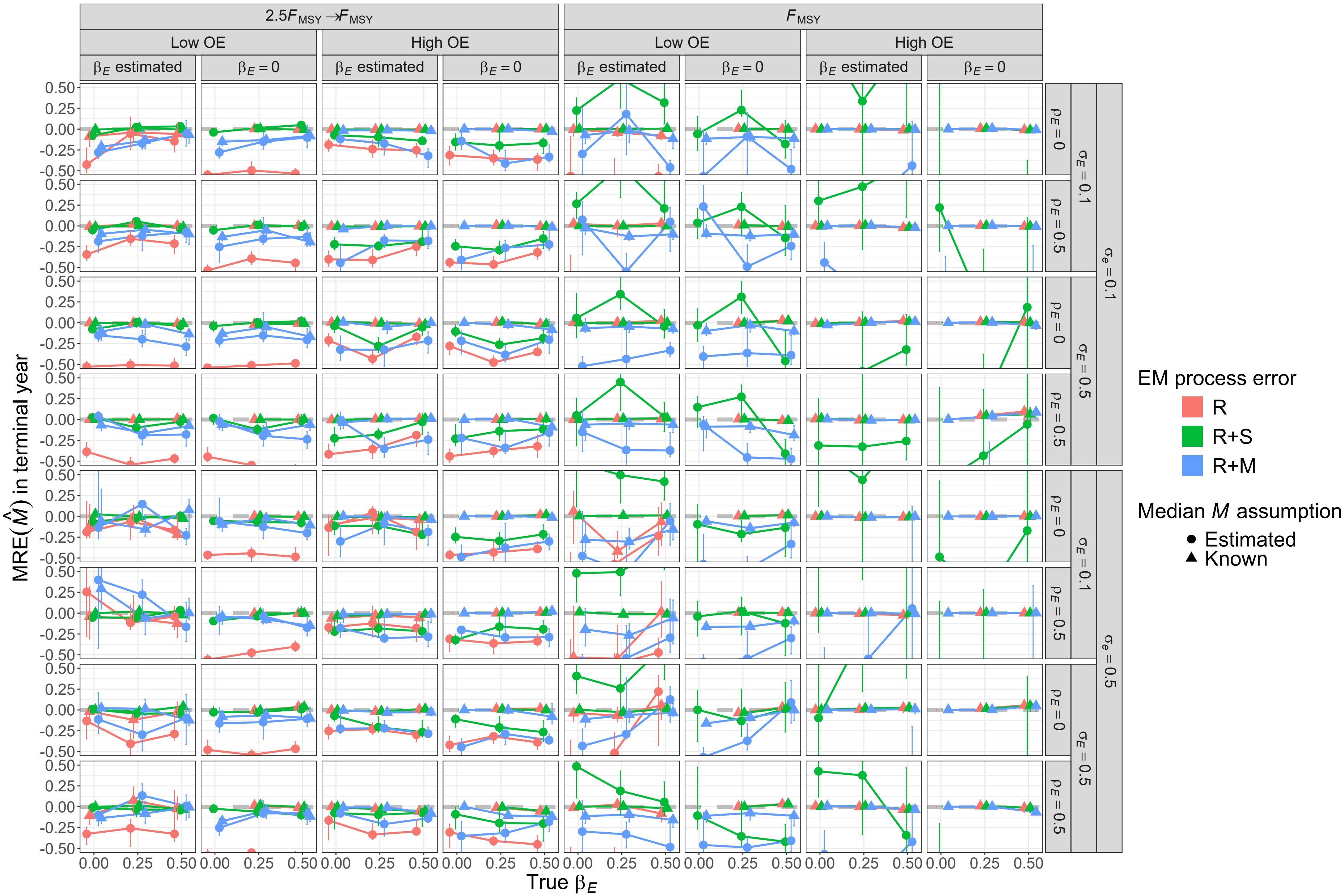

Fig. S26. For R+S operating models, median relative error (MRE) of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

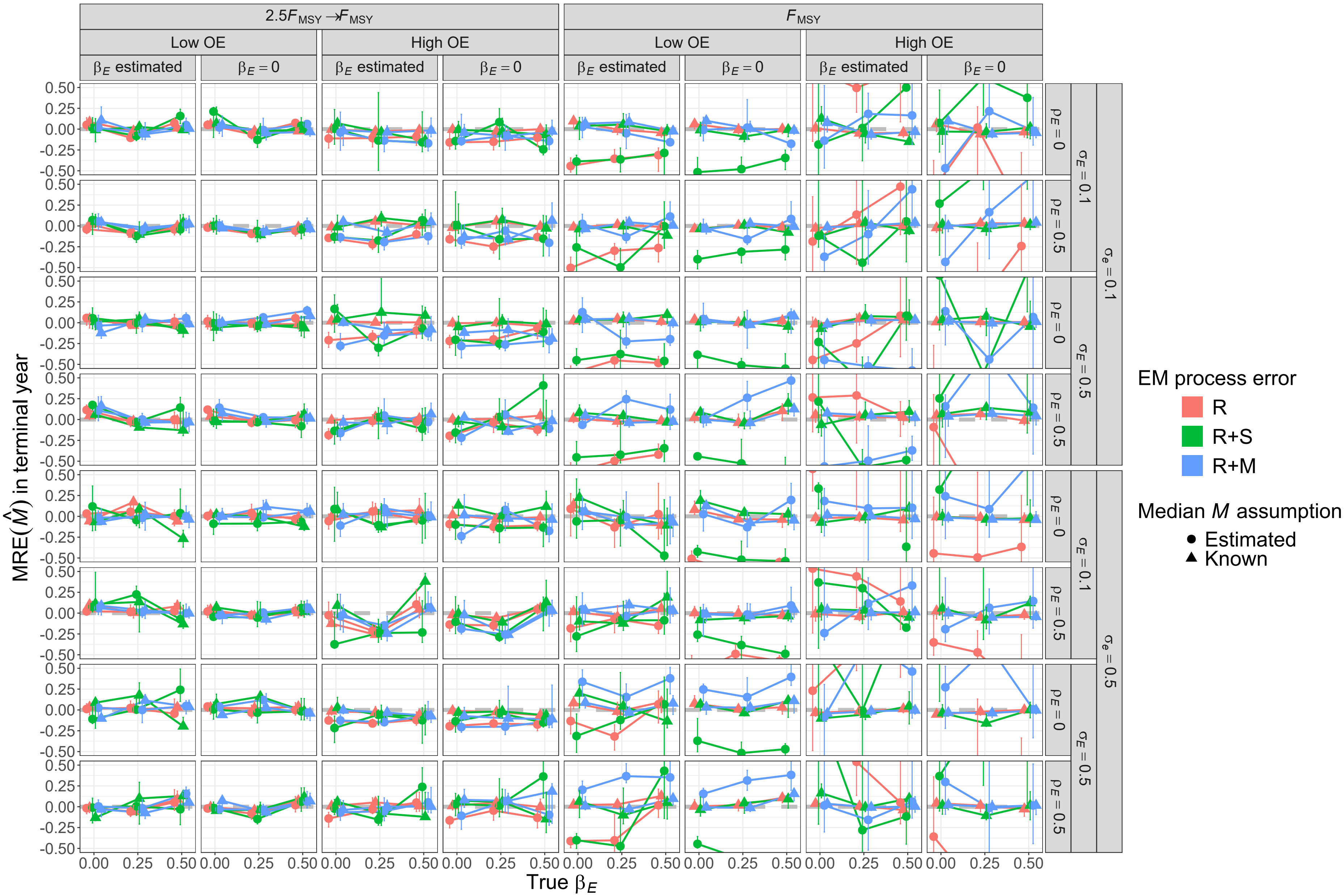

Fig. S27. For R+M operating models, median relative error (MRE) of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

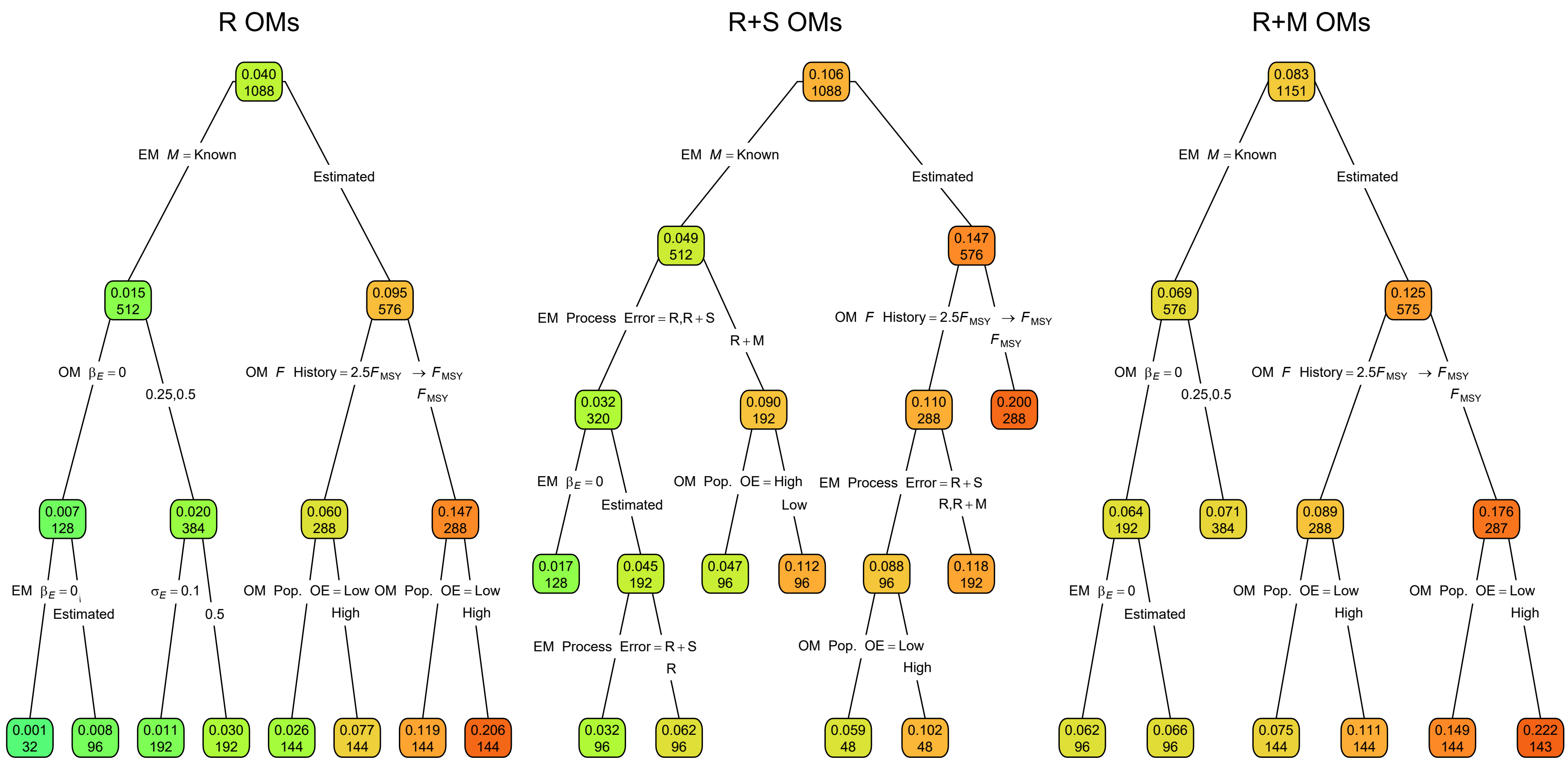

Fig. S28. Regression trees for the log-transformed RMSE of terminal year M. Each node shows the median RMSE (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

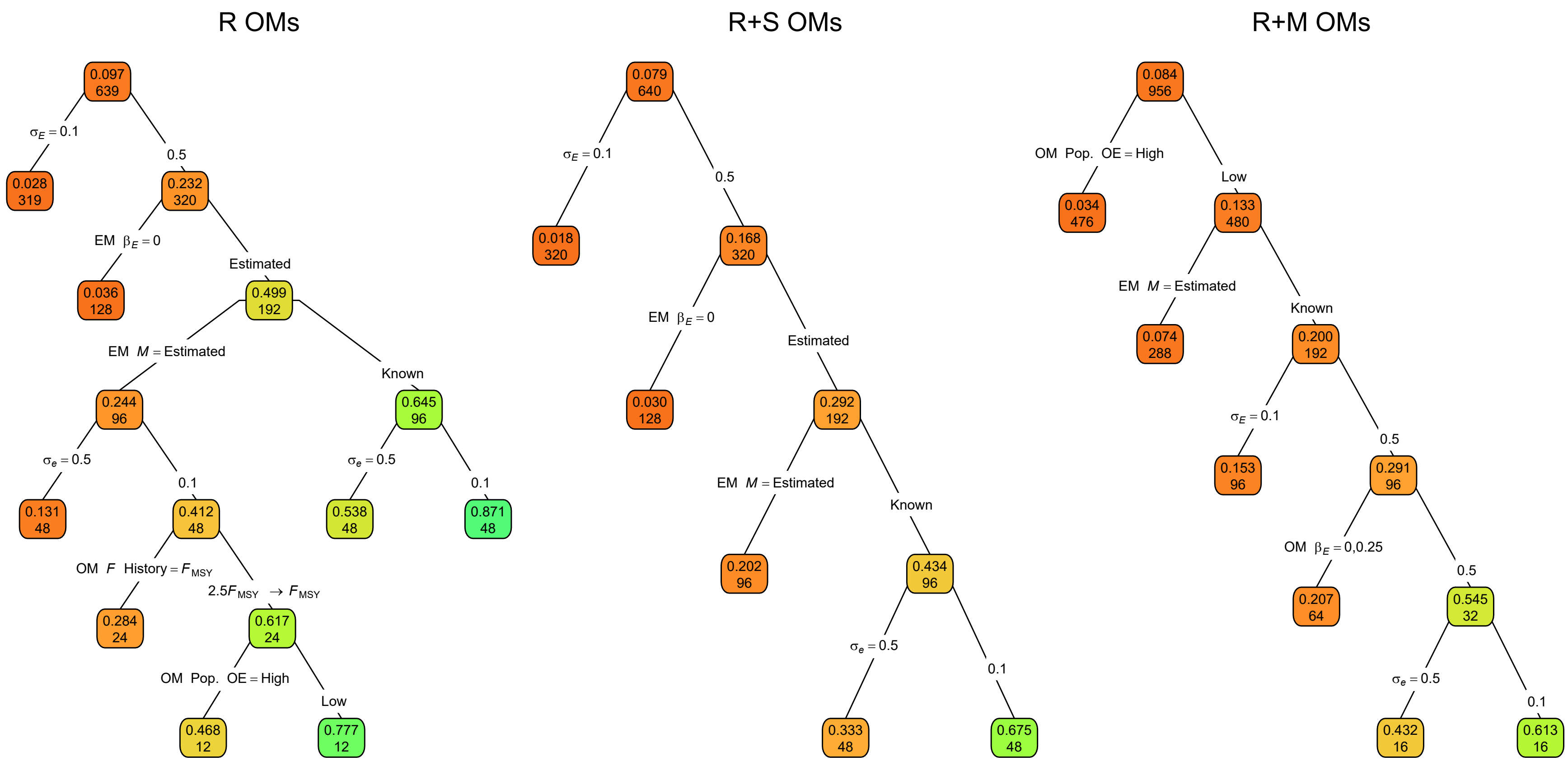

Fig. S29. Regression trees for the correlation of terminal year M estimates and true values. Each node shows the median correlation for the corresponding subset. Lower or higher median correlations are indicated by more red or green polygons, respectively. See Figure 1 caption and Table S1 for further details.

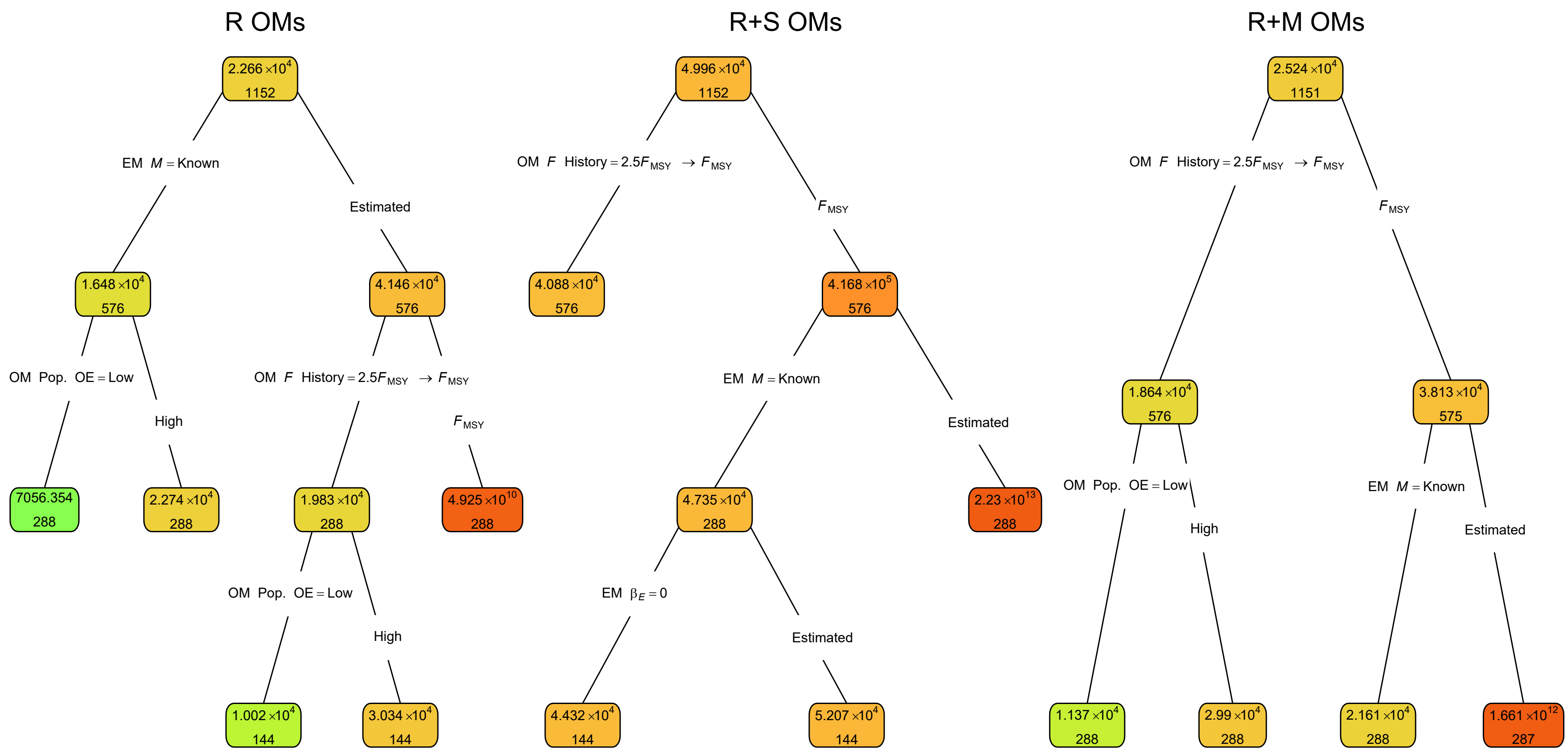

Fig. S30. Regression trees for the log-transformed RMSE of terminal year SSB. Each node shows the median RMSE (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more red or green polygons, respectively.

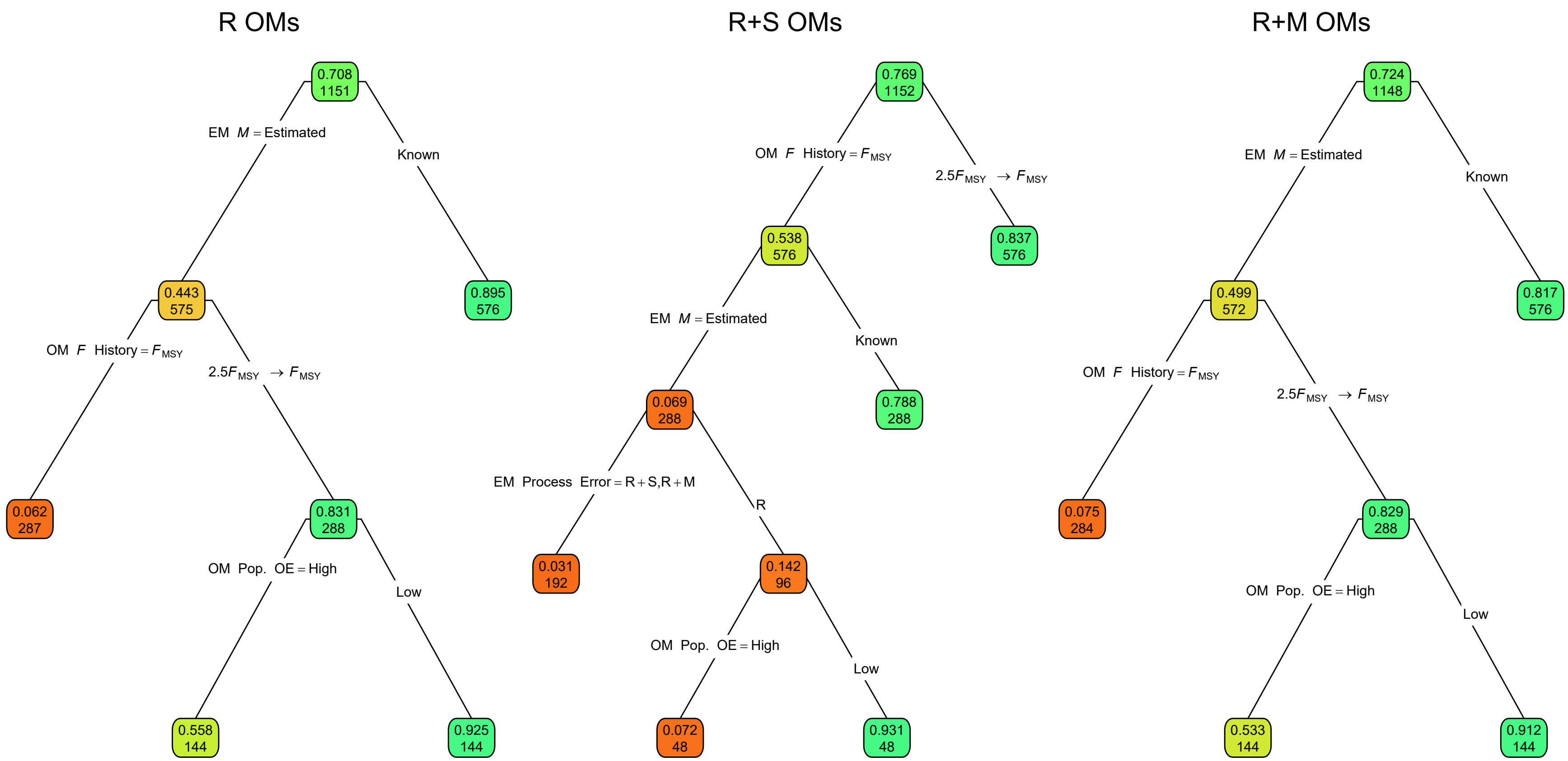

Fig. S31. Regression trees for the correlation of terminal year SSB estimates and true values. Each node shows the median correlation for the corresponding subset. Lower or higher median correlations are indicated by more red or green polygons, respectively. See Figure 1 caption and Table S1 for further details.

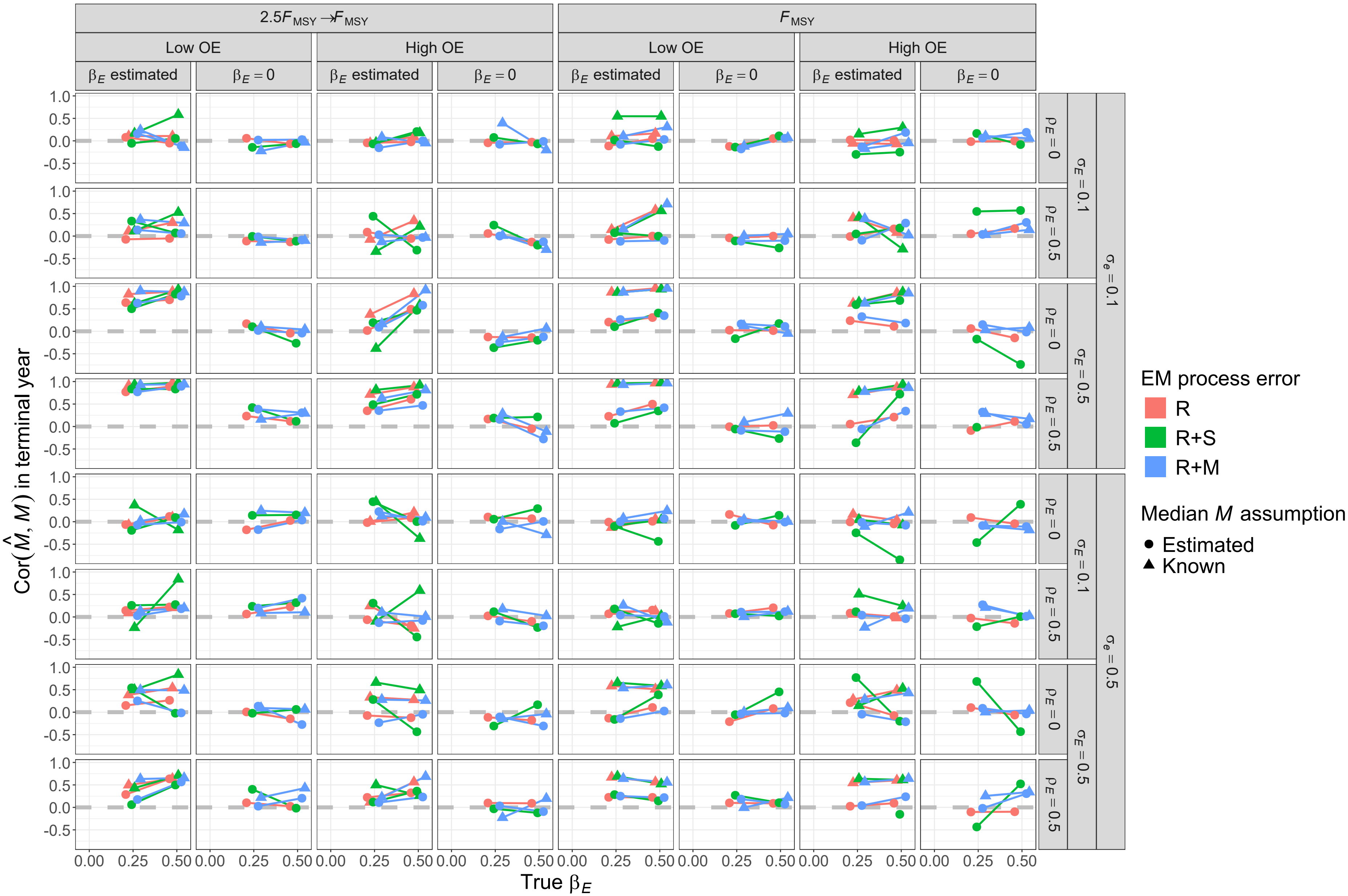

Fig. S32. For R operating models, correlation of estimates and true values of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

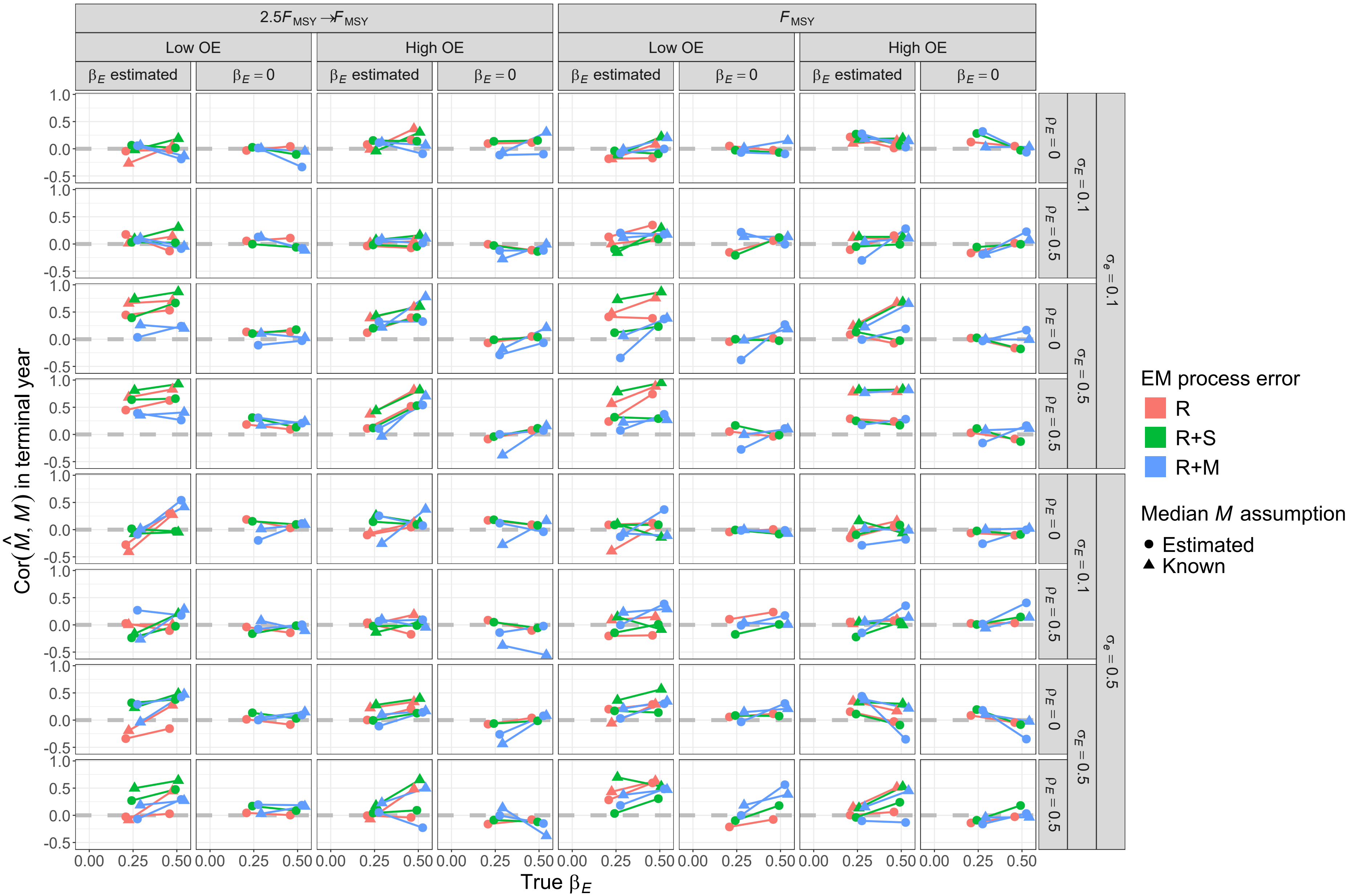

Fig. S33. For R+S operating models, correlation of estimates and true values of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

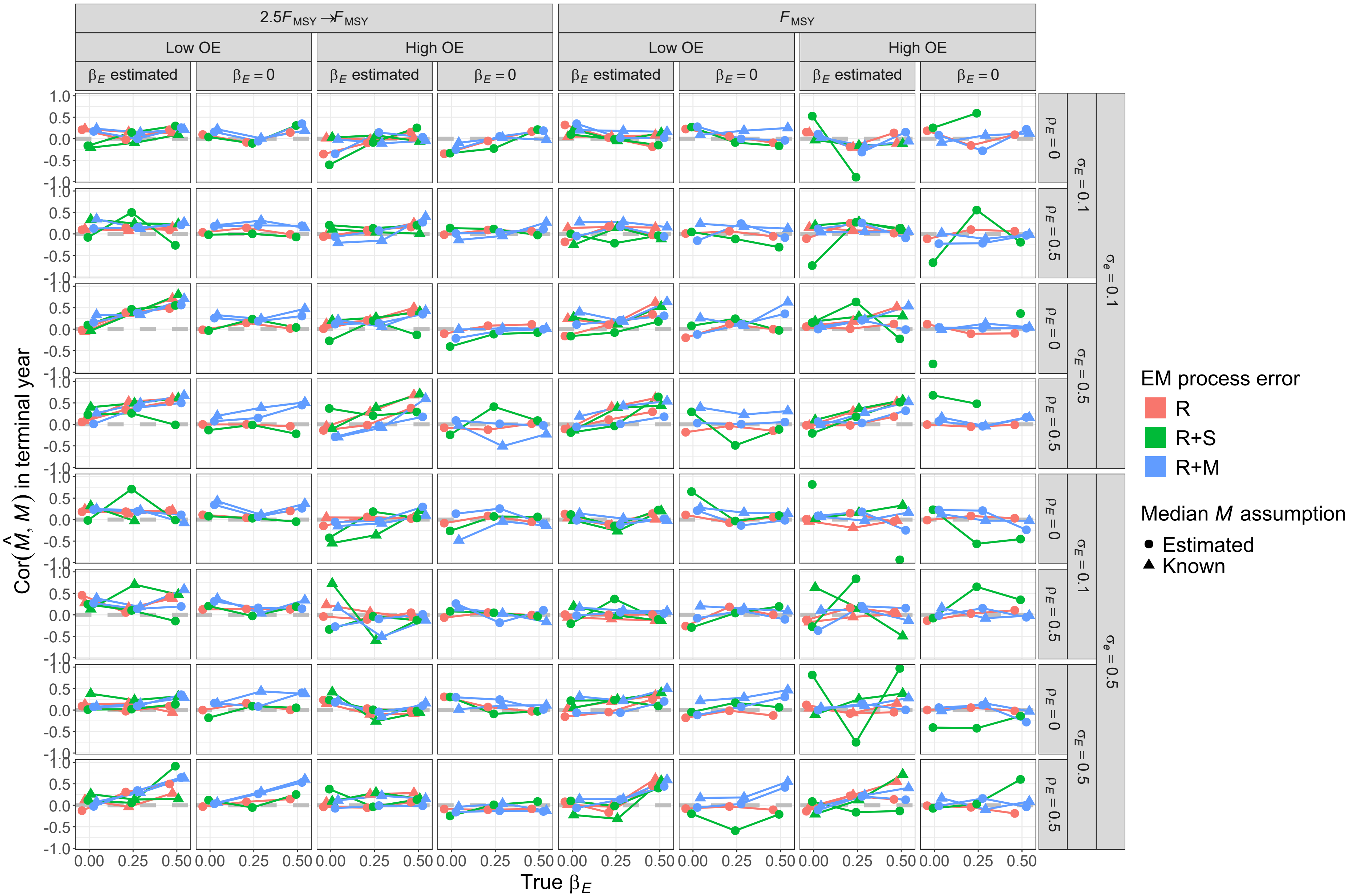

Fig. S34. For R+M operating models, correlation of estimates and true vales of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

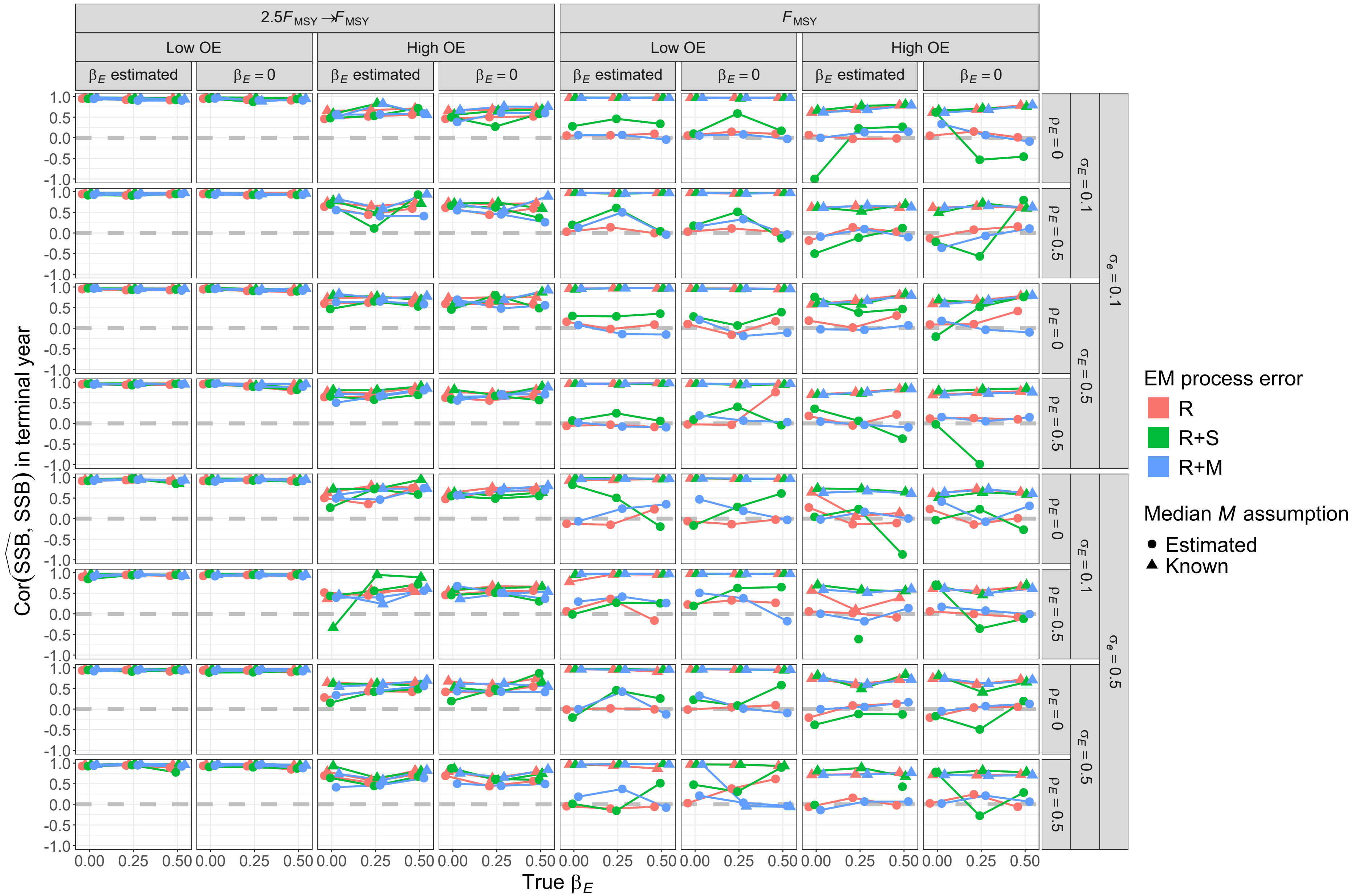

Fig. S35. For R operating models, correlation of estimates and true values of terminal year SSB for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

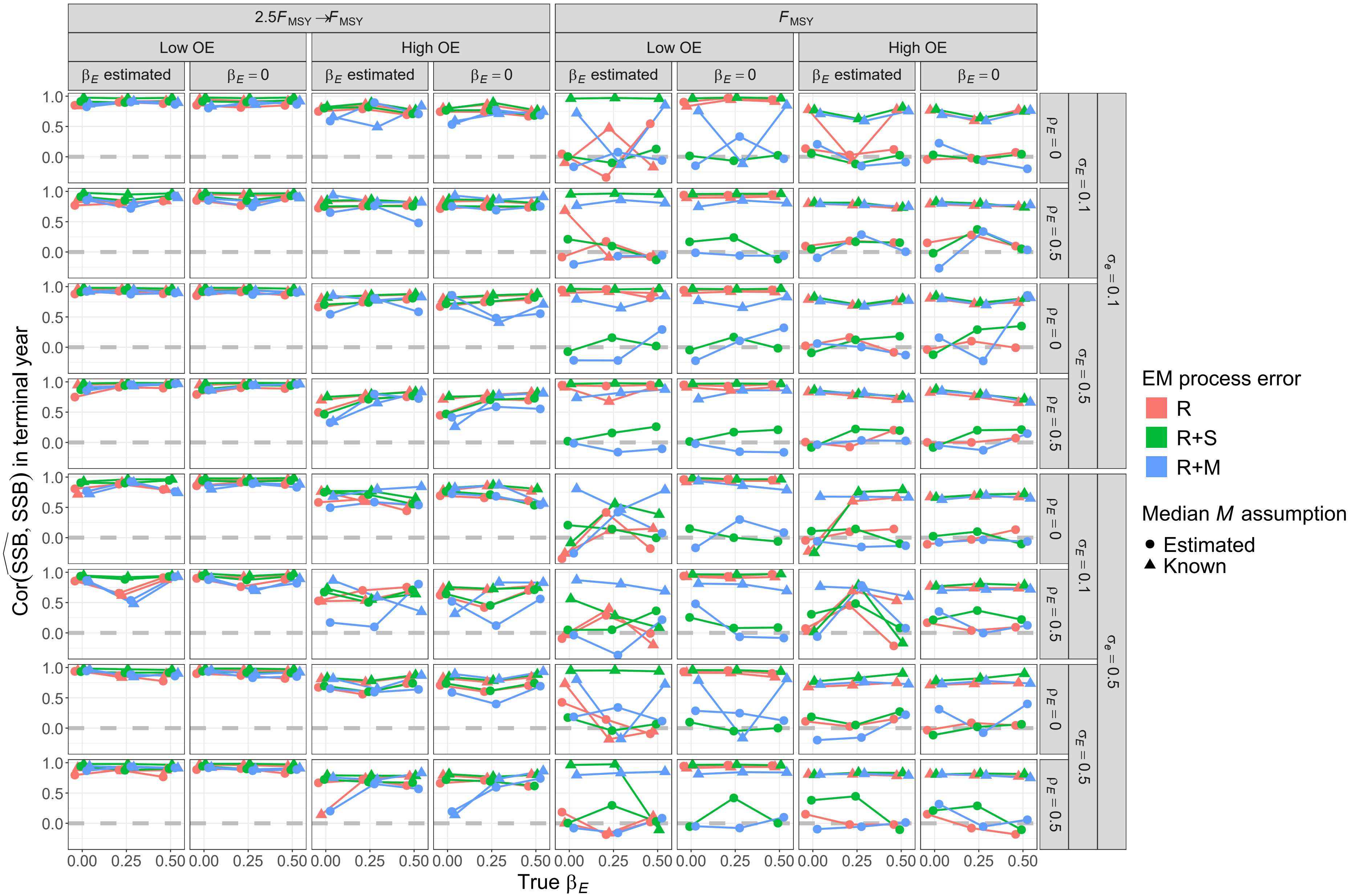

Fig. S36. For R+S operating models, correlation of estimates and true values of terminal year SSB for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

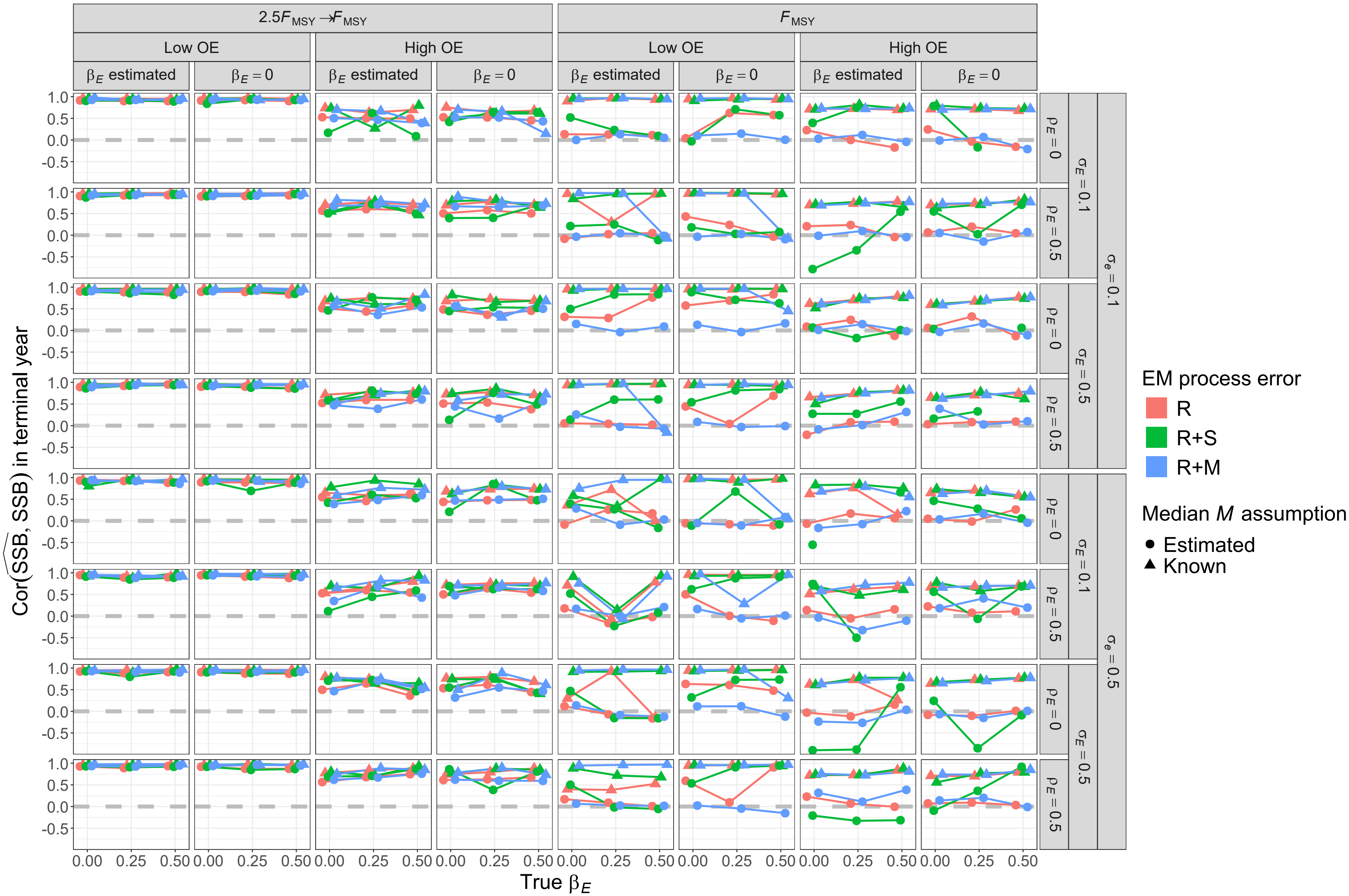

Fig. S37. For R+M operating models, correlation of estimates and true vales of terminal year SSB for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

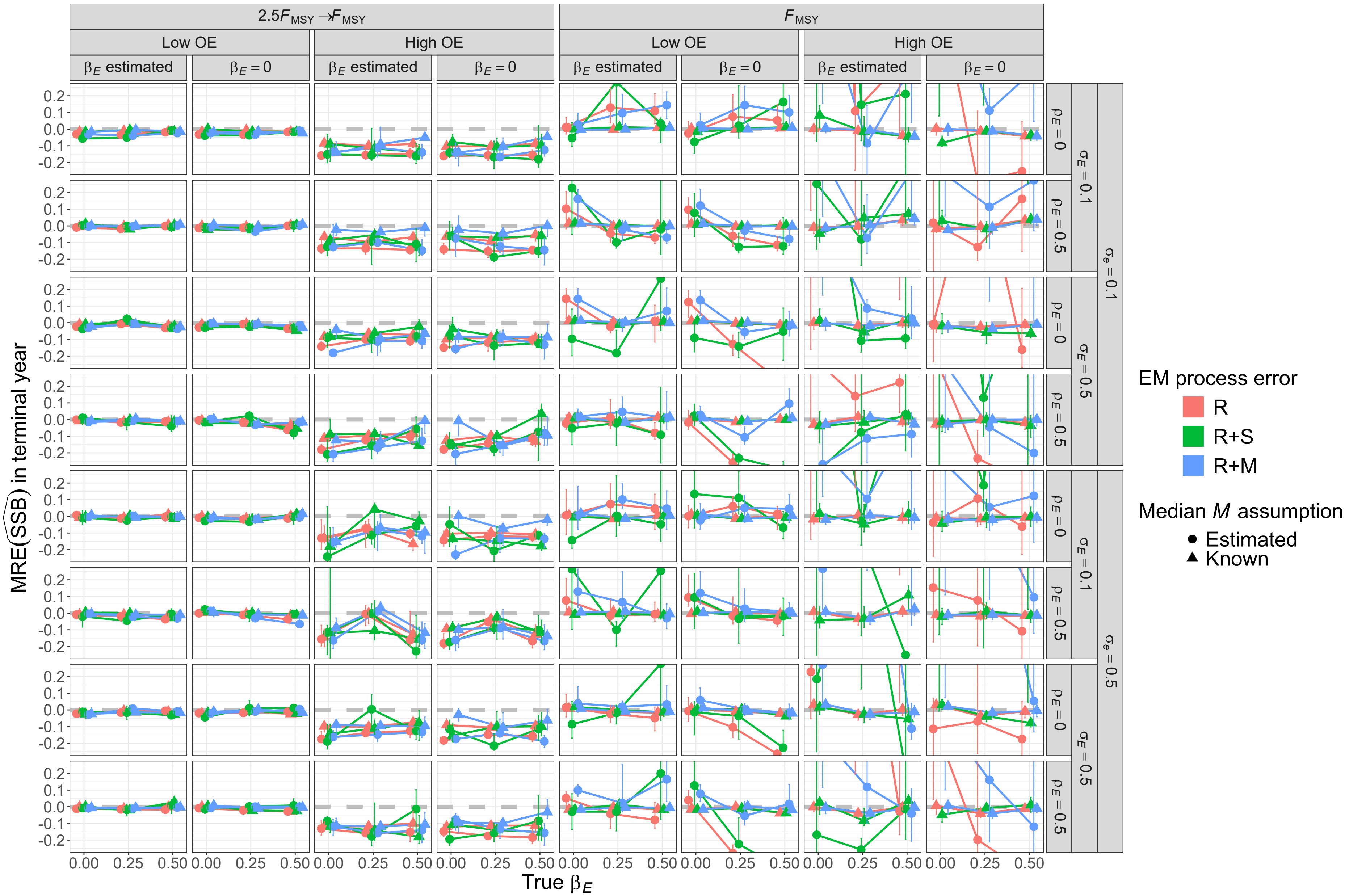

Fig. S38. For R operating models, median relative error (MRE) of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

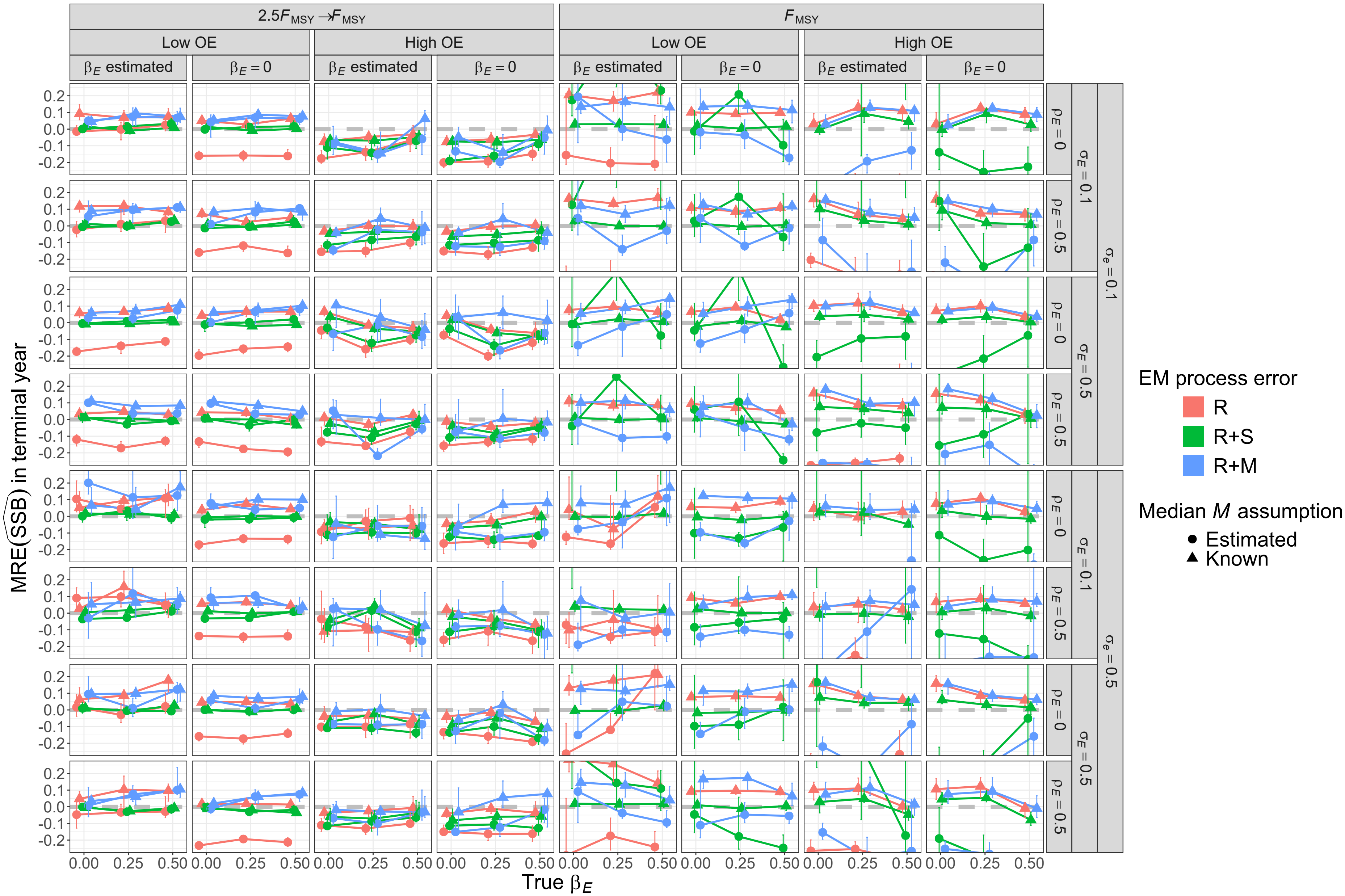

Fig. S39. For R+S operating models, median relative error (MRE) of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

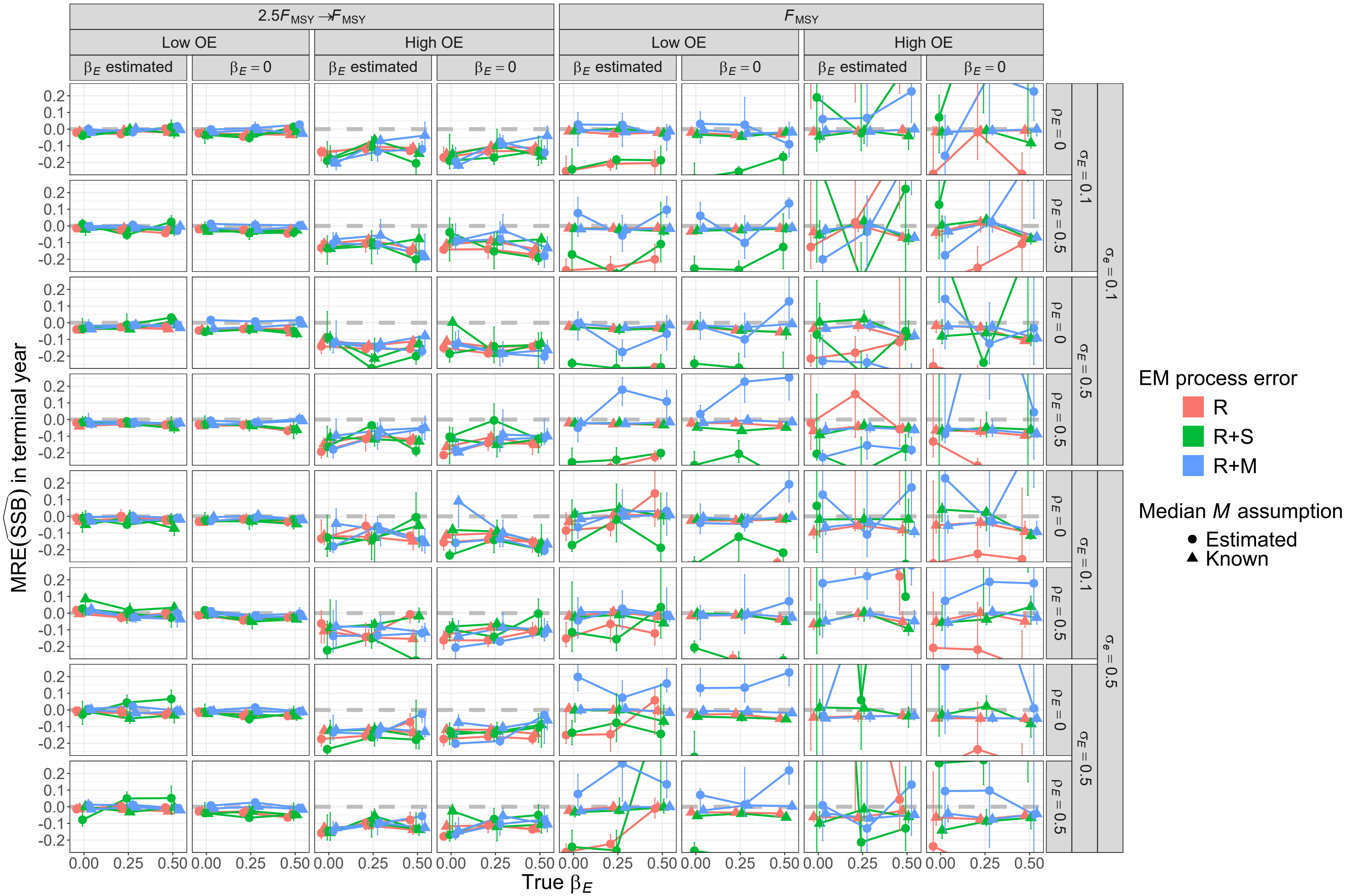

Fig. S40. For R+M operating models, median relative error (MRE) of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

Table S1. Operating (OM) and estimating model (EM) attributes used in analyses of deviance reduction and classification and regression trees. R, R+S, and R+M, denote models (OM or EM) with process error on recruitment only, recruitment and apparent survival, and recruitment and natural mortality, respectively. SD denotes standard deviation and F_{MSY} is the fishing mortality rate that acheives maximum sustainable yield.

Factor	Levels
OM Process error	R, R+S, R+M
OM Fishing History	F_{MSY}, $2.5F_{\mathrm{MSY}} \rightarrow F_{\mathrm{MSY}}$
OM Covariate effect size (β_E)	0, 0.25, 0.5
OM Population Observation Error	Low, High
OM Covariate Observation Error	Low ($\sigma_e = 0.1$), High ($\sigma_e = 0.5$)
OM Covariate Process Error SD (σ_E)	0.1, 0.5
OM Covariate Process Error Correlation (ρ_E)	0, 0.5
EM Process Error	R, R+S, R+M
EM Covariate Effect	None ($\beta_E = 0$), β_E estimated
EM Median Natural Mortality Rate Parameter (β_M)	Known, Estimated

Percent deviance reduction results

Estimating model convergence

For fitted logistic regression models to indicators of convergence, the estimating model process error assumption provided the largest percent reduction in deviance (9-25%) of all operating model and estimating model attributes for all three operating model process error types (Table S2). Including second order interactions of operating model and estimating model factors also provided further reduction of residual deviance (between 7 and 11%), indicating successful convergence depended on a combination of operating model and estimating model attributes.

AIC performance of process error source selection

For all OM process error types, the level of error in both the population and covariate observations were the attributes that resulted in the largest percent reductions in deviance for multinomial regression fits to indicators of AIC selection of the process error configuration (Table S3). For R+S and R+M OMs, level of error in population observations provided deviance reduction (19% and 13%, respectively) much larger than any other OM and EM factors whereas error in covariate observations provided the largest reduction (8%) for R OMs. Lesser deviance reduction (2-3%) occurred for the true covariate effect size and whether the EM made the correct assumption about including a covariate effect for R OMs and error in covariate observations for R+M OMs. Including second and third order interactions, provided the largest reductions in residual deviance for R OMs (24%) and little reduction for R+S and R+M OMs (4% and 6%, respectively).

AIC performance of covariate effect selection

The factors explaining the largest reduction in deviance for logistic regression models fitted to indicators of AIC selection of the correct covariate effect assumption depended on the magnitude of the true effect assumed in the OM (Table S4). When OMs had no covariate effect ($\beta_E = 0$) the OM process errror type provided the largest reduction in deviance (4%). When OMs assumed a covariate effect ($\beta_E > 0$), level of error in population observations and magnitude of temporal variability in the true covariate provided the largest reductions in deviance (4% and 8-21%, respectively). The reduction in deviance provided by the magnitude of covariate temporal variability also increased with larger true covariate effect. Including second and third order interactions provided a modest reduction in deviance (5-7%).

Median natural mortality rate bias

Across all OM process error types, regression models fit to errors in $\widehat{\beta}_M$ showed largest reductions in deviance from the type of OM fishing pressure (4-17%), EM process error assumption (2-12%), and the EM covariate effect assumption (3-7%, Table S5). The level of observation error also provided a similar reduction in deviance for R OMs (3%). Relatively large reductions in deviance also occurred with second and third order interactions (7-16%).

Median natural mortality rate standard error bias

The EM process error assumption provided the largest reduction in deviance for regression models fitted to errors in $\widehat{\mathrm{SE}}\left(\widehat{\beta}_M\right)$ from R (3%) and R+M (7%) OM simulations (Table S6). The type of OM fishing pressure provided the largest reduction (13%) for R+S OMs and a lesser reduction in devaince for R+M OMs (4%). The EM covariate effect assumption also provided similar reductions in deviance for R+S (9%) and R+M (4%) OMs.

Median natural mortality rate confidence interval bias

For logistic regressions of indicators of CIs including β_M, factors that provided the largest reductions in deviance were the OM fishing history and level of error in population observations for all OM process error types (0.3-6%) (Table S7). Including second and third order interactions provided the largest relative increase in deviance reduction for R+S (8-11%) and R+M OMs (5-8%) and lesser increases for R OMs (3-5%).

Covariate effect bias

For regression models fit to errors in estimation of β_E, we found very little reduction in deviance for any of the OM and EM factors (Table S8). For the normal model in these regressions, deviance is the sum of squares and there were extremely large estimates of β_E (and errors) for many simulations which resulted in extremely large sums of squares and relative differences from the null model were therefore small (<1%). Nevertheless, observation error of the covariate provided either the largest or second largest reductions in deviance of any of the single OM and EM attributes for OM process error configuration types. OM fishing history provided reductions of similar scale for R and R+S OMs. Other factors providing similar reductions were true covariate effect size for R OMs and level of error in population observations and EM process error assumption for R+S OMs. Including second and third order interactions also provided further reductions in residual deviance similar in scale to the individual factors.

Covariate effect standard error bias

Factors providing the largest reductions in deviance for estimation error of $\widehat{\mathrm{SE}}\left(\widehat{\beta}_E\right)$ were generally similar to those that were most important for estimation of β_E, but deviance reductions were one or two orders of magnitude larger (Table S9). Level of error in covariate observations and EM process error assumption was important for all OM process error configurations (6-13% and 3-4%, respectively). Level of error in population observations provided relatively large reductions in deviance for R+S and R+M OMs (3% and 8%, respectively). The OM fishing pressure history provided a lesser reduction in deviance for all OM process error types (2-4%). Like estimation of β_E, additional large reductions in deviance also occurred when second and third order interactions of OM and EM attributes were included in the regression models (11-31%).

Covariate effect confidence interval bias

For logistic regressions of indicators of CIs including the true value of β_E, factors that provided the largest reductions in deviance were true covariate effect size for R and R+M OMs (11% and 3%, respectively) and EM process error configuration for R+S OMs (7%) (Table S10). Covariate observation error also provided relatively large reduction in deviance for R OMs (8%). Lesser reductions of 1-2% occurred for level of error in population observations and covariate temporal variability for all OM process error types. Including second and third order interactions also provided relatively large further reductions in deviance (5-9%).

Terminal year natural morality rate bias

Regressions of estimation errors show largest reductions in deviance from whether the EM treats β_M as known (Table S11). For R and R+S OMs, annual M was constant, and therefore, terminal M was known when the true β_M was assumed known and there will be no error in any of those EMs. Beyond the β_M assumption, the OM fishing history, level of error in population observations, and EM assumptions for covariate effect and process error all provided notable deviance reductions.

Terminal year natural morality rate accuracy

Like bias of terminal year M, the EM assumption for β_M provided the largest reduction in deviance (22-45%) for accuracy as measured by RMSE for all OM process error types (Table S12). Including second and third order interactions also provided large further reductions in deviance (21-38%).

Using correlation to measure accuracy of terminal year M estimation provides a different perspective. Treatment of β_M provided relatively large reductions in deviance for all OM process error types (3-12%), but level of variability in the covariate was equally or more important (3-15%) as was whether the covariate effect was included in the EM (12%) for R and R+S OMs (Table S13). Including second and third order interactions also provided relatively large ruther reductions in deviance (15 to 24%).

Terminal year spawning stock biomass bias

Regression model fits showed largest reductions in deviance for OM fishing history (2-3%), error in population observations (1% for R and R+M OMs), and EM β_M assumption (2%; Table S14). The type of EM process error assumed also provided a deviance reduction similar in scale for R+S OMs (1%). Including second and third order interactions of OM and EM attributes provided further deviance reductions of 5-10%.

Terminal year spawning stock biomass accuracy

Largest reductions in deviance were provided by whether β_M was known or estimated (20-21%) and whether there was temporal contrast in fishing pressure (20-33%) across all OM process error types (Table S15). Lesser reductions were also provided by the EM process error assumption for R and R+M OMs. Inclusion of second and third order interactions of OM and EM attributes provided further reductions in deviance of 26-40%.

Table S2. Percent reduction in deviance for logistic regression models fit to indicators of convergence (providing Hessian-based standard errors). Table rows give each operating model (OM) and estimating model (EM) factor included individually, combined, and with second and third order interactions. Table columns categorize operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). See Table S1 for further details.

Factor	R	R+S	R+M
OM F history	0.86	<0.01	0.24
OM Obs. Error	0.56	0.03	0.26
OM σ_e	0.66	2.99	0.93
OM σ_E	0.53	2.22	0.62
OM ρ_E	<0.01	0.02	<0.01
OM β_E	0.02	<0.01	<0.01
EM Process Error	24.53	9.33	23.06
EM β_E assumption	0.16	2.96	0.51
EM M Assumption	0.18	2.83	0.40
All factors	28.92	22.67	27.34
+ All Two Way	36.15	33.54	34.03
+ All Three Way	37.95	36.58	35.57

Using correlation to measure accuracy, temporal contrast in fishing pressure and EM treatment of β_M provided relatively important reductions in deviance like RMSE, but level of error in population observations was important rather than EM process error type (Table S16). Including second and third order interactions also provided relatively large ruther reductions in deviance (6 to 20%).

Table S3. Percent reduction in deviance for multinomial logistic regression models fit to indicators of process error assumption in the estimating models with lowest AIC. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.23	0.06	0.30
OM Obs. Error	0.54	18.58	13.43
OM σ_e	8.41	0.05	2.23
OM σ_E	0.17	0.08	0.59
OM ρ_E	1.29	0.03	0.06
OM β_E	3.42	0.04	0.38
EM M Assumption	0.35	0.34	0.48
EM β_E Assumption Correct	2.35	0.22	0.78
All factors	21.97	19.29	19.07
+ All Two Way	38.25	21.13	24.73
+ All Three Way	46.26	22.85	26.97

Further detailed results

Table S4. Percent reduction in deviance for logistic regression models fit to indicators of correct estimating model (EM) covariate effect assumption (no effect for operating models assuming $\beta_E = 0$ or estimated effect for operating models assuming $\beta_E > 0$) with lowest AIC. Table columns categorize operating models with different true values for β_E. See Tables S2 and S1 for further details.

Factor	$\beta_E = 0$	$\beta_E = 0.25$	$\beta_E = 0.5$
OM F history	<0.01	0.09	0.18
OM Obs. Error	1.19	4.06	4.64
OM Process Error	4.11	0.71	0.25
OM σ_e	0.66	2.00	1.97
OM σ_E	0.35	7.97	20.66
OM ρ_E	0.21	1.23	1.48
EM M Assumption	0.23	0.15	0.15
EM Process Error Assumption Correct	1.95	0.39	0.06
All factors	6.95	17.65	33.44
+ All Two Way	10.58	23.21	38.36
+ All Three Way	11.83	24.95	39.79

Table S5. Percent reduction in deviance for linear regression models fit to errors in estimation of the median natural mortality rate parameter ($\widehat{\beta}_M$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	4.03	16.52	6.51
OM Obs. Error	2.74	0.54	0.64
OM σ_e	0.01	<0.01	0.01
OM σ_E	0.18	0.02	0.15
OM ρ_E	0.07	0.05	<0.01
OM β_E	0.37	0.04	0.10
EM Process Error	2.31	11.99	2.65
EM β_E assumption	2.83	7.20	3.25
All factors	12.14	33.71	12.81
+ All Two Way	21.74	46.46	19.84
+ All Three Way	26.38	49.41	22.24

Table S6. Percent reduction in deviance for linear regression models fit to errors in estimation of the Hessian-based standard error of the median M parameter ($\widehat{\mathrm{SE}}\left(\widehat{\beta}_M\right)$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.55	12.91	3.65
OM Obs. Error	0.98	0.01	0.07
OM σ_e	<0.01	0.03	0.11
OM σ_E	0.13	0.03	0.19
OM ρ_E	<0.01	0.02	<0.01
OM β_E	0.12	0.07	0.23
EM Process Error	3.16	8.79	6.83
EM β_E assumption	0.18	8.96	3.81
All factors	5.32	28.39	14.71
+ All Two Way	11.80	44.67	22.70
+ All Three Way	17.83	51.54	29.46

Table S7. Percent reduction in deviance for logistic regression models fit to indicators of whether CIs for estimates of the median natural mortality rate parameter ($\widehat{\beta}_M$) includes the true value. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	6.08	1.46	0.14
OM Obs. Error	1.48	0.41	0.29
OM σ_e	0.01	0.06	0.01
OM σ_E	0.13	0.02	0.04
OM ρ_E	<0.01	<0.01	0.02
OM β_E	0.08	0.11	0.09
EM Process Error	0.53	0.08	0.04
EM β_E assumption	<0.01	0.19	0.01
All factors	8.46	2.32	0.62
+ All Two Way	11.97	10.57	5.99
+ All Three Way	13.35	13.51	8.44

Table S8. Percent reduction in deviance for linear regression models fit to errors in estimation of the covariate effect on natural mortality rate ($\widehat{\beta}_E$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.03	0.06	0.02
OM Obs. Error	<0.01	0.06	0.02
OM σ_e	0.04	0.08	0.06
OM σ_E	<0.01	0.02	0.01
OM ρ_E	<0.01	0.01	<0.01
OM β_E	0.05	<0.01	<0.01
EM Process Error	0.01	0.06	0.01
EM M assumption	0.02	0.02	<0.01
All factors	0.18	0.37	0.14
+ All Two Way	0.50	1.11	0.35
+ All Three Way	1.38	1.88	0.66

Table S9. Percent reduction in deviance for linear regression models fit to errors in estimation of the Hessian-based standard error of the covariate effect on M ($\widehat{\mathrm{SE}}\left(\widehat{\beta}_E\right)$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	1.84	2.68	4.00
OM Obs. Error	0.06	2.74	8.04
OM σ_e	6.27	7.28	12.83
OM σ_E	0.48	1.05	0.46
OM ρ_E	0.17	0.01	0.02
OM β_E	0.37	0.01	0.05
EM Process Error	2.90	3.89	2.98
EM M assumption	0.15	0.34	0.47
All factors	13.01	20.95	29.24
+ All Two Way	28.58	32.25	42.77
+ All Three Way	44.44	42.83	51.84

Table S10. Percent reduction in deviance for logistic regression models fit to indicators of whether CIs for covariate effect estimates includes the true value. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.24	0.05	0.01
OM Obs. Error	1.36	1.91	1.78
OM σ_e	8.26	0.24	0.92
OM σ_E	0.92	1.70	1.56
OM ρ_E	0.03	0.05	0.06
OM β_E	11.10	0.68	2.72
EM Process Error	0.01	6.51	0.27
EM M assumption	0.25	0.29	0.06
All factors	23.61	12.16	7.78
+ All Two Way	30.17	18.08	12.63
+ All Three Way	32.58	20.62	14.87

Table S11. Percent reduction in deviance for linear regression models fit to errors in terminal year natural mortality rate (M) as measured by Eq. 2. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
Convergence	<0.01	0.18	0.01
OM F history	0.66	2.69	1.00
OM Obs. Error	0.29	0.43	0.16
OM σ_e	<0.01	<0.01	<0.01
OM σ_E	0.02	<0.01	<0.01
OM ρ_E	<0.01	0.01	<0.01
OM β_E	0.03	0.01	0.01
EM Process Error	0.22	1.42	0.43
EM β_E assumption	0.35	1.07	0.39
EM M assumption	0.84	5.10	1.22
All factors	2.56	10.83	3.37
+ All Two Way	5.41	18.75	6.38
+ All Three Way	7.93	22.49	8.76

Table S12. Percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year natural mortality (M). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.26	2.21	14.65
OM Obs. Error	1.65	0.71	4.68
OM σ_e	0.02	0.16	0.01
OM σ_E	1.06	1.23	0.84
OM ρ_E	0.02	0.01	<0.01
OM β_E	3.73	1.94	1.14
EM Process Error	2.04	3.85	0.87
$EM\beta_E$ assumption	0.19	0.59	0.07
EM M assumption	22.28	45.15	44.70
All factors	32.83	56.71	66.99
+ All Two Way	57.65	84.17	87.96
+ All Three Way	74.93	94.57	91.94

Table S13. Percent reduction in deviance for linear regression models fit to correlation of terminal year natural mortality (M) estimates and true values. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.05	0.02	0.37
OM Obs. Error	2.00	0.89	3.73
OM σ_e	2.30	2.56	0.37
OM σ_E	14.91	13.78	3.03
OM ρ_E	1.98	0.33	0.14
OM β_E	0.41	3.76	3.87
EM Process Error	0.43	1.52	0.62
$EM\beta_E$ assumption	12.72	12.30	1.56
EM M assumption	12.34	7.16	3.03
All factors	42.11	38.31	15.76
+ All Two Way	63.38	62.40	31.99
+ All Three Way	72.79	75.98	46.62

Table S14. Percent reduction in deviance for linear regression models fit to errors in terminal year spawning stock biomass (SSB) as measured by Eq. 2. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
Convergence	0.02	0.13	<0.01
OM F history	2.08	2.74	1.75
OM Obs. Error	1.09	0.18	1.19
OM σ_e	0.01	<0.01	<0.01
OM σ_E	<0.01	<0.01	<0.01
OM ρ_E	<0.01	<0.01	<0.01
OM β_E	<0.01	0.01	<0.01
EM Process Error	0.52	1.05	0.57
EM β_E assumption	<0.01	0.05	0.01
EM M assumption	1.76	2.12	1.51
All factors	5.97	6.58	5.21
+ All Two Way	12.81	13.00	11.97
+ All Three Way	16.41	15.57	15.73

Table S15. Percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year spawning stock biomass (SSB). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	19.88	32.67	20.92
OM Obs. Error	2.21	0.11	1.48
OM σ_e	0.05	0.16	0.03
OM σ_E	0.08	0.16	0.02
OM ρ_E	0.03	<0.01	0.02
OM β_E	0.05	0.03	0.14
EM Process Error	5.72	0.13	5.73
$EM\beta_E$ assumption	0.03	0.79	0.05
EM M assumption	20.05	21.11	19.43
All factors	48.09	55.17	47.82
+ All Two Way	79.09	80.98	78.40
+ All Three Way	88.30	89.49	87.67

Table S16. Percent reduction in deviance for linear regression models fit to correlation of terminal year spawning stock biomass (SSB) estimates and true values. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	19.27	25.70	19.59
OM Obs. Error	12.53	3.60	9.46
OM σ_e	0.08	0.44	0.09
OM σ_E	0.10	0.44	0.11
OM ρ_E	<0.01	<0.01	0.09
OM β_E	0.09	0.05	<0.01
EM Process Error	0.07	0.92	0.71
$EM\beta_E$ assumption	<0.01	1.32	0.38
EM M assumption	28.94	19.74	25.36
All factors	61.19	52.21	56.09
+ All Two Way	81.48	72.16	74.98
+ All Three Way	85.23	83.22	80.48

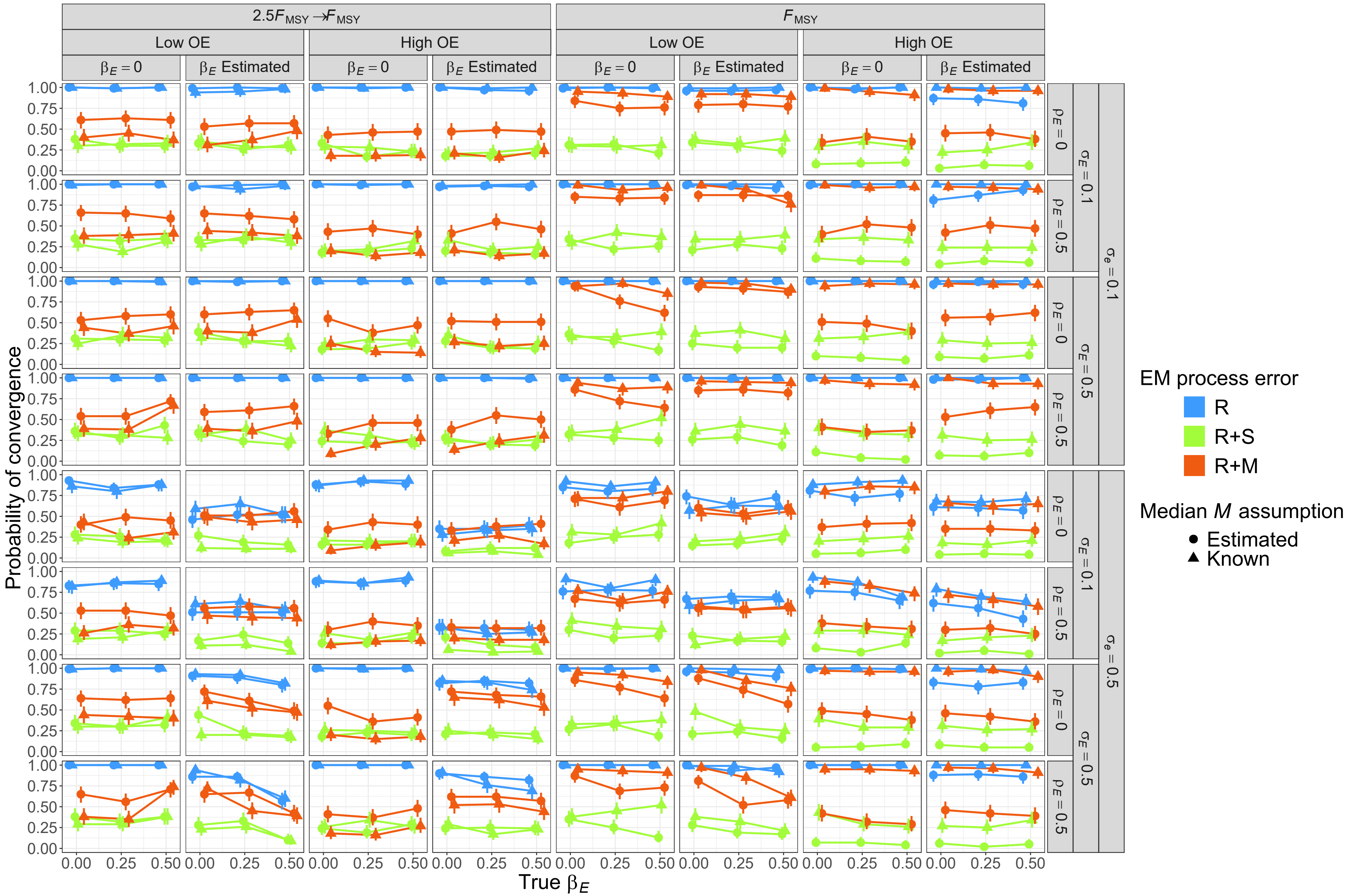

Fig. S41. Estimated probability of fits providing Hessian-based standard errors for estimating models assuming alternative process error, that estimate or assume known median M, and that estimate or assume no covariate effect on median M when fitted to R operating models and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

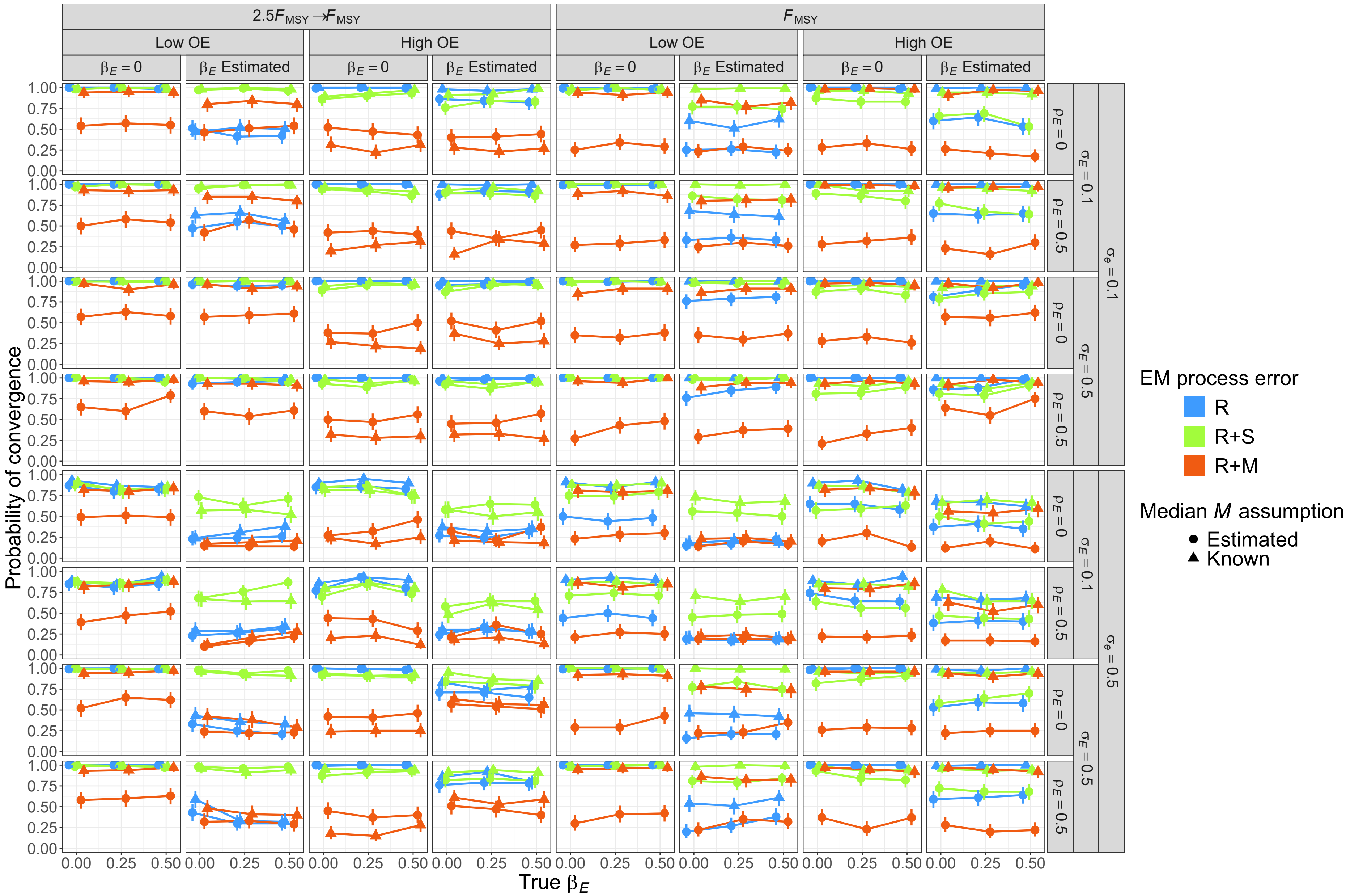

Fig. S42. Estimated probability of fits providing Hessian-based standard errors for estimating models assuming alternative process error, that estimate or assume known median M, and that estimate or assume no covariate effect on median M when fitted to R+S operating models and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

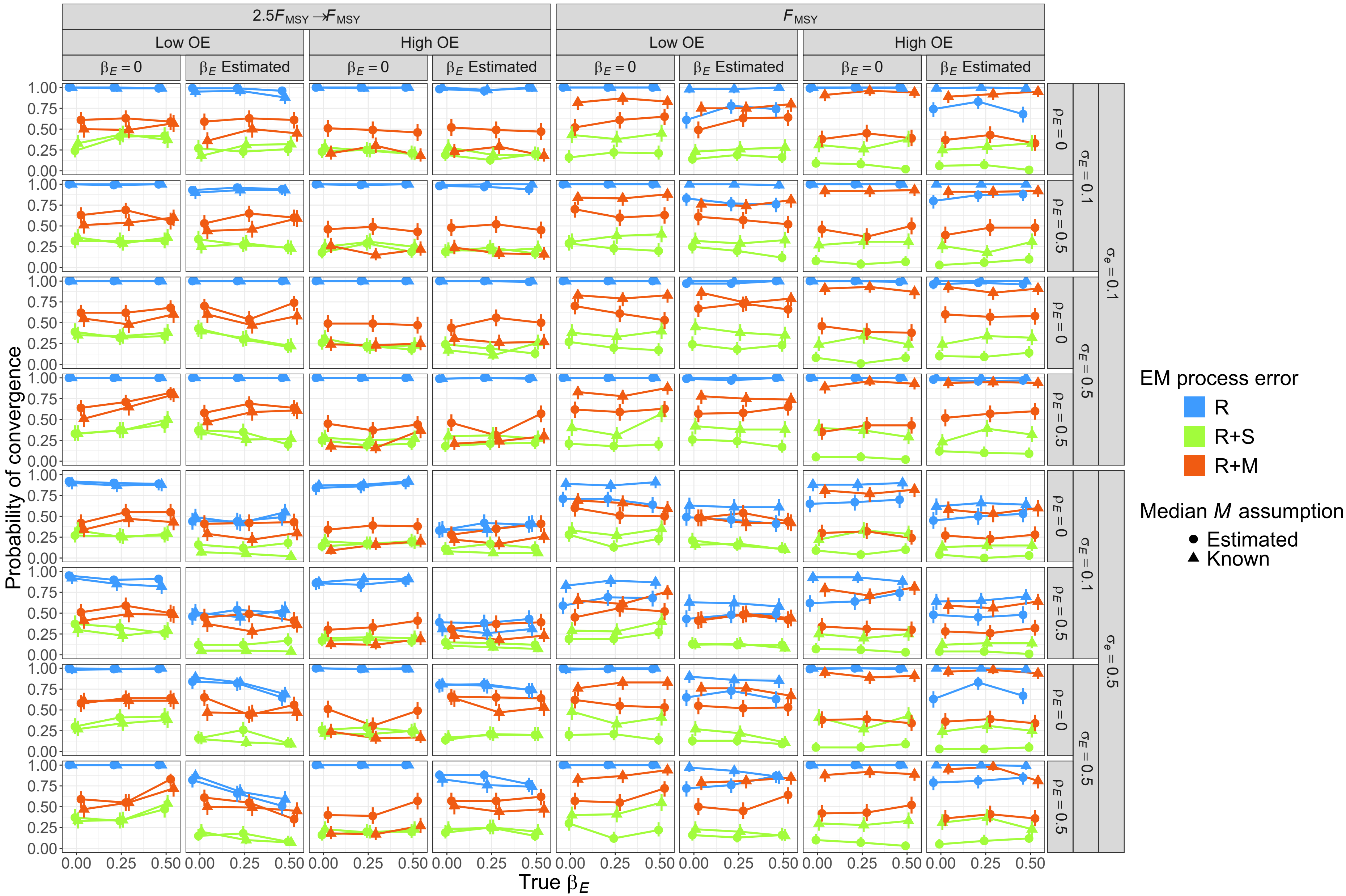

Fig. S43. Estimated probability of fits providing Hessian-based standard errors for estimating models assuming alternative process error, that estimate or assume known median M, and that estimate or assume no covariate effect on median M when fitted to R+M operating models and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

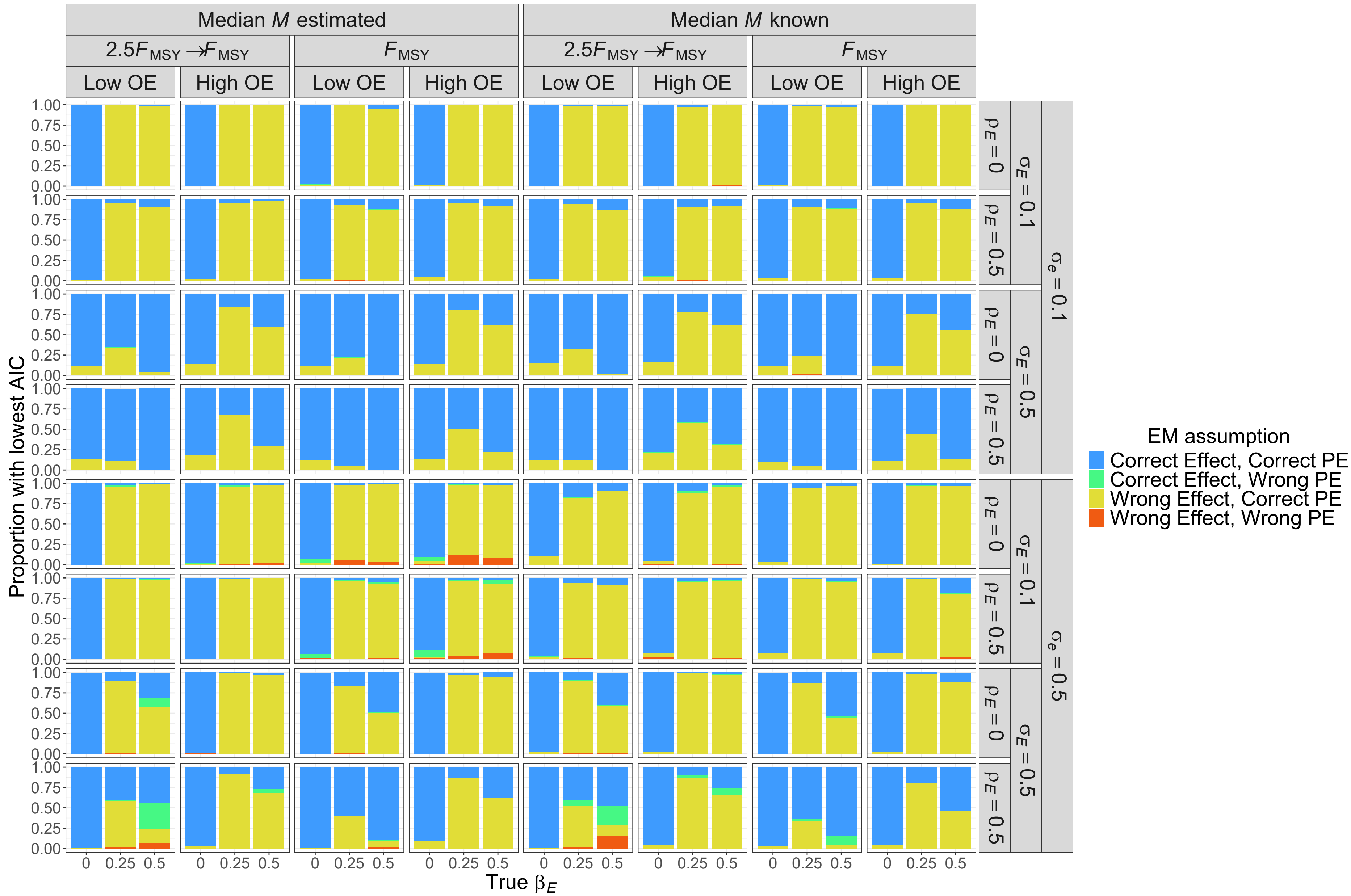

Fig. S44. Proportion of simulated data sets for R operating models where the estimating model type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

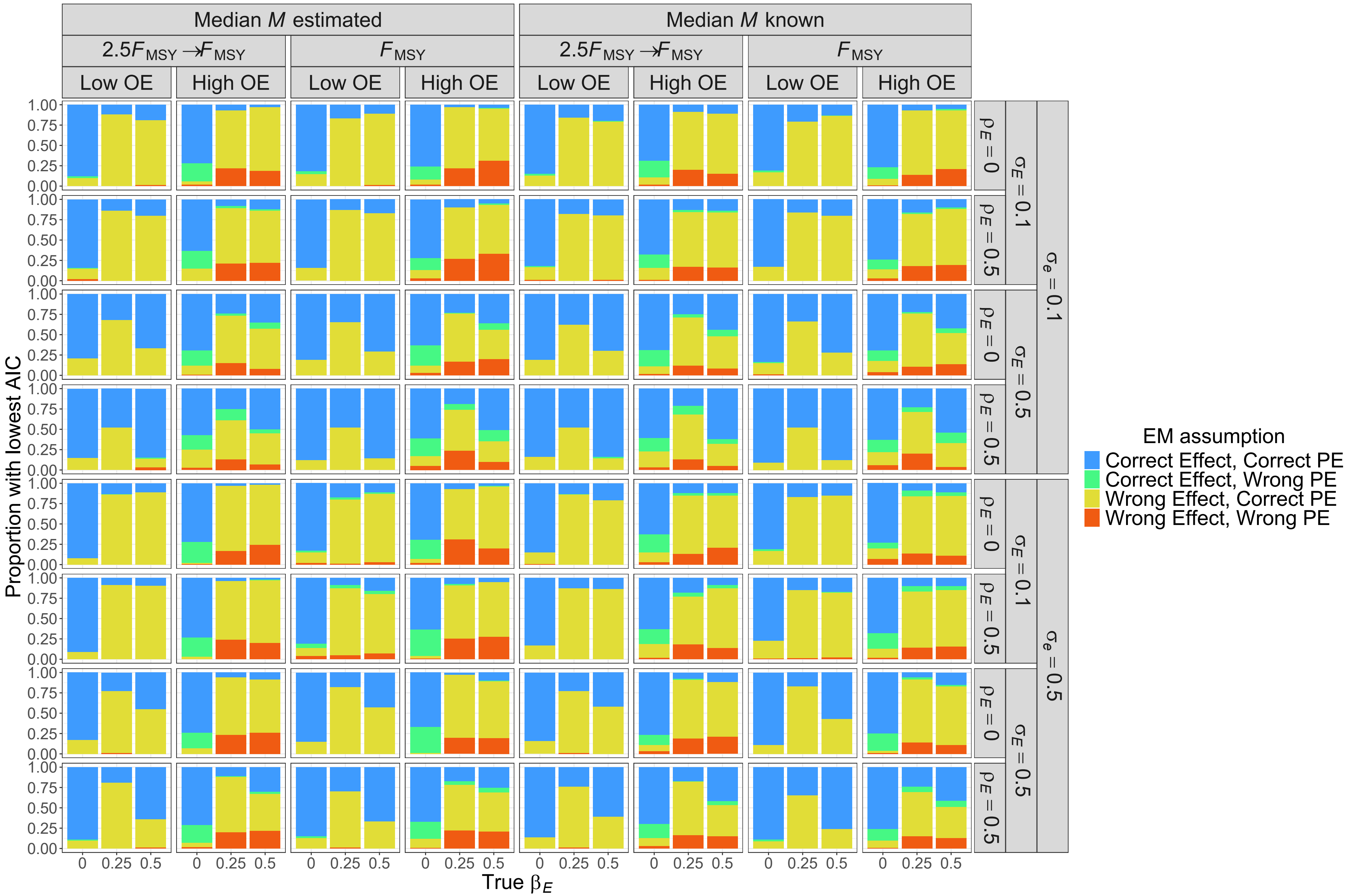

Fig. S45. Proportion of simulated data sets for R+S operating models where the estimating model type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

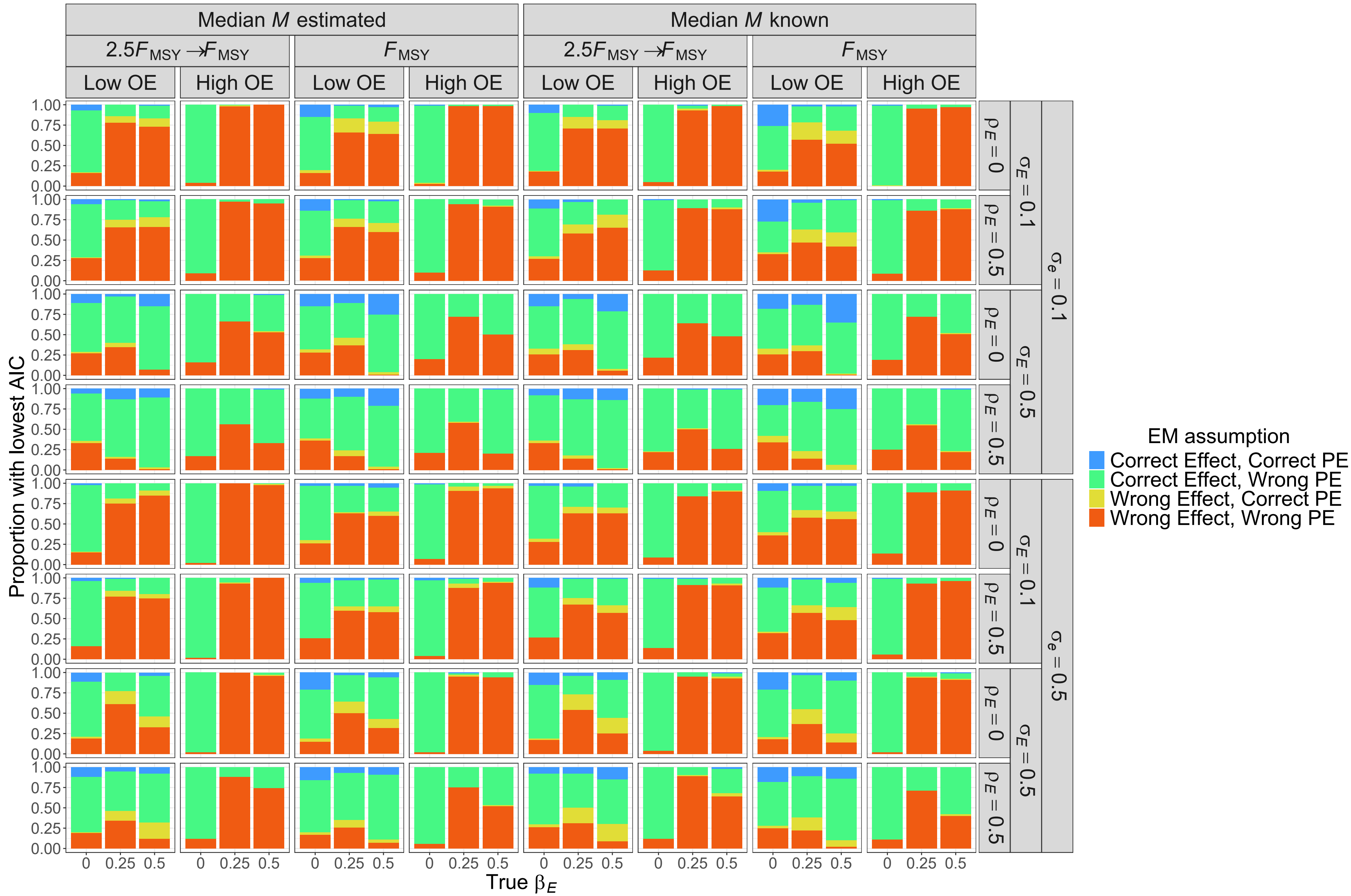

Fig. S46. Proportion of simulated data sets for R+M operating models where the estimating model type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

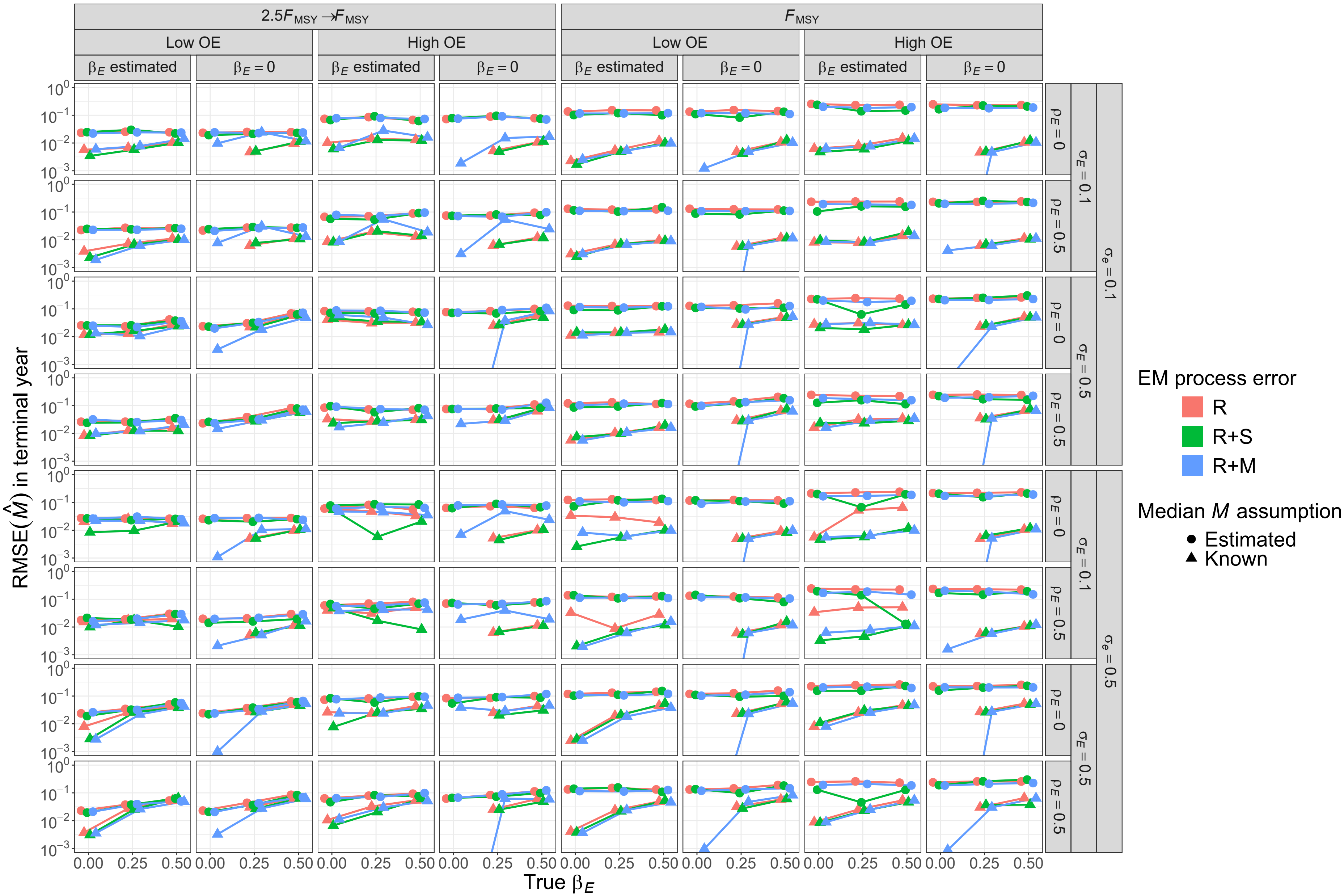

Fig. S47. For R operating models, RMSE of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

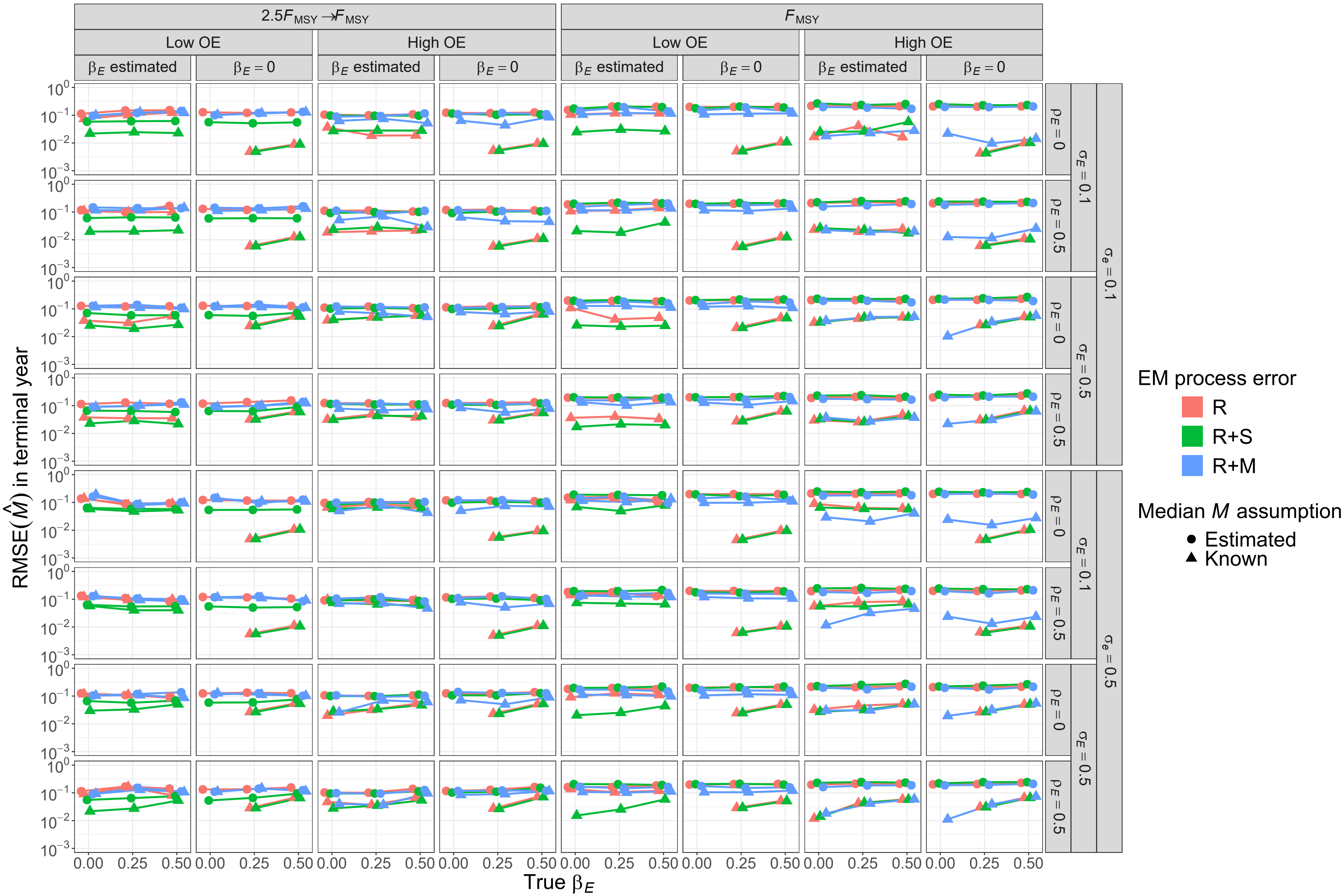

Fig. S48. For R+S operating models, RMSE of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

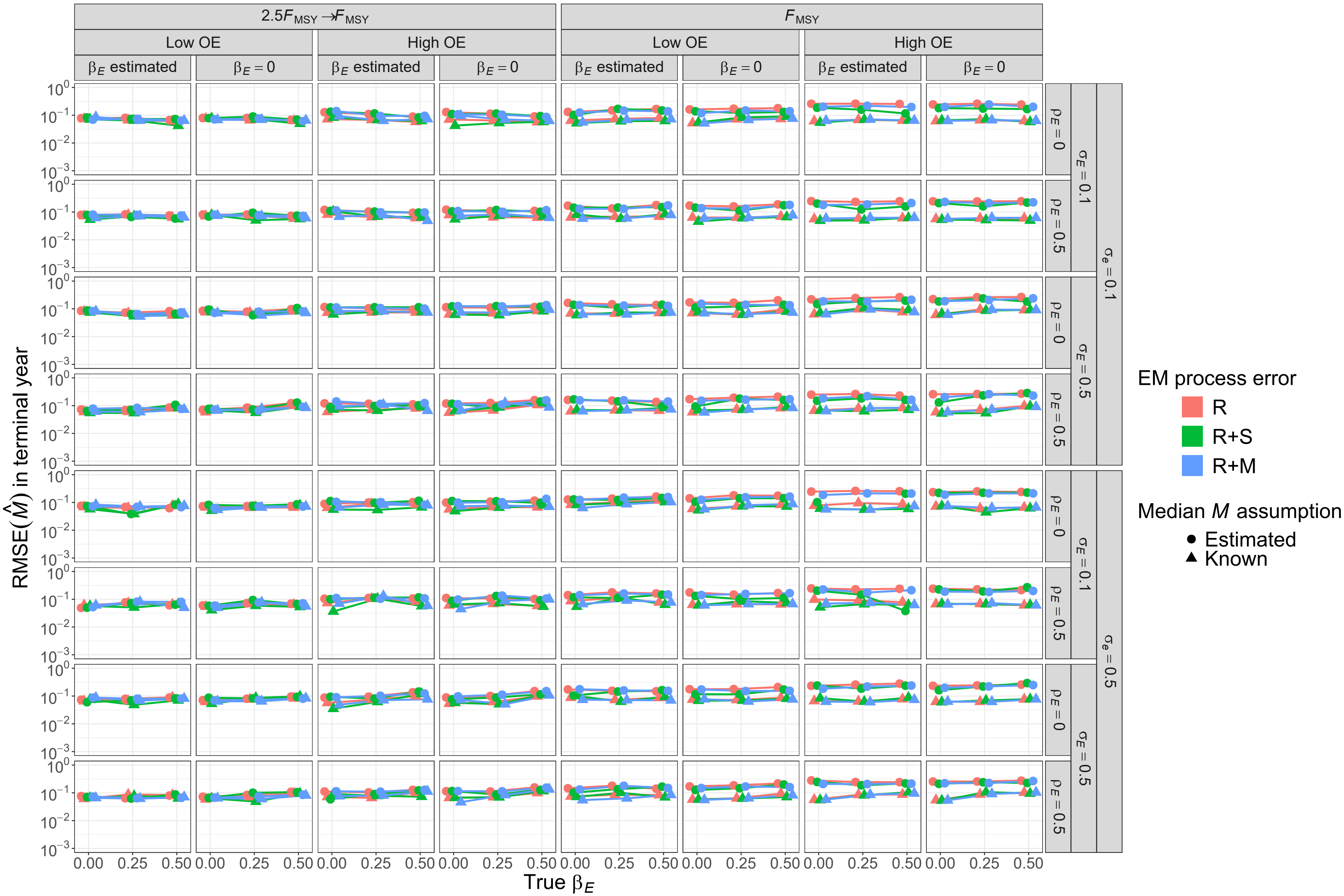

Fig. S49. For R+M operating models, RMSE of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

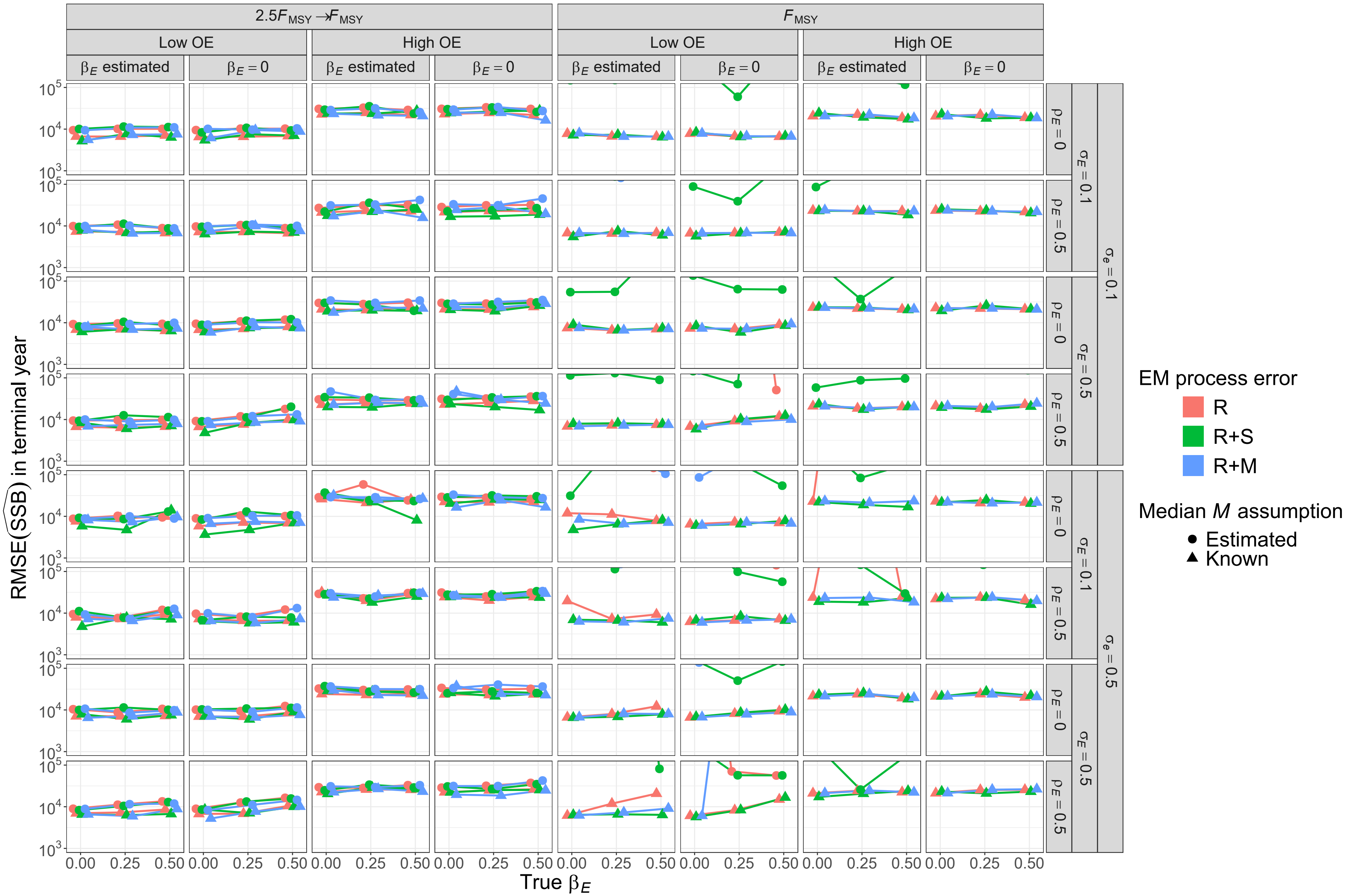

Fig. S50. For R operating models, RMSE of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

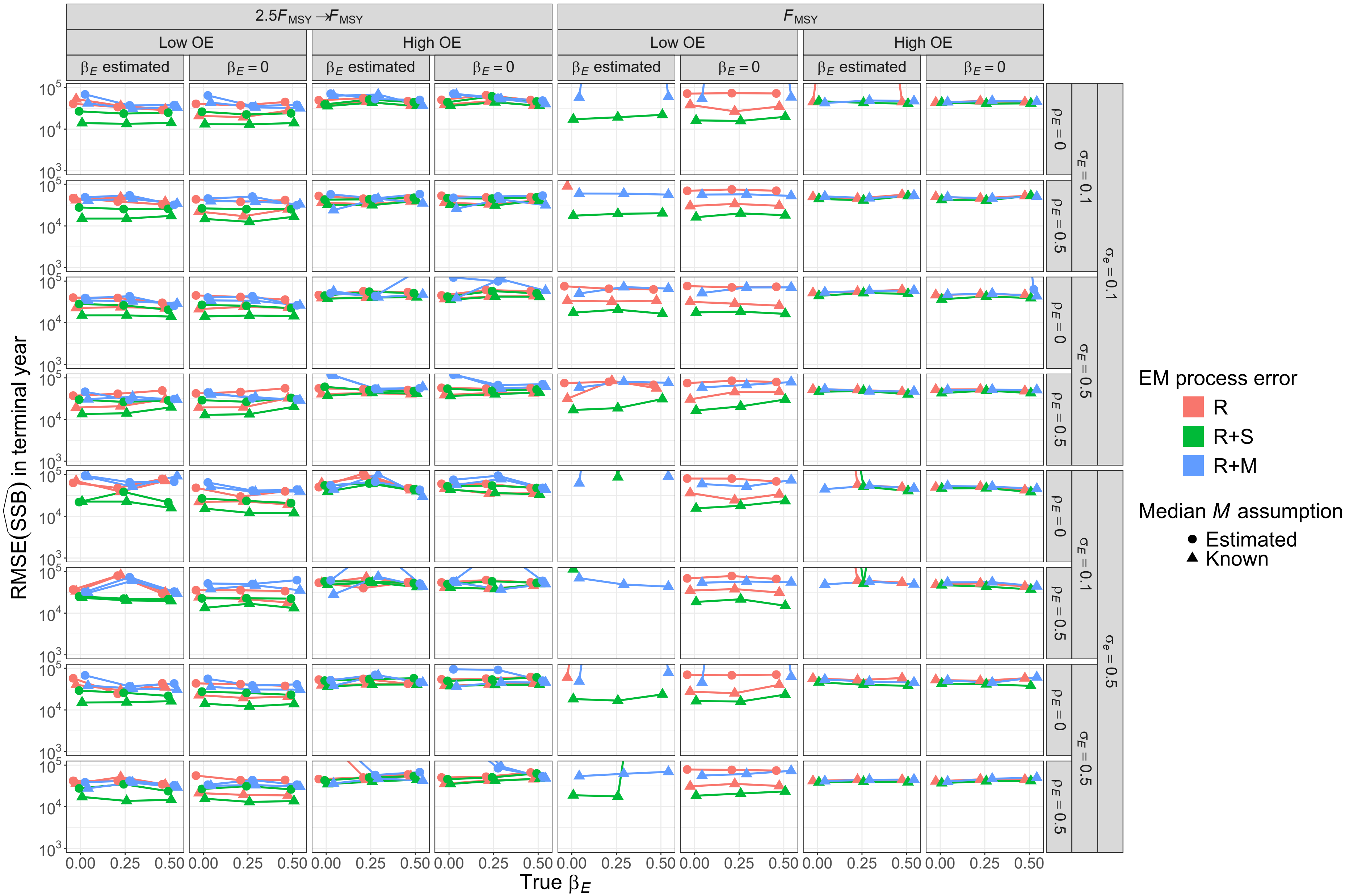

Fig. S51. For R+S operating models, RMSE of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

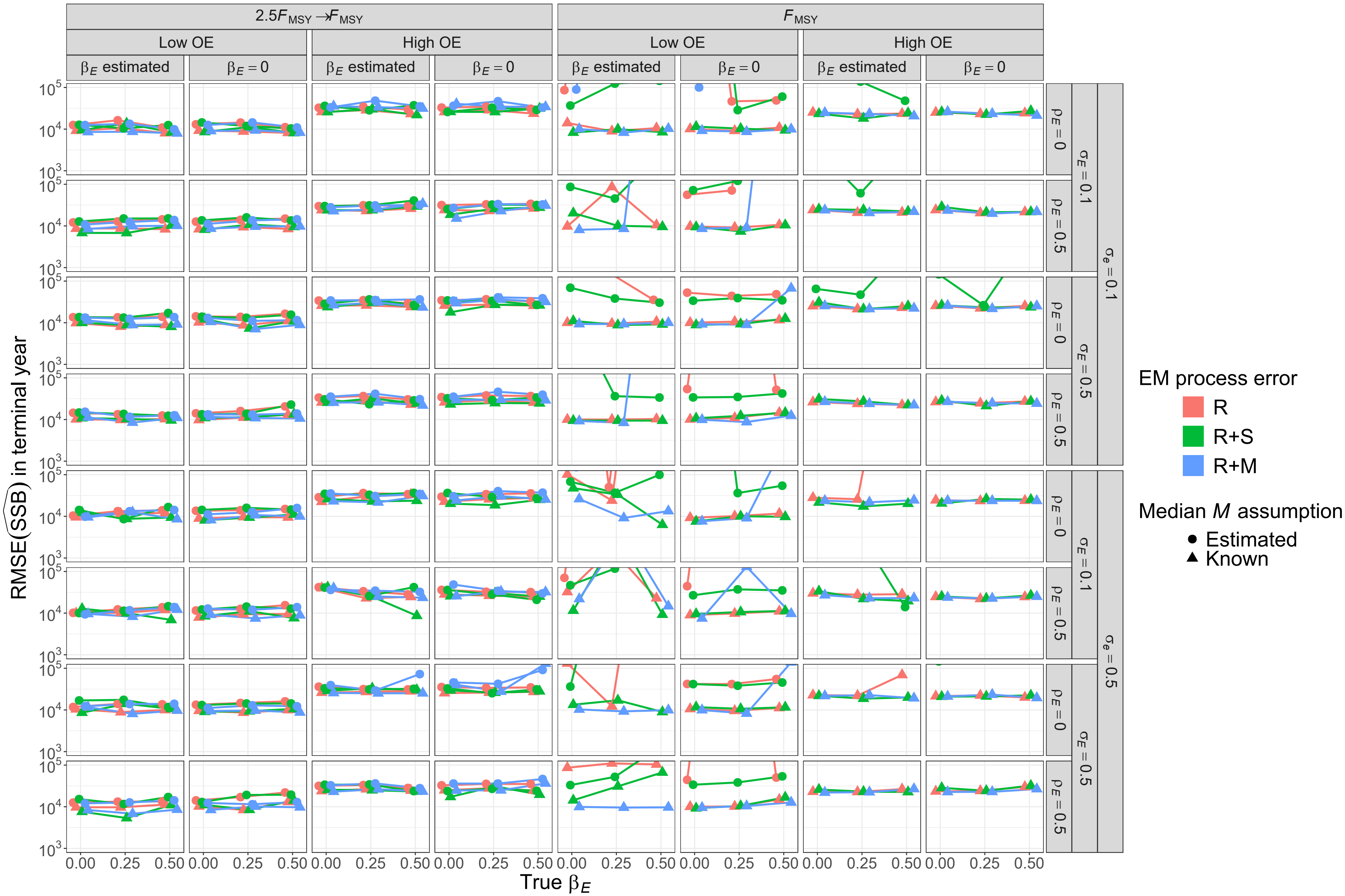

Fig. S52. For R+M operating models, RMSE of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

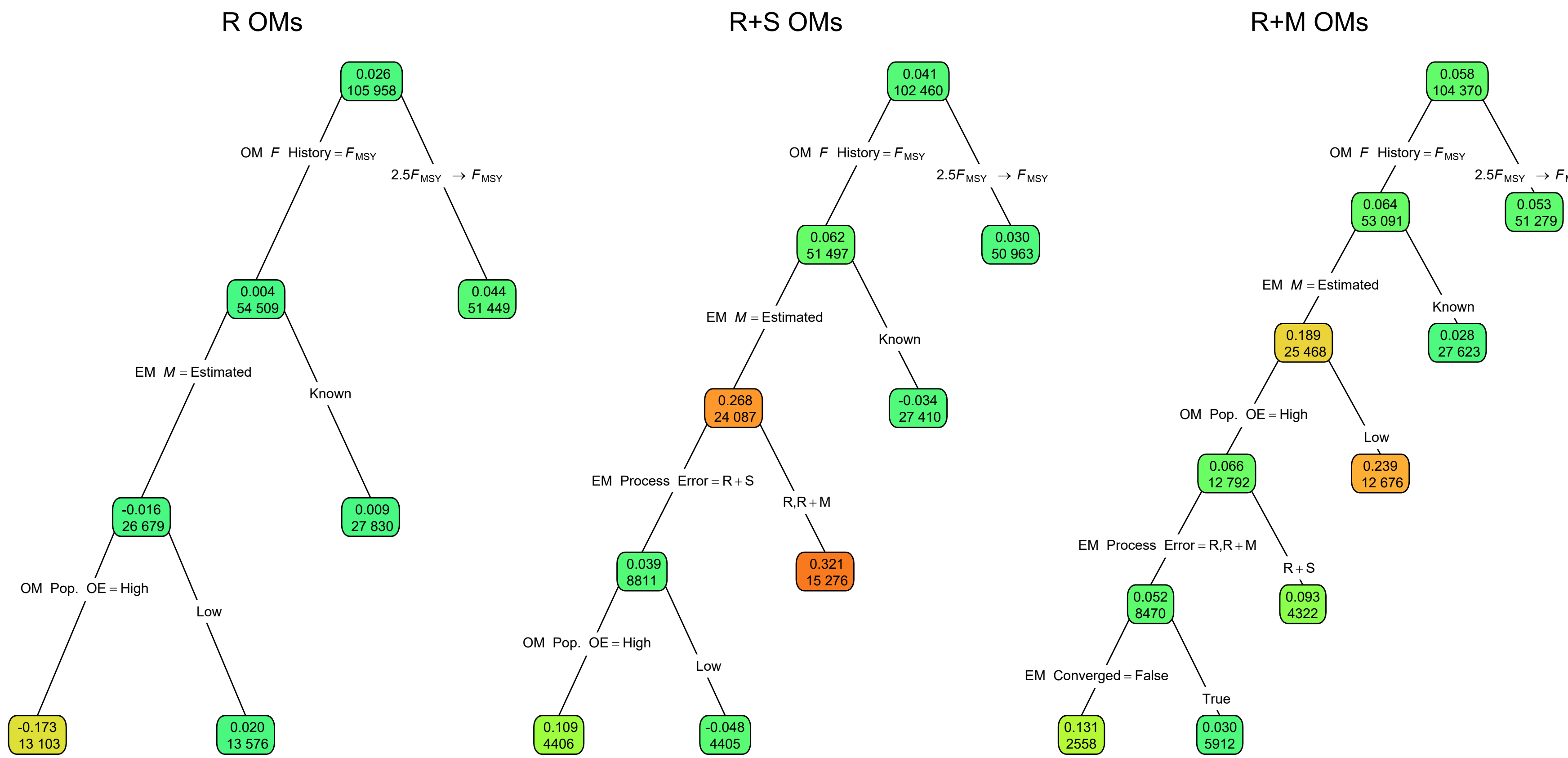

Fig. S53. Regression trees for errors terminal year fishing mortality rate as measured by Eq. 2. Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

Table S17. Percent reduction in deviance for linear regression models fit to errors in terminal year fishing mortality rate as measured by Eq. 2. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
Convergence	<0.01	0.19	0.01
OM F history	1.92	2.42	1.94
OM Obs. Error	0.74	0.12	1.12
OM σ_e	0.01	<0.01	<0.01
OM σ_E	<0.01	<0.01	<0.01
OM ρ_E	<0.01	<0.01	<0.01
OM β_E	<0.01	0.01	<0.01
EM Process Error	0.40	0.92	0.48
EM β_E assumption	0.01	0.03	0.02
EM M assumption	1.58	1.85	1.68
All factors	4.95	5.82	5.35
+ All Two Way	10.91	11.03	12.23
+ All Three Way	14.16	13.24	16.19

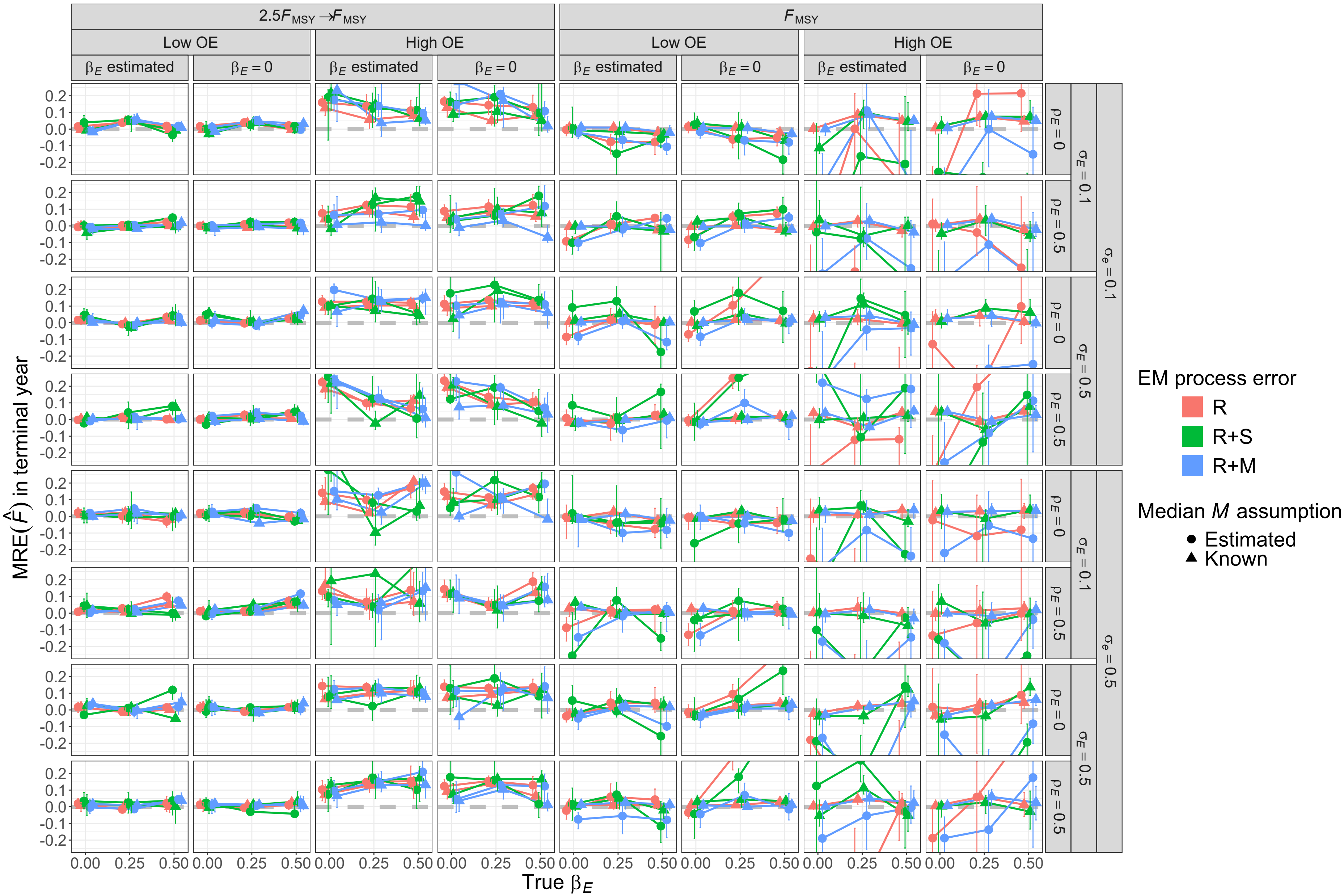

Fig. S54. For R operating models, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

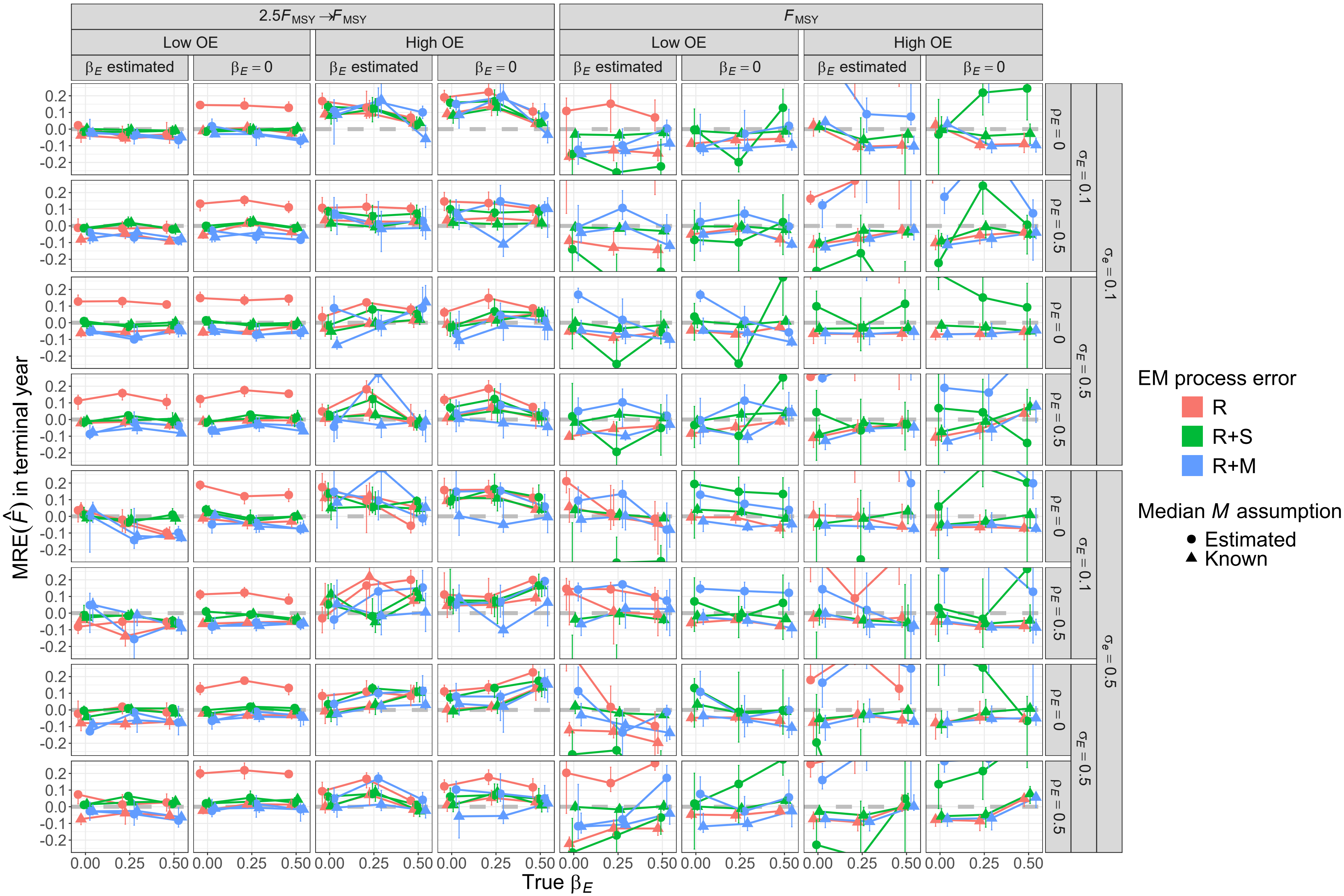

Fig. S55. For R+S operating models, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

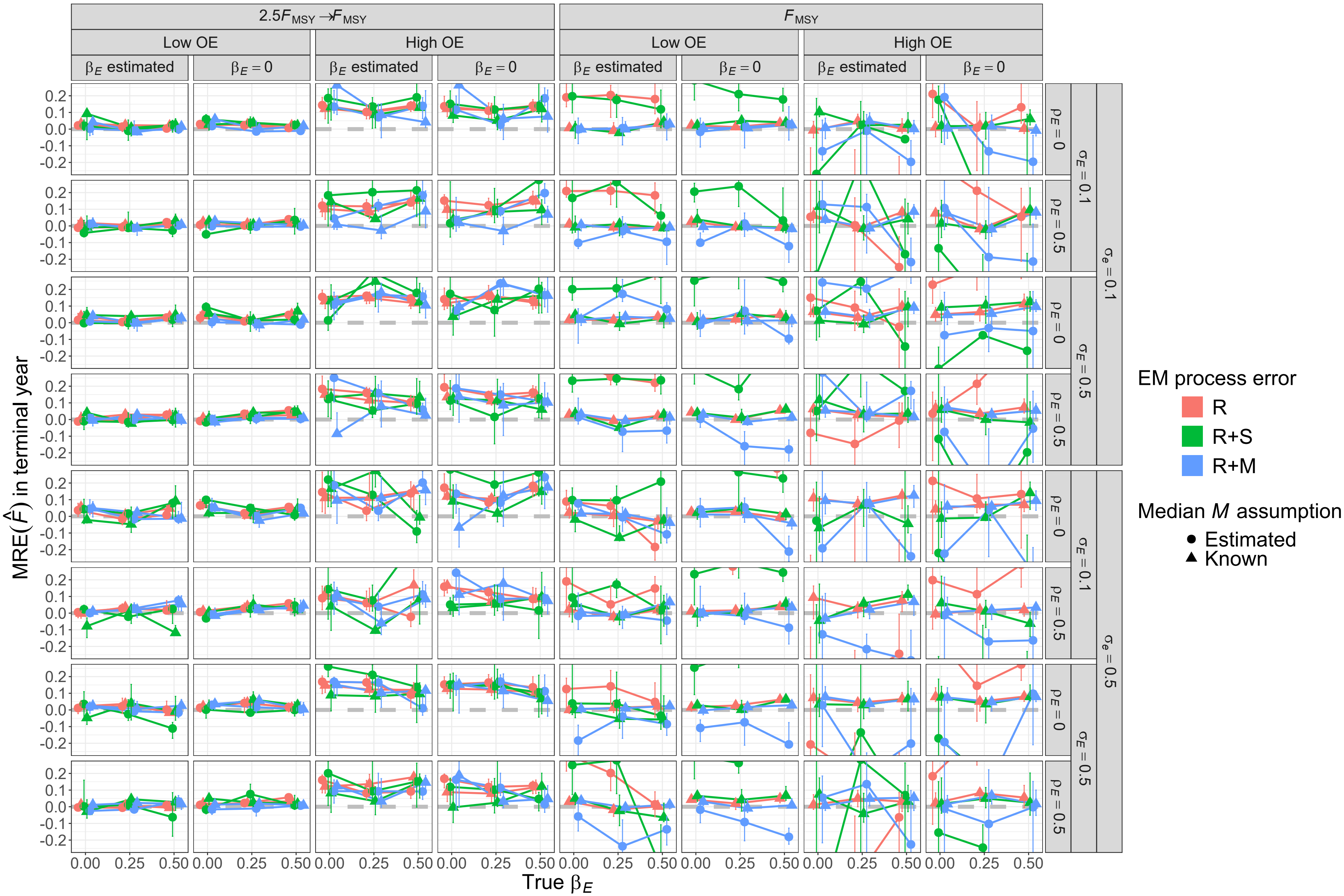

Fig. S56. For R+M operating models, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

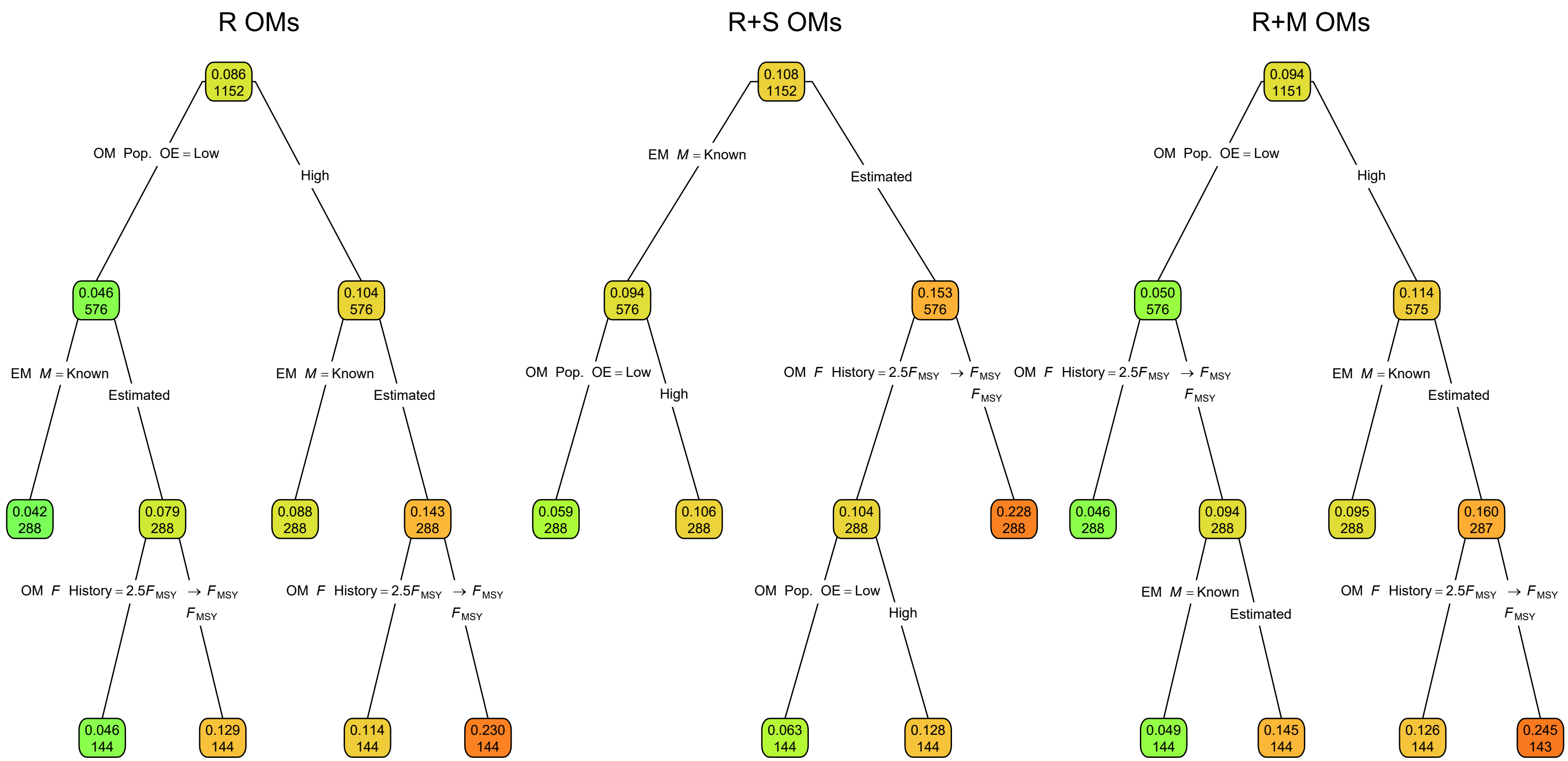

Fig. S57. Regression trees for the log-transformed RMSE of terminal year fishing mortality rate. Each node shows the median error for the corresponding subset. Lower or higher median RMSE is indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

Table S18. Percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year fishing mortality rate. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	11.29	25.63	13.46
OM Obs. Error	37.27	21.42	35.47
OM σ_e	<0.01	0.02	0.02
OM σ_E	0.09	0.03	0.14
OM ρ_E	0.02	0.01	0.04
OM β_E	0.15	0.03	0.21
EM Process Error	0.42	0.75	0.63
$EM\beta_E$ assumption	0.04	<0.01	0.04
EM M assumption	25.38	26.10	24.68
All factors	74.65	73.99	74.76
+ All Two Way	92.28	91.29	91.64
+ All Three Way	94.31	95.51	93.73

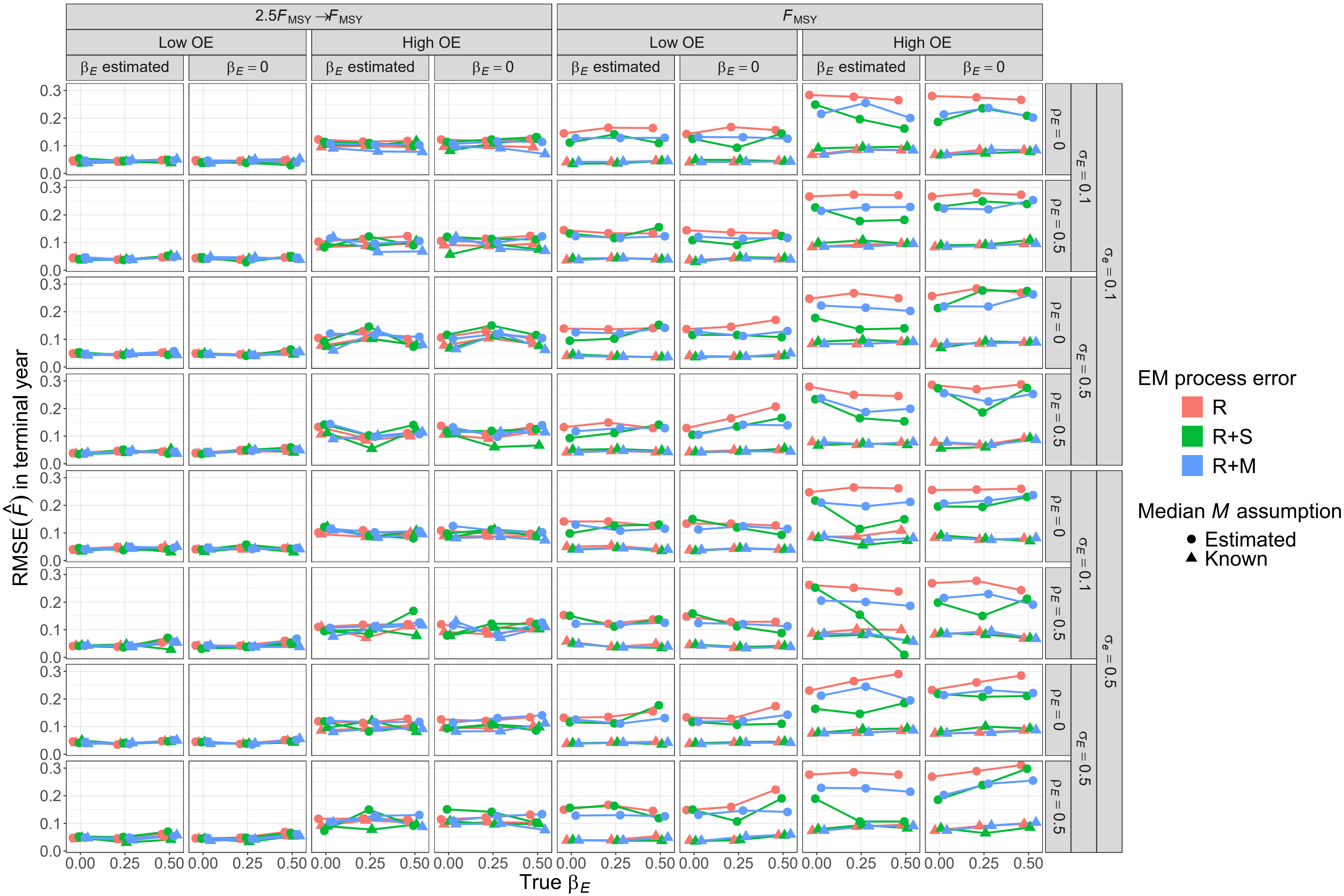

Fig. S58. For R operating models, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

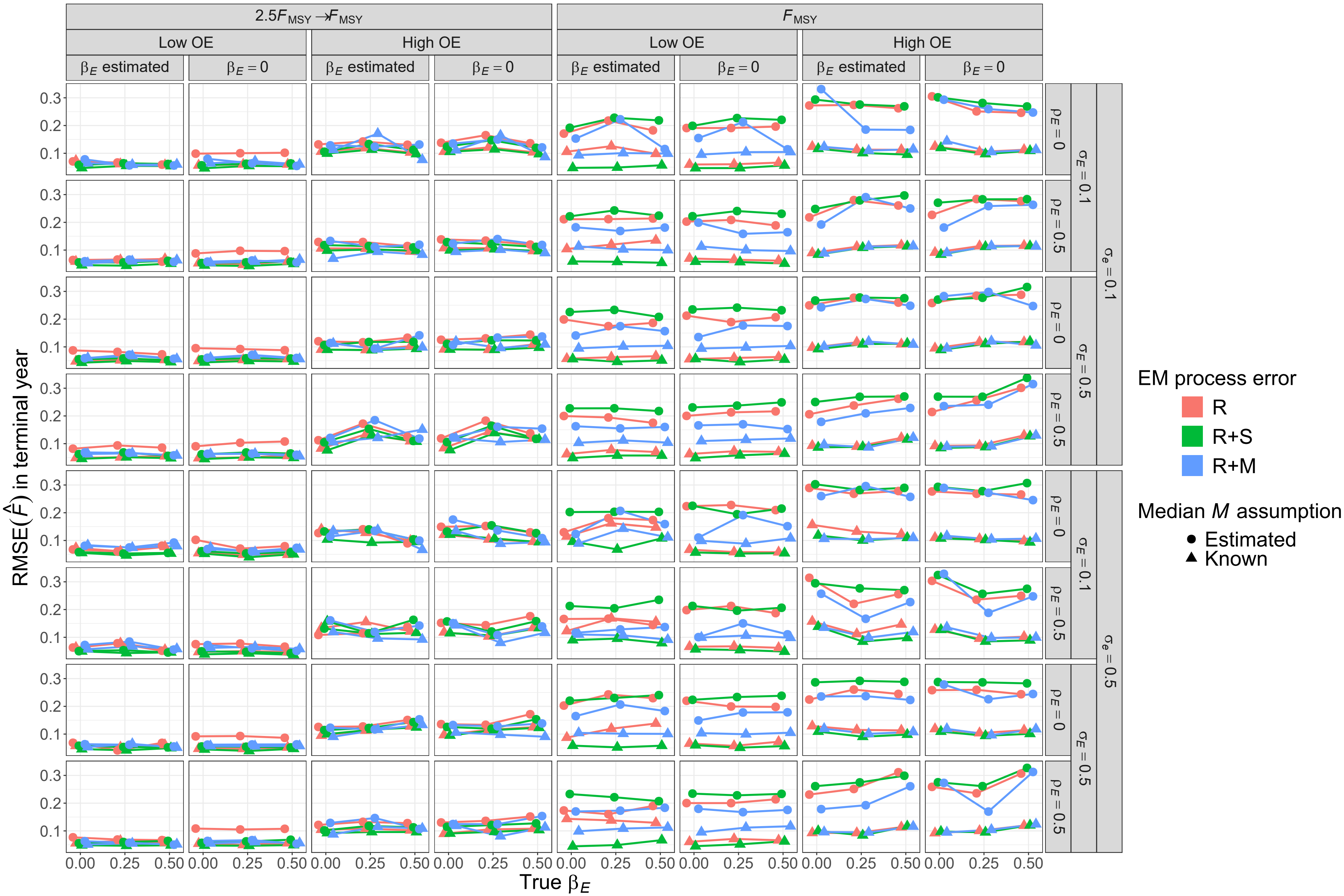

Fig. S59. For R+S operating models, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

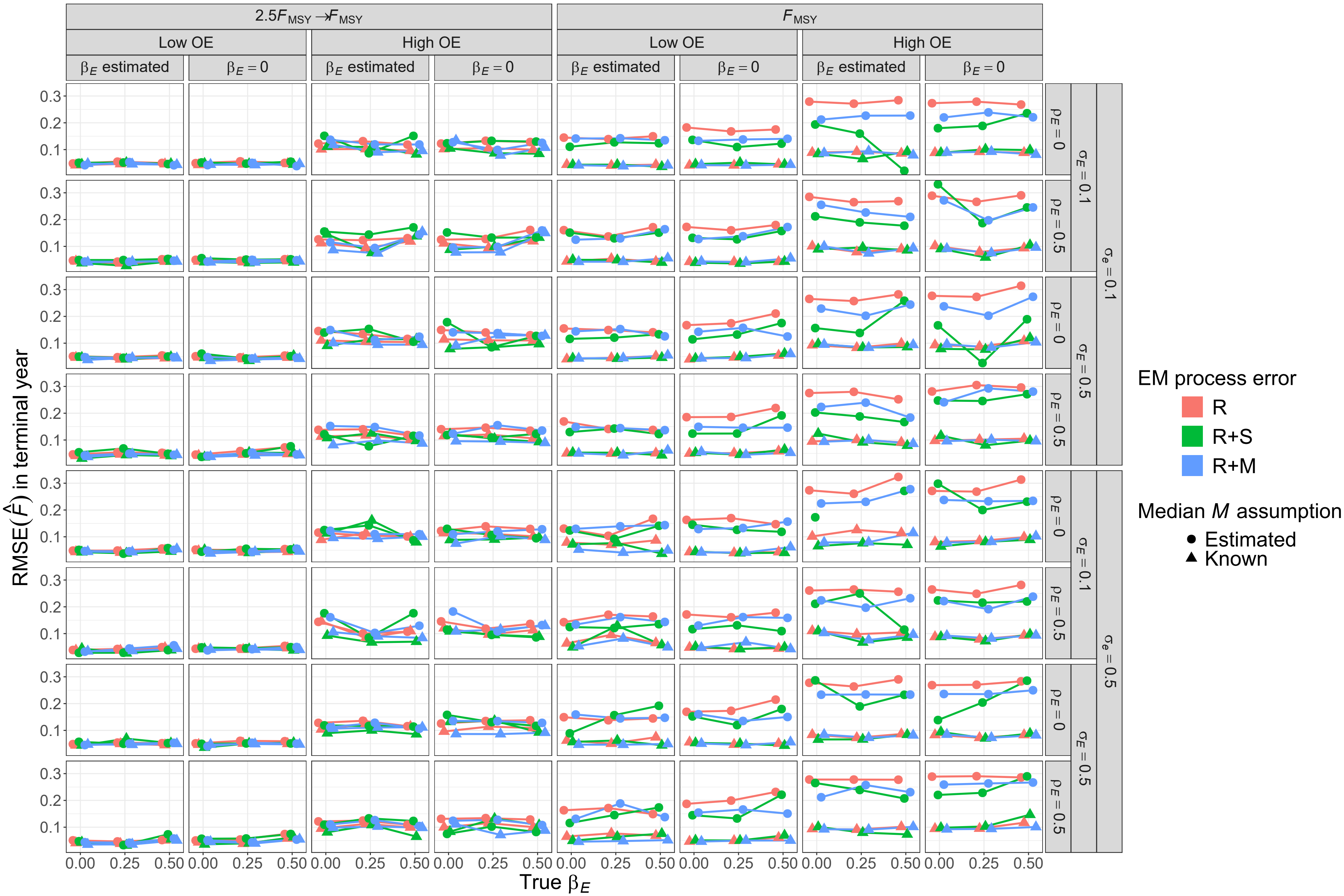

Fig. S60. For R+M operating models, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).